

IMO Shortlists

2000-2016

Contestant's Compilation

IMO Training-Hong Kong Team

All problems used in this document are from shortlists of International Mathematics Olympiad, and thus are works of their respective proposers.

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Preface

During my own preparation for the IMO, practicing IMO is the defining method, which is evidently not a common practice. I feel extra support should be given to those who have the determination to actively seek problems for training, which sets my motivation for compiling this document. This document aims to enhance shortlist-practicing experience.

IMO shortlist is not always correctly ordered. One moment you can solve a C6, the next you cannot solve a C2. Unless one can consistently solve all problems, this can lead to serious frustration.

This arrangement aims to provide a smooth sense of progression. The feelings of slowly solving higher rated questions is highly rewarding. Indeed, sense of progression is what keeps everyone in what they desire. With similar ideas grouped together, students can reinforce memory and learn more out of each question by pairwise comparisons.

An additional benefit is that it serves as a fine reference for training materials preparations, possibly enhancing training quality.

Unfortunately, the final IMO did not go my way(*) . This document would have been much more convincing had it been otherwise. I could only ask for readers' trust on this aspect.

I apologize in advance for probable grammatical mistakes, due to my poor linguistic abilities. Having the list compiled by one person, mistakes and inaccuracies are unavoidable. We welcome all types of feedback , even as minor as inaccurate ratings.

Arvin Leung(#)

August 2017

(*) The Author entered the team with rank 1 and spent the last two months full-time alike professional athletes for final preparation. No one, seriously, no one, saw that coming.

(#) It is a common practice not to sign notes. However, trainees have to know who to find when they do have feedback.

Students

The booklet contains all recent shortlist problems, organized for self-practices. The first and most important step is (Colour) print it out (in A5 book form). (Trust me, a beginning is half of the work done) Readers can attempt questions whenever time allows. One may ask how could there be so much time? Answer: **Dedication**. Regardless of how busy you are, there are some ways in the world to squeeze such time out, which may or may not include doing problems during *inappropriate* circumstances and/or “create” free time. Secondly, there are long summer holidays, and two months in-between DSE and IMO.

Through a rough approximation, it takes 400 hours to solve every last question (assuming gold strength). This is daunting and is a serious amount. Putting things into perspective however, the overall time you spent in trainings is not any lower.

Of course, training shortlists is only suitable for some, those selected few that enter the very late phrases of IMO trainings and stuck around for at least a year. You are strongly recommended to check out “**Rating and Guidelines**” before proceeding if you have no idea what IMO shortlist is.

There are some rough insights for each category, which is intended give readers a basic idea about the topic. However, they are not refined and may not fully cover the ideas on the respective topic. For this purpose, Readers should prepare himself with a future document.

Allow up to 3 hours for each question. It may be tempting to attempt high difficulty questions, to see if one is up there. However, one learns nothing and is of waste of time if the problems are over his abilities, for in general he cannot make a meaningful attempt.

The difficulty rating and your self perception should give you an idea of whether you are expected to solve a particular problem. If you can't solve a problem which is rated below your level, then likely you are oblivious to certain ideas. You should read solution, actively think and identify your weakness, then follow up with relevant notes on the concepts/ skills. Highly skilled contestants are capable of deducing the missing ideas on the spot, however.

For year 2006-2016, solutions can be found on official website <https://www.imo-official.org> , Problem , shortlist. Solutions are official and well organized. For all other years, only community solutions, which may or may not be correct, are available, in <https://artofproblemsolving.com> , contest collections, International.

If you desire more problems, of various levels, art of problem solving is also a good site for its collection of problems.

If you can't solve a problem that is at or above your level, then it is somewhat to be expected. These problems should be left for another day.

As long as you keep on training, there will be improvements. You won't learn enough to solve a 4 immediately after solving a 3, but everything adds up.

In general, problem difficulty increases much faster than the improvements one can make doing them. It is natural to feel stuck at a certain level. Be willing to acknowledge this and find notes and problems available online for more ideas.

If you think you are prepared, such as by solving all problems here, attempt the Mastery section. Afterwards, prepare for greatness, but keep in mind of the **Contest Curse**.

Trainers/Teachers

The shortlist problems are grouped according to ideas. This allows for a direct search in the shortlist for relevant questions. Considering these problems and find important ideas would be ideal when preparing training notes.

The problems also give a comprehensive description on what problems could be asked in the IMO, as well as how problems of different difficulty feels like.

An appropriate sub collection of this document can produce some high quality mock exam and problem sets, or aid the search of questions with similar nature.

Rating and Guidelines

IMO shortlist is a list of problems prepared for each year of IMO. 6 problems are chosen for the actual IMO contest (Frequently ranked around 1,4,6,2,5,7). To ensure existence of choice, the shortlist represents a wide range of difficulty, spreading from strength of 1 to 8. This feature makes the list ideal for practicing for any high level competitor.

How Difficult Is The Shortlist

In brief, **UNIMAGINABLY DIFFICULT**.

Fortunately by having this document, you should have some success in the local math subjects, maybe M2, maybe a prize in the local Pui ching or IMO-Prelim. You can probably imagine, at the very least.

IMO shortlist is the hardest problems that can be realistically solved within 1-2 hours by human. Seriously, this is of international championship level. At least 7 orders of magnitude harder than local math-subjects. A gold medal in IMO prelim is only equivalent to level 1 in this list, out of 8.

If you cannot solve any problems in the **Hong Kong Team Selection Tests**, you are absolutely **NOT** prepared for even the easiest problems in this list. Problems in this list is at least twice as hard. Other problem sets (To be constructed), or contests problems, found in Art of Problem Solving, such as China Girls Math Olympiad / China Western Mathematical Olympiad may be more suitable. Also, you can still learn a lot from trainings organized by HKAGE and IMOHKC.

Rating Descriptions

Each problem is rated a number from 1 to 8, representing the general expected strength of a problem with such shortlisted number. However, shortlist number may deviate from actual difficulty by a lot, examples include 16G2,05N6. Therefore, the numbers are simply my own perception of an 'average'.

The only reasonable definition would be by examples, i.e., by the ratings given to the shortlist problems.... There is no choice but to include this reference loop in

ratings.

General Grade descriptions:

- 1: Standard problems with straight forward and simple solution.
- 4: Include some special and non-conventional ideas. Require non-intuitive steps.
- 7: Exotic and unseen ideas. Long and complicated solutions. Require multiple discoveries of hidden ideas.

The ratings is generally chosen to represent a wide range of feasible solutions, and also how realistic it is to come up with such solution. Difficulty of the problem is clearly subjective. However, since almost all problems (Not less than the solve rate shown in Goals:Perfect score) have been solved, I have a comprehensive idea of difficulty which can at least provide a consistent ratings.

I regret to say that it is not possible to responsibly give a rating to certain classes of geometry problems, reason simply being I have not solved them myself. To indicate this, an underscore next to rating implies it is a 'Natural' rating, i.e., the shortlist number, and relative difficulty of the shortlist from the respective year. We would be grateful if any individuals who have the abilities and determination to complete all the geometry list would provide ratings based on his perceptions.

Training Targets

Time

In contest conditions, taking too long to solve a question would be meaningless, for it eliminated the opportunity to attempt more questions. Frequently, a tedious problem 2/5 would prevent even the best contestants from solving 3/6, regardless of its actual difficulty. The very best contestant must keep solving time down for every single question, difficult or easy.

These only apply when you are preparing for IMO, of course. In general, solving a problem, no matter how long it takes, is still better than reading solutions. Remember, math is more than contest prizes.

The following recommended timings are for reference. Note that time for organizing solutions is not accounted.

Solving Rate

The following gives an approximation on your ability based on ratio of problems solved. Note that this applies for 2016/17 mostly. IMO difficulty, as always, is projected to increase.

ACGN (Time)	Bronze	Silver	Gold	Perfect score/ HM
1	90	60	20	5
2/3	120	90	45	10
4	--	120	60	15
5	--	180	90	30
6	--	--	150	45
7	--	--	180	60
8	--	--	--	90

ACGN	B	S	G	42/ HM
1	70%	99%	99%	99%
2/3	50%	70%	90%	99%
4	--	50%	70%	90%
5	--	30%	50%	90%
6	--	--	20%	80%
7	--	--	20%	70%
8	--	--	--	50%

Curse of Contest

Observant readers may have wondered about such high requirements for honorable mentions.

It is hard to believe that the world is not deterministic, that high abilities do not always lead to high performance in contest. But now I believe. Those who are aiming very high should do the same.

Being capable is a necessary condition for an insane performance. Indeed, that's where most contestants fall. However, it is not sufficient. Regardless of how well you are prepared, you can only be prepared for 99% of the circumstances, and there is no guarantee that 6 problems do not all lie on exactly the 1%. This does not mean one have to accept the fate and give up, but instead, properly acknowledge this possibility and prepare accordingly. That means, one must take extra steps to minimize impacts of bad luck. The following items are of the greatest importance in these aspects.

(1) **State of mind.** To perform at the highest level, the body must be in optimal condition. Plan sufficient sleep time, and adjust for jet lag early if possible. Furthermore, exercise regularly for improved circulation. 4.5 hours of excess brain activity is hard to bear otherwise.

(2) **Stress.** Under extreme contest stress, one could forget everything he had learnt. There is no effective way to reduce stress on the spot- it is already too late. Instead, during preparations one must simulate contest environments as much as possible. The more times you experience contest conditions, the less likely you will panic.

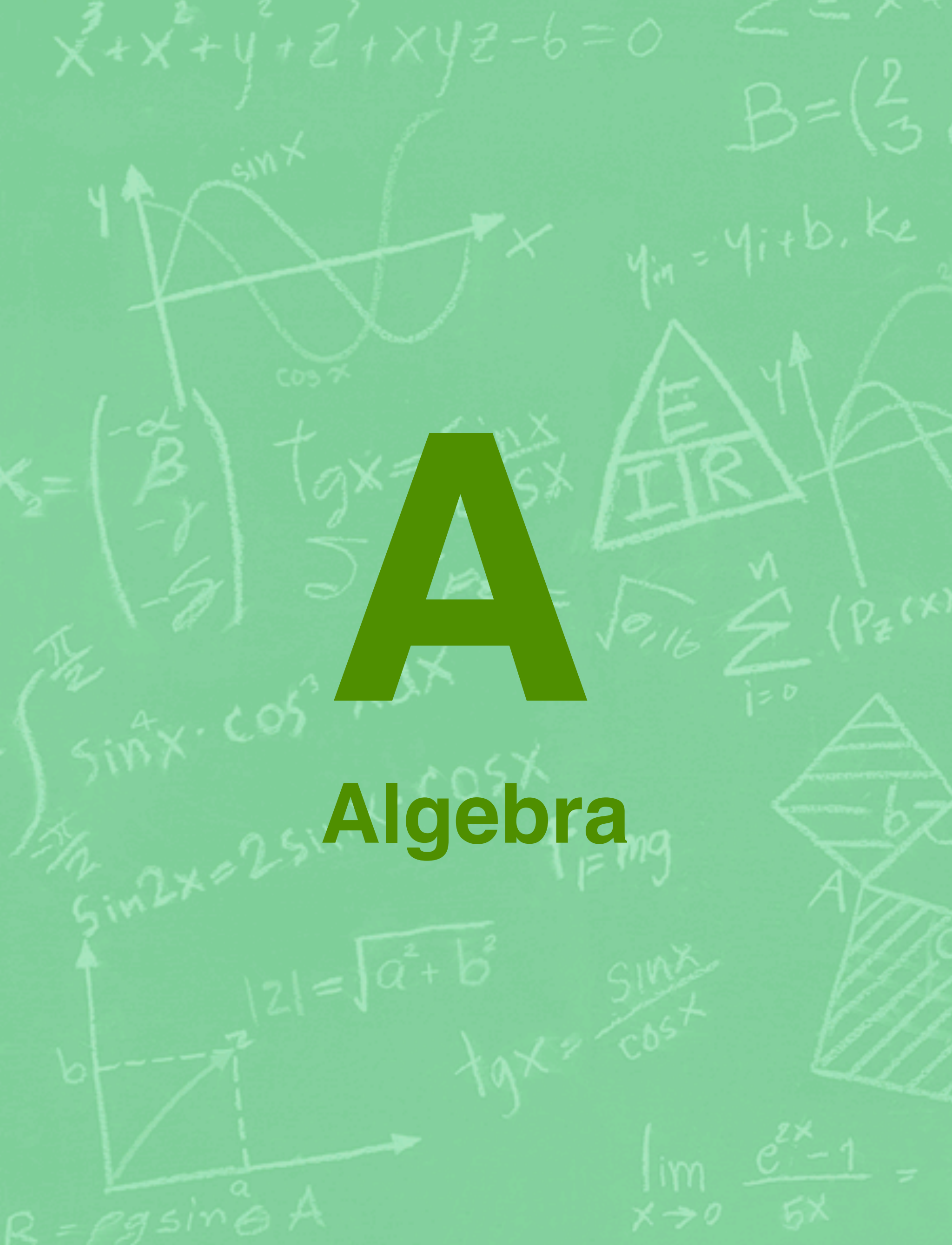
(3) **Contest skills.** One must prepare for the worst case scenario - Let's say you are aiming perfect score but yet you can't solve any of the 3 questions 2 hours in. Now what? We outline the most important items for countering this:

(A) Think about what did you assume to be a necessary step. Take a step backward.

(B) Write down what methods have you tried. It is probably none of those. Think of more before coming back. Think of new perspective to make use of each of the method.

(C) Read the questions once again, think about if you have properly make use of all conditions, and focus on those you haven't.

(More information will be available in a future separate document)



A

Algebra

Introductory Level

Description

A Collection of problems that showcase elementary ideas in algebra.

Main Idea

Most algebra problems ask for instinctive sense of sizes, including growth rate of an expression, behaviour of when variable changes. ideas of linear, polynomial, exponential functions.

Although less prominent than in combinatorics problems, algebra problems also ask for combinatorial ideas. pigeonhole principle, proof by contradiction, appropriate case splitting, induction, counting, bounding, invariant. are all essential.

Ways To Improve

To build an image of these concepts in brain, solving sufficiently many problems would do, but likely more than problems in this list. To accelerate this process, any university year level 1 analysis textbook would be an alternative.

A combinatorial introduction to algebra

1

Real numbers $a_1, a_2, \dots, a_n$ are given. For each $i, (1 \leq i \leq n)$ define $d_i = \max\{a_j \mid 1 \leq j \leq i\} - \min\{a_j \mid i \leq j \leq n\}$, and let $d = \max\{d_i \mid 1 \leq i \leq n\}$

07A1 (a) Prove that, for any real numbers $x_1 \leq x_2 \leq \dots \leq x_n$,

$$\max\{|x_i - a_i| \mid 1 \leq i \leq n\} \geq \frac{d}{2}. \quad (*)$$

(b) Show that there are real numbers $x_1 \leq x_2 \leq \dots \leq x_n$ such that the equality holds in (*).

2

Let $a_1, a_2, \dots, a_n$ be distinct positive integers, $n \geq 3$. Prove that there exist distinct indices i and j such that $a_i + a_j$ does not divide any of the numbers $3a_1, 3a_2, \dots, 3a_n$.

08N2

2

Given any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers, we denote the sum $a_1 + a_2 + a_3 + a_4$ by s_A . Let n_A denote the number of pairs (i, j) with $1 \leq i < j \leq 4$ for which $a_i + a_j$ divides s_A . Find all sets A of four distinct positive integers which achieve the largest possible value of n_A .

11A1

2

Prove that in any set of 2000 distinct real numbers there exist two pairs $a > b$ and $c > d$ with $a \neq c$ or $b \neq d$, such that $\left| \frac{a-b}{c-d} - 1 \right| < \frac{1}{100000}$.

13A2

4

Find all positive integers n such that the following statement holds: Suppose real numbers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ satisfy $|a_k| + |b_k| = 1$ for all $k = 1, \dots, n$. Then there exists $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$, each of which is either -1 or 1 , such that

16A3

$$\left| \sum_{i=1}^n \varepsilon_i a_i \right| + \left| \sum_{i=1}^n \varepsilon_i b_i \right| \leq 1.$$

Trivial Inequalities

1

00A1

Let a, b, c be positive real numbers so that $abc = 1$. Prove that

$$\left(a - 1 + \frac{1}{b}\right) \left(b - 1 + \frac{1}{c}\right) \left(c - 1 + \frac{1}{a}\right) \leq 1.$$

2

08A2

Prove that $\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1$

(a) For all real numbers x, y, z each different from 1, and satisfying $xyz = 1$

(b) Prove that equality holds above for infinitely many triples of rational numbers x, y, z , each different from 1, and satisfying $xyz = 1$.

“Obvious” Functional

Find all functions

$f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that $\frac{(f(w))^2 + (f(x))^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$ for all positive real numbers

w, x, y, z satisfying $wx = yz$

1

08A1

Find all function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$ the following equality holds

$$f(\lfloor x \rfloor y) = f(x) \lfloor f(y) \rfloor$$

where $\lfloor a \rfloor$ is greatest integer not greater than a

1

10A1

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$, satisfying

$$f(xy)(f(x) - f(y)) = (x - y)f(x)f(y) \text{ for all } x, y.$$

3

01A4

Existential Combinatorics with algebra essence

2
00A2

Let a, b, c be positive integers satisfying the conditions $b > 2a$ and $c > 2b$. Show that there exists a real number λ with the property that all the three numbers $\lambda a, \lambda b, \lambda c$ have their fractional parts lying in the interval $\left(\frac{1}{3}, \frac{2}{3}\right]$.

2
03A1

Let a_{ij} (with the indices i and j from the set $\{1, 2, 3\}$) be real numbers such that $a_{ij} > 0$ for $i = j$; $a_{ij} < 0$ for $i \neq j$. Prove the existence of positive real numbers c_1, c_2, c_3 such that the numbers

$$a_{11}c_1 + a_{12}c_2 + a_{13}c_3$$
$$a_{21}c_1 + a_{22}c_2 + a_{23}c_3$$
$$a_{31}c_1 + a_{32}c_2 + a_{33}c_3$$

are either all negative, or all zero, or all positive.

Proving the obvious

A sequence of real numbers $a_0, a_1, a_2, \dots$ is defined by the formula $a_{i+1} = [a_i] \cdot \langle a_i \rangle$ for $i \geq 0$; here a_0 is an arbitrary real number, $[a_i]$ denotes the greatest integer not exceeding a_i , and $\langle a_i \rangle = a_i - [a_i]$. Prove that $a_i = a_{i+2}$ for i sufficiently large.

1
06A1

Let $a_0, a_1, a_2, \dots$ be an infinite sequence of real numbers satisfying the equation $a_n = |a_{n+1} - a_{n+2}|$ for all $n \geq 0$, where a_0 and a_1 are two different positive reals.

2
04A2

Can this sequence $a_0, a_1, a_2, \dots$ be bounded?

Define the function $f : (0, 1) \rightarrow (0, 1)$ by $f(x) = x + \frac{1}{2}$ if $x < \frac{1}{2}$, x^2 if $x \geq \frac{1}{2}$. Let a and b be two real numbers such that $0 < a < b < 1$. We define the sequences a_n and b_n by $a_0 = a, b_0 = b$, and $a_n = f(a_{n-1}), b_n = f(b_{n-1})$ for $n > 0$. Show that there exists a positive integer n such that $(a_n - a_{n-1})(b_n - b_{n-1}) < 0$.

2
14A2

Let T denote the set of all ordered triples (p, q, r) of nonnegative integers. Find all functions $f : T \rightarrow \mathbb{R}$ satisfying

$$f(p, q, r) = \begin{cases} 0 & \text{if } pqr = 0, \\ 1 + \frac{1}{6}(f(p+1, q-1, r) + f(p-1, q+1, r) \\ + f(p-1, q, r+1) + f(p+1, q, r-1) \\ + f(p, q+1, r-1) + f(p, q-1, r+1)) & \text{otherwise} \end{cases}$$

2
01A1

for all nonnegative integers p, q, r .

Construct, or Contradict

2

08A3

Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a Spanish Couple on S , if they satisfy the following conditions:

- (i) Both functions are strictly increasing,
- (ii) The inequality $f(g(g(x))) < g(f(x))$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple

- (a) on the set $S = \mathbb{N}$ of positive integers;
- (b) on the set $S = \{a - \frac{1}{b} : a, b \in \mathbb{N}\}$

2

12A2

a) Does there exist a partition of $\mathbb{Z}$ into three non-empty subsets A, B, C such that the sets $A + B, B + C, C + A$ are disjoint?

b) Does there exist a partition of $\mathbb{Q}$ into three non-empty subsets A, B, C such that the sets $A + B, B + C, C + A$ are disjoint?

Here $X + Y$ denotes the set $\{x + y : x \in X, y \in Y\}$, for $X, Y \subseteq \mathbb{Z}$ and for $X, Y \subseteq \mathbb{Q}$.

2

03A3

Consider two monotonically decreasing sequences (a_k) and (b_k) , where $k \geq 1$, and a_k and b_k are positive real numbers for every k . Now, define the sequences

$$c_k = \min(a_k, b_k)$$

$$A_k = a_1 + a_2 + \dots + a_k$$

$$B_k = b_1 + b_2 + \dots + b_k$$

$$C_k = c_1 + c_2 + \dots + c_k$$

for all natural numbers k .

(a) Do there exist two monotonically decreasing sequences (a_k) and (b_k) of positive real numbers such that the sequences (A_k) and (B_k) are not bounded, while the sequence (C_k) is bounded?

(b) Does the answer to problem (a) change if the sequence (b_k) must be

$$b_k = \frac{1}{k} \text{ for all } k?$$

3

04A3

Does there exist a function $s: \mathbb{Q} \rightarrow \{-1, 1\}$ such that if x and y are distinct rational numbers satisfying $xy = 1$ or $x + y \in \{0, 1\}$, then $s(x)s(y) = -1$?

4

11A5

Prove that for every positive integer n , the set $\{2, 3, 4, \dots, 3n + 1\}$ can be partitioned into n triples in such a way that the numbers from each triple are the lengths of the sides of some obtuse triangle.

4

16A6

The equation $(x - 1)(x - 2) \cdots (x - 2016) = (x - 1)(x - 2) \cdots (x - 2016)$ is written on the board, with 2016 linear factors on each side. What is the least possible value of k for which it is possible to erase exactly k of these 4032 linear factors so that at least one factor remains on each side and the resulting equation has no real solutions?

Insights

A combinatorial introduction to algebra

Most problems have combinatorial elements as backbone, while sometimes algebraic elements set the theme, mostly comparison and bounding of sizes. These problems therefore demand instinctive sense to sizes.

Strategy: As with most combinatorics questions, general problem solving strategy applies. Explore small cases. Think about results that could make use of the given conditions the most. Try to go with the flow of the problem, i.e. when something special is found, think of a natural way to make use of it. These are guided by Experience and abilities.

Trivial Inequalities

These inequalities can be handled purely by tricks from books or standard manipulations. This is purely a test of knowledge and simple concepts about inequalities. Accurate and swift algebraic manipulation will result in quick solution.

Strategy: There are incredible amount of resources available for inequalities, due to its predictable and systematic nature. Try to read up on more techniques and tools. The solving of inequalities itself is supposed to be trivial.

“Obvious” Functional

These functional have very strong problem statements. The catch is illustrating exactly why it is obvious following the flow of the question. Repeatedly, it requires one to find simple implication that give information.

Existential Combinatorics with algebra essence

Existential Combinatorics require the proof something satisfying the requirements exist. As seen in combinatorics section, there are two general methods: derive a contradiction if such configurations fail to exist, or describe an algorithm, that must lead to construction of the required configuration.

Proving the obvious

“Proving the obvious” problems requires the proving of a rather weak statement- a statement that is true as many difficult requirements must be satisfied otherwise, likely infinitely many times. Contradiction problems require ones to illustrate these requirements. The key lies on observing what would go wrong otherwise.

Algebra aspect simply sets the theme and ideas for these combinatorial problems.

Construct or Contradict

Instead of asking for a proof, these questions ask a question. If one is convinced the answer is yes, then he in general should give a construction. Otherwise, he must derive a contradiction. These two paths do not cross, thus having the correct answer in mind is essential to success. Therefore, an essential quality is to not assume or fixate on an answer without a good reason to support.

It is generally said that deriving a contradiction is more interesting of a contest problem than a construction. Nonetheless, similar to “unexpected hanging paradox”, we cannot assume this always holds.

Algebra aspect simply sets the theme and ideas for these combinatorial problems.

Conventional Algebra

Description

A collection of mainstream middle-tier algebra problems.

Main Idea

Conventional algebra problems consist of mainly functional equations and inequalities. Both functional equations and inequalities require cumbersome steps and likely non-intuitive. One cannot count on his problem solving skills and wishful/intuition to reliably figure out these problems, except with excess practices and understanding every last idea. Otherwise, The only counter is to improve on calculation fortitudes.

Building upon the tier 1 algebra problems, these problems now demand decent calculation fortitude, that is , the ability to accurately and swiftly carry out algebraic manipulations. Calculation fortitudes enable one to attempt more directions within the same amount of time, ultimately leading to a greater probability of solving.

Ways To Improve

Improve calculation fortitude, which relies on practicing these conventional problems (Functional Equations are strongly recommended) and actively focus on improving speed of algebraic manipulation.

Maximum-Possible-combinatorial problems with algebraic settings

1

09A1

Find the largest possible integer k , such that the following statement is true:

Let 2009 arbitrary non-degenerated triangles be given. In every triangle the three sides are coloured, such that one is blue, one is red and one is white.

Now, for every colour separately, let us sort the lengths of the sides. We

obtain $b_1 \leq b_2 \leq \dots \leq b_{2009}$ the lengths of the blue sides

$r_1 \leq r_2 \leq \dots \leq r_{2009}$ the lengths of the red sides

$w_1 \leq w_2 \leq \dots \leq w_{2009}$ the lengths of the white sides

Then there exist k indices j such that we can form a non-degenerated triangle with side lengths b_j, r_j, w_j .

2

13C1

Let n be a positive integer. Find the smallest integer k with the following property: Given any real numbers $a_1, \dots, a_d$ such that $a_1 + a_2 + \dots + a_d = n$ and $0 \leq a_i \leq 1$ for $i = 1, 2, \dots, d$, it is possible to partition these numbers into k groups (some of which may be empty) such that the sum of the numbers in each group is at most 1.

2

16A2

Find the smallest constant $C > 0$ for which the following statement holds: among any five positive real numbers a_1, a_2, a_3, a_4, a_5 (not necessarily distinct), one can always choose distinct subscripts i, j, k, l such that

$$\left| \frac{a_i}{a_j} - \frac{a_k}{a_l} \right| \leq C.$$

For a sequence $x_1, x_2, \dots, x_n$ of real numbers, we define its price as $\max_{1 \leq i \leq n} |x_1 + \dots + x_i|$. Given n real numbers, Dave and George want to arrange them into a sequence with a low price. Diligent Dave checks all possible ways and finds the minimum possible price D . Greedy George, on the other hand, chooses x_1 such that $|x_1|$ is as small as possible; among the remaining numbers, he chooses x_2 such that $|x_1 + x_2|$ is as small as possible, and so on. Thus, in the i -th step he chooses x_i among the remaining numbers so as to minimise the value of $|x_1 + x_2 + \dots + x_i|$. In each step, if several numbers provide the same value, George chooses one at random. Finally he gets a sequence with price G .

Find the least possible constant C such that for every positive integer n , for every collection of n real numbers, and for every possible sequence that George might obtain, the resulting values satisfy the inequality $G \leq cD$.

Conventional Inequalities

4

09A2

Let a, b, c be positive real numbers such that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = a + b + c$. Prove that: $\frac{1}{(2a + b + c)^2} + \frac{1}{(a + 2b + c)^2} + \frac{1}{(a + b + 2c)^2} \leq \frac{3}{16}$.

4

01A6

Prove that for all positive real numbers a, b, c , $\frac{a}{\sqrt{a^2 + 8bc}} + \frac{b}{\sqrt{b^2 + 8ca}} + \frac{c}{\sqrt{c^2 + 8ab}} \geq 1$.

5

04A5

Let a, b, c be positive real numbers such that $ab + bc + ca = 1$, prove that $\sqrt[3]{\frac{1}{a} + 6b} + \sqrt[3]{\frac{1}{b} + 6c} + \sqrt[3]{\frac{1}{c} + 6a} \leq \frac{1}{abc}$.

5

05A5

Let x, y, z be positive real numbers such that $xyz \geq 1$. Prove that $\frac{x^5 - x^2}{x^5 + y^2 + z^2} + \frac{y^5 - y^2}{x^2 + y^5 + z^2} + \frac{z^5 - z^2}{x^2 + y^2 + z^5} \geq 0$.

5 Let a, b, c be positive real numbers such that $ab + bc + ca \leq 3abc$. Prove that

09A4
$$\sqrt{\frac{a^2 + b^2}{a + b}} + \sqrt{\frac{b^2 + c^2}{b + c}} + \sqrt{\frac{c^2 + a^2}{c + a}} + 3 \leq \sqrt{2} \left(\sqrt{a + b} + \sqrt{b + c} + \sqrt{c + a} \right)$$

5 If a, b, c are the sides of a triangle, prove that

06A5
$$\frac{\sqrt{b + c - a}}{\sqrt{b} + \sqrt{c} - \sqrt{a}} + \frac{\sqrt{c + a - b}}{\sqrt{c} + \sqrt{a} - \sqrt{b}} + \frac{\sqrt{a + b - c}}{\sqrt{a} + \sqrt{b} - \sqrt{c}} \leq 3$$

5 Given six positive numbers a, b, c, d, e, f such that $a < b < c < d < e < f$. Let $a + c + e = S$ and $b + d + f = T$. Prove that

10A8
$$2ST > \sqrt{3(S + T)(S(bd + bf + df) + T(ac + ae + ce))}.$$

Standard Functional

3 Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

($f(x) + f(z)$) ($f(y) + f(t)$) = $f(xy - zt) + f(xt + yz)$ for all real x, y, z, t .

02A4

3 Determine all pairs of functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$g(f(x + y)) = f(x) + (2x + y)g(y)$ for all real numbers x and y .

11A3

4 Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$f(x)f(y) = 2f(x + yf(x))$

for all positive real numbers x and y .

05A2

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(xy) + 2xy + 1 \text{ for all real numbers } x \text{ and } y.$$

5

05A4

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(-1) \neq 0$ and

$$f(1+xy) - f(x+y) = f(x)f(y) \text{ for all } x, y \in \mathbb{R}.$$

5

12A5

Conventional Multivariable Inequalities

1

04A1

Let $n \geq 3$ be an integer. Let $t_1, t_2, \dots, t_n$ be positive real numbers such that

$$n^2 + 1 > (t_1 + t_2 + \dots + t_n) \left(\frac{1}{t_1} + \frac{1}{t_2} + \dots + \frac{1}{t_n} \right).$$

Show that t_i, t_j, t_k are side lengths of a triangle for all i, j, k with $1 \leq i < j < k \leq n$

1

01A3

Let $x_1, x_2, \dots, x_n$ be arbitrary real numbers. Prove the inequality

$$\frac{x_1}{1+x_1^2} + \frac{x_2}{1+x_1^2+x_2^2} + \dots + \frac{x_n}{1+x_1^2+\dots+x_n^2} < \sqrt{n}.$$

3

07A3

Let n be a positive integer, and let x and y be positive real numbers such that

$$x^n + y^n = 1. \text{ Prove that } \left(\sum_{k=1}^n \frac{1+x^{2k}}{1+x^{4k}} \right) \cdot \left(\sum_{k=1}^n \frac{1+y^{2k}}{1+y^{4k}} \right) < \frac{1}{(1-x) \cdot (1-y)}.$$

3

03A4

Let n be a positive integer and let $x_1 \leq x_2 \leq \dots \leq x_n$ be real numbers.

$$\text{Prove that } \left(\sum_{i,j=1}^n |x_i - x_j| \right)^2 \leq \frac{2(n^2-1)}{3} \sum_{i,j=1}^n (x_i - x_j)^2.$$

Show that the equality holds if and only if $x_1, \dots, x_n$ is an arithmetic sequence.

Prove the inequality

$$\sum_{i < j} \frac{a_i a_j}{a_i + a_j} \leq \frac{n}{2(a_1 + a_2 + \dots + a_n)} \cdot \sum_{i < j} a_i a_j \text{ for positive reals } a_1, a_2, \dots, a_n.$$

4

06A4

Fully-Enclosed Functional

3

03A2

Find all nondecreasing functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

(i) $f(0) = 0, f(1) = 1$;

(ii) $f(a) + f(b) = f(a)f(b) + f(a + b - ab)$ for all real numbers a, b such that $a < 1 < b$.

3

13A3

Let $\mathbb{Q}_{>0}$ be the set of all positive rational numbers. Let $f : \mathbb{Q}_{>0} \rightarrow \mathbb{R}$ be a function satisfying the following three conditions:

(i) for all $x, y \in \mathbb{Q}_{>0}$, we have $f(x)f(y) \geq f(xy)$;

(ii) for all $x, y \in \mathbb{Q}_{>0}$, we have $f(x + y) \geq f(x) + f(y)$;

(iii) there exists a rational number $a > 1$ such that $f(a) = a$.

Prove that $f(x) = x$ for all $x \in \mathbb{Q}_{>0}$.

4

03A5

Let $\mathbb{R}^+$ be the set of all positive real numbers. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ that satisfy the following conditions:

$f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$ for all $x, y, z \in \mathbb{R}^+$;

$f(x) < f(y)$ for all $1 \leq x < y$.

Additive Functional

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$f(f(x) + y) = 2x + f(f(y) - x)$ for all real x, y

2

02A1

Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$f(x + f(y)) = f(x + y) + f(y)$ for all positive reals x and y .

4

07A4

Insights

Maximum-Possible-combinatorial problems with algebraic theme

These problems require the maximum possible element to be determined, which include the following three aspects: Guessing the answer, Provide a proof, likely a construction, that this is possible, and show that this is the maximum. The catch is flow for a proof in both directions is very different

Therefore one have to keep asking himself: Why am I convinced that this answer must be it? Could I have improved through a better construction/ counter example? For if the guessed answer is wrong, one can never prove the statement.

Conventional inequalities

Conventional Inequalities require the application of the few well known inequalities, in increasing order of strength, AM-GM, cauchy, Holder's, except under some very specific manipulation of terms. Similar to geometry, it requires construction of certain terms, that somehow bridges the LHS and RHS. Since the only way to find out whether a construction would work or not is to go through calculation, the ability to handle a lot of calculations, accurately, in a short time is the most essential quality. Through which, one can experiment with a large number of construction and likely find a successful one.

Fortunately, Classical Inequalities are out of the play field since the powerful Murihead, Schur's, and Sum-of-squares render them mechanical. However, sequence inequality and miraculous types are not.

Standard Functional

Functional equations centers around substituting suitable values, and applying the list of standard ideas such as injective and surjective. For a difficult functional equation, only a few specific substitution is useful. A quality to achieve this is be exhaustive , experiment with everything that is said on books. Along with great calculation fortitudes, time taken to solve a functional would be manageable.

Workspace organization is important for functionals. There are results that are useful down the line with support of future statements, but meaningless at the moment. It would be a significant time cost if these ideas are forgotten when they become useful.

Conventional Multivariable Inequalities

Multivariable inequalities share the same general skills with conventional inequalities. Techniques that handle a sequence of terms, such as summing formula, or induction, will be helpful.

Fully-Enclosed Functional

Fully-Enclosed functionals mean every single variable is wrapped inside one or more function - there is no exposed terms on the equation itself. It is therefore almost impossible to directly conclude the images, under general substitution at least.

Therefore, it worthwhile to attempt the one of two proven line of play, which is likely the only two. 1. Injectivity, 2. Bounded - Denseness (refer to Cauchy equation for details). A special case is applying results from cauchy equation, which simply save a few lines on method 2.

Additive Functional

Additive Functional means that multiplication never appear in any part of the functional. There are 2 implications: Firstly, there are no critical values such as 1,-1, which had value in functionals with multiplication as a multiplicative identity. This implies the entire solution revolves around substituting variables, instead of trivial values.

Unconventional Algebra

Description

This collection includes types of problems that are less main stream, and make use of less well known ideas.

Main Idea

These problems make use of less well-known ideas. Therefore, there are no specific skills to cover all of them. In general, algebra themed problems demand high calculation fortitudes, patience and willingness to carry out tedious calculations, whereas combinatorial themed problems demand great problem solving skills.

Ways To Improve

Be sure to meet all knowledge requirements: standard inequalities and functional tricks, polynomials, and general problem solving skills. Afterwards, one can only improve by practicing, enhancing mental capability.

Recursive-Sequence-Inequality

1
01A2

Let $a_0, a_1, a_2, \dots$ be an arbitrary infinite sequence of positive numbers. Show that the inequality $1 + a_n > a_{n-1} \sqrt[n]{2}$ holds for infinitely many positive integers n .

2
01A5

Find all positive integers $a_1, a_2, \dots, a_n$ such that $\frac{99}{100} = \frac{a_0}{a_1} + \frac{a_1}{a_2} + \dots + \frac{a_{n-1}}{a_n}$, where $a_0 = 1$ and $(a_{k+1} - 1)a_{k-1} \geq a_k^2(a_k - 1)$ for $k = 1, 2, \dots, n-1$.

2
06A2

The sequence of real numbers $a_0, a_1, a_2, \dots$ is defined recursively by $a_0 = -1, \sum_{k=0}^n \frac{a_{n-k}}{k+1} = 0$ for $n \geq 1$. Show that $a_n > 0$ for all $n \geq 1$.

2
14A1

Let $a_0 < a_1 < a_2 \dots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that $a_n < \frac{a_0 + a_1 + a_2 + \dots + a_n}{n} \leq a_{n+1}$.

4
15A1

Suppose that a sequence $a_1, a_2, \dots$ of positive real numbers satisfies $a_{k+1} \geq \frac{ka_k}{a_k^2 + (k-1)}$ for every positive integer k . Prove that $a_1 + a_2 + \dots + a_n \geq n$ for every $n \geq 2$.

Tedious Functional

Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + f(x + y)) + f(xy) = x + f(x + y) + yf(x)$ for all real numbers x and y .

3
15A4

Find all functions

$f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that for any $x, y \in \mathbb{R}^+$,
 $xf(x^2)f(f(y)) + f(yf(x)) = f(xy)(f(f(x^2)) + f(f(y^2)))$.

3

16A4

Find all pairs of functions $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$, such that

$f(x + g(y)) = xf(y) - yf(x) + g(x)$ for all $x, y \in \mathbb{R}$.

4

00A3

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$f(xf(x + y)) = f(yf(x)) + x^2$ for all $x, y \in \mathbb{R}$.

5

09A7

Functional inequalities

5

09A5

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Prove that there exist real numbers x and y such that $f(x - f(y)) > yf(x) + x$

6

11A6

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that satisfies

$f(x + y) \leq yf(x) + f(f(x))$

for all real numbers x and y . Prove that $f(x) = 0$ for all $x \leq 0$

Integer Functional

Determine all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$f(x - f(y)) = f(f(x)) - f(y) - 1$ holds for all $x, y \in \mathbb{Z}$.

2

15A2

Determine all functions f from the set of positive integers to the set of positive integers such that, for all positive integers a and b , there exists a non-degenerate triangle with sides of lengths $a, f(b)$ and $f(b + f(a) - 1)$.

2

09A3

3

07A2

Consider those functions $f : \mathbb{N} \mapsto \mathbb{N}$ which satisfy the condition $f(m+n) \geq f(m) + f(f(n)) - 1$ for all $m, n \in \mathbb{N}$. Find all possible values of $f(2007)$.

3

10A5

Denote by $\mathbb{Q}^+$ the set of all positive rational numbers. Determine all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ which satisfy the following equation for all $x, y \in \mathbb{Q}^+$:

$$f(f(x)^2 y) = x^3 f(xy).$$

4

12A1

Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that, for all integers a, b, c that satisfy $a + b + c = 0$, the following equality holds:

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a).$$

Sequence-Combin-Algebra

1

00A4

The function F is defined on the set of nonnegative integers and takes nonnegative integer values satisfying the following conditions: for every $n \geq 0$,

- (i) $F(4n) = F(2n) + F(n)$,
- (ii) $F(4n + 2) = F(4n) + 1$,
- (iii) $F(2n + 1) = F(2n) + 1$.

Prove that for each positive integer m the number of integers with $0 \leq n < 2^m$ and $F(4n) = F(3n)$ is $F(2^{m+1})$.

2

13A1

Let n be a positive integer and let $a_1, \dots, a_{n-1}$ be arbitrary real numbers. Define the sequences $u_0, \dots, u_n$ and $v_0, \dots, v_n$ inductively by $u_0 = u_1 = v_0 = v_1 = 1$, and $u_{k+1} = u_k + a_k u_{k-1}$, $v_{k+1} = v_k + a_{n-k} v_{k-1}$ for $k = 1, \dots, n-1$.

Prove that $u_n = v_n$.

3

10A4

A sequence $x_1, x_2, \dots$ is defined by $x_1 = 1$ and $x_{2k} = -x_k$, $x_{2k-1} = (-1)^{k+1} x_k$ for all $k \geq 1$. Prove that $\forall n \geq 1$, $x_1 + x_2 + \dots + x_n \geq 0$

Let n be a positive integer, and consider a sequence $a_1, a_2, \dots, a_n$ of positive integers. Extend it periodically to an infinite sequence $a_1, a_2, \dots$ by defining $a_{n+i} = a_i$ for all $i \geq 1$. If $a_1 \leq a_2 \leq \dots \leq a_n \leq a_1 + n$ and $a_{a_i} \leq n + i - 1$ for $i = 1, 2, \dots, n$, prove that $a_1 + \dots + a_n \leq n^2$.

3

13A4

Let n be a positive integer. Given a sequence $\varepsilon_1, \dots, \varepsilon_{n-1}$ with $\varepsilon_i = 0$ or $\varepsilon_i = 1$ for each $i = 1, \dots, n-1$, the sequences $a_0, \dots, a_n$ and $b_0, \dots, b_n$ are constructed by the following rules: $a_0 = b_0 = 1$, $a_1 = b_1 = 7$,

4

09C3

$$a_{i+1} = \begin{cases} 2a_{i-1} + 3a_i, & \text{if } \varepsilon_i = 0, \\ 3a_{i-1} + a_i, & \text{if } \varepsilon_i = 1 \end{cases} \text{ for each } i = 1, \dots, n-1,$$

$$b_{i+1} = \begin{cases} 2b_{i-1} + 3b_i, & \text{if } \varepsilon_{n-i} = 0, \\ 3b_{i-1} + b_i, & \text{if } \varepsilon_{n-i} = 1 \end{cases} \text{ for each } i = 1, \dots, n-1.$$

Prove that $a_n = b_n$.

For an integer m , denote by $t(m)$ the unique number in $\{1, 2, 3\}$ such that $m + t(m)$ is a multiple of 3. A function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ satisfies $f(-1) = 0, f(0) = 1, f(1) = -1$ and $f(2^n + m) = f(2^n - t(m)) - f(m)$ for all integers $m, n \geq 0$ with $2^n > m$. Prove that $f(3p) \geq 0$ holds for all integers $p \geq 0$.

4

08A4

.Polynomials

2

05A1

Find all pairs of integers a, b for which there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that product $(x^2 + ax + b) \cdot P(x)$ is a polynomial of a form $x^n + c_{n-1}x^{n-1} + \dots + c_1x + c_0$ where each of $c_0, c_1, \dots, c_{n-1}$ is equal to 1 or -1.

3

02A3

Let P be a cubic polynomial with integer coefficients and non-zero leading coefficient. Suppose that for infinitely many pairs (x, y) of integers with $x \neq y$. Prove that the equation $P(x) = 0$ has an integer root.

3

04A4

Find all polynomials f with real coefficients such that for all reals a, b, c such that $ab + bc + ca = 0$ we have the following relations
 $f(a - b) + f(b - c) + f(c - a) = 2f(a + b + c)$.

4

12A4

Let f and g be two nonzero polynomials with integer coefficients and $\deg f > \deg g$. Suppose that for infinitely many primes p the polynomial $pf + g$ has a rational root. Prove that f has a rational root.

4

02N6

Find all pairs of positive integers $m, n \geq 3$ for which there exist infinitely many positive integers a such that $\frac{a^m + a - 1}{a^n + a^2 - 1}$ is itself an integer.

5

13A6

Let $m \neq 0$ be an integer. Find all polynomials $P(x)$ with real coefficients such that

$$(x^3 - mx^2 + 1)P(x + 1) + (x^3 + mx^2 + 1)P(x - 1) = 2(x^3 - mx + 1)P(x)$$

for all real numbers x .

6

14A5

Consider all polynomials $P(x)$ with real coefficients that have the following property: for any two real numbers x and y one has
 $|y^2 - P(x)| \leq 2|x|$ if and only if $|x^2 - P(y)| \leq 2|y|$. Determine all possible values of $P(0)$.

Insights

Recursive-Sequence-Inequality

No two problems of this type share the same solution. Therefore, ways to improve are general problem solving skills and algebraic manipulation skills.

Tedious Functional

A chaotic functional equation that makes no sense to human brains. One must make sure every substitution is accurate since a mistake is difficult to detect. Consequently, only speed and accuracy of substitution is important.

Functional Inequality

Share the same general ideas with normal functionals except problems are more versatile. One must be sensitive to inequality sign for all time, as some substitutions are not valid due to reversed direction.

Integer-Functional

Functionals that set in integer to integer. General skills remains, however the playing field is very different. Firstly, there are no potential for any non-continuous cauchy-equation solution-like functions. Secondly, an important feature about integer is well-ordering, that is, one can take a minimum over the whole function. Moreover, number theoretical properties of integer may also play a role.

Sequence-combin-Algebra

Algebraic sequences, even though it showcases some very important concepts about combinatorics, mainly the use of induction and small cases explorations. These questions require one to figure out subtle relationships between terms and construct auxiliary statements

Polynomials

These questions require making use of strong properties of polynomials. 1) growth of

Polynomials are very steady. 2) the ability to interpolate a polynomial: that if two polynomial matches enough times, they are identical. 3) Factor theorem and roots related properties. Indeed, the set of roots (including complex, with multiplicity) determine a polynomial.

Advanced Algebra

Description

Problems that is the hardest of all contest problems. Problems that separate the very good contestants from insane ones.

Main Idea

These problems make use of a wide ranges of ideas, skills and techniques, possibly including completely original ones. These are final test of Contest Math Abilities (algebra)

Equip yourself with **GOLD** strength before proceeding. Otherwise one have no realistic chances against most problems, while wasting time and efforts.

Ways To Improve

Unfortunately , there is no reliable ways to improve as these problems are close to human limits. Do make sure you are well familiar with all fundamentals however.

Analytic Algebra

2

02A2

Let $a_1, a_2, \dots$ be an infinite sequence of real numbers, for which there exists a real number c with $0 \leq a_i \leq c$ for all i , such that

$$|a_i - a_j| \geq \frac{1}{i+j} \quad \text{for all } i, j \text{ with } i \neq j.$$

Prove that $c \geq 1$.

3

11A2

Determine all sequences $(x_1, x_2, \dots, x_{2011})$ of positive integers, such that for

every positive integer n there exists an integer a with $\sum_{j=1}^{2011} jx_j^n = a^{n+1} + 1$

6

07A5

Let $c > 2$ and let $a_1, a_2, \dots$ be a sequence of nonnegative real numbers such that $a_{m+n} \leq 2 \cdot a_m + 2 \cdot a_n$ for all $m, n \geq 1$, and $a_{2^k} \leq \frac{1}{(k+1)^c}$ for all $k \geq 0$. Prove

that the sequence a_n is bounded.

6

02A6

Let A be a non-empty set of positive integers. Suppose that there are positive integers $b_1, \dots, b_n$ and $c_1, \dots, c_n$ such that for each i the set

$b_i A + c_i = \{b_i a + c_i : a \in A\}$ is a subset of A , and

the sets $b_i A + c_i$ and $b_j A + c_j$ are disjoint whenever $i \neq j$. Prove that

$$\frac{1}{b_1} + \dots + \frac{1}{b_n} \leq 1.$$

Advanced Integer Functional

Determine all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ satisfying

$$f(f(m) + n) + f(m) = f(n) + f(3m) + 2014 \quad \text{for all integers } m \text{ and } n.$$

4

14A4

Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy $f^{g(n)+1}(n) + g^{f(n)}(n) = f(n+1) - g(n+1) + 1$ for every positive integer n . Here, $f^k(n)$ means $\underbrace{f(f(\dots f(n)\dots))}_k$

4

11A4

Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$n^2 + 4f(n) = f(f(n))^2 \text{ for all } n \in \mathbb{Z}$$

4

14A6

Let $2\mathbb{Z} + 1$ denote the set of odd integers. Find all functions $f : \mathbb{Z} \rightarrow 2\mathbb{Z} + 1$ satisfying $f(x + f(x) + y) + f(x - f(x) - y) = f(x + y) + f(x - y)$ for every $x, y \in \mathbb{Z}$.

5

15A5

Let $\mathbb{Z}_{\geq 0}$ be the set of all nonnegative integers. Find all the functions $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$ satisfying the relation

$$f(f(f(n))) = f(n + 1) + 1 \text{ for all } n \in \mathbb{Z}_{\geq 0}.$$

6

13A5

Advanced Inequalities

7

06A6

Determine the least real number M such that the inequality $|ab(a^2 - b^2) + bc(b^2 - c^2) + ca(c^2 - a^2)| \leq M(a^2 + b^2 + c^2)^2$

holds for all real numbers a, b, c

7

11A7

Let a, b and c be positive real numbers satisfying $\min(a + b, b + c, c + a) > \sqrt{2}$ and $a^2 + b^2 + c^2 = 3$. Prove that

$$\frac{a}{(b + c - a)^2} + \frac{b}{(c + a - b)^2} + \frac{c}{(a + b - c)^2} \geq \frac{3}{(abc)^2}.$$

7

08A7

Prove that for any four positive real numbers a, b, c, d the inequality

$$\frac{(a - b)(a - c)}{a + b + c} + \frac{(b - c)(b - d)}{b + c + d} + \frac{(c - d)(c - a)}{c + d + a} + \frac{(d - a)(d - b)}{d + a + b} \geq 0 \text{ holds.}$$

Determine all cases of equality.

Insane Functional

6

08A6

Let $f : \mathbb{R} \rightarrow \mathbb{N}$ be a function which satisfies $f\left(x + \frac{1}{f(y)}\right) = f\left(y + \frac{1}{f(x)}\right)$ for all $x, y \in \mathbb{R}$. Prove that there is a positive integer which is not a value of f .

6

17Q2
[A6]

Let $\mathbb{R}$ be the set of real numbers. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that, for any real numbers x and y , $f(f(x)f(y)) + f(x + y) = f(xy)$.

7

04A6

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation $f(x^2 + y^2 + 2f(xy)) = (f(x + y))^2$ for all $x, y \in \mathbb{R}$.

7

16A7

Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) \neq 0$ and for all $x, y \in \mathbb{R}$, $f(x + y)^2 = 2f(x)f(y) + \max\{f(x^2 + y^2), f(x^2) + f(y^2)\}$.

8

10A6

Suppose that f and g are two functions defined on the set of positive integers and taking positive integer values. Suppose also that the equations $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ hold for all positive integer n . Prove that $f(n) = g(n)$ for all positive integer n .

Miraculous inequality

Let a, b, c be positive real numbers such that $\min\{ab, bc, ca\} \geq 1$. Prove that

$$\sqrt[3]{(a^2 + 1)(b^2 + 1)(c^2 + 1)} \leq \left(\frac{a + b + c}{3}\right)^2 + 1.$$

3

16A1

Let the real numbers a, b, c, d satisfy the relations $a + b + c + d = 6$ and $a^2 + b^2 + c^2 + d^2 = 12$. Prove that

$$36 \leq 4(a^3 + b^3 + c^3 + d^3) - (a^4 + b^4 + c^4 + d^4) \leq 48.$$

3

10A2

3

08A5

Let a, b, c, d be positive real numbers such that $abcd = 1$ and

$a + b + c + d > \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$. Prove that $a + b + c + d < \frac{b}{a} + \frac{c}{b} + \frac{d}{c} + \frac{a}{d}$

3

05A3

Four real numbers p, q, r, s satisfy $p + q + r + s = 9$ and $p^2 + q^2 + r^2 + s^2 = 21$.

Prove that there exists a permutation (a, b, c, d) of (p, q, r, s) such that $ab - cd \geq 2$.

3

12A3

Let $n \geq 3$ be an integer, and let $a_2, a_3, \dots, a_n$ be positive real numbers such that $a_2 a_3 \cdots a_n = 1$. Prove that

$(1 + a_2)^2 (1 + a_3)^3 \cdots (1 + a_n)^n > n^n$.

4

10A3

Let $x_1, \dots, x_{100}$ be nonnegative real numbers such that $x_i + x_{i+1} + x_{i+2} \leq 1$ for all $i = 1, \dots, 100$ (we put $x_{101} = x_1, x_{102} = x_2$). Find the maximal possible value of the sum $S = \sum_{i=1}^{100} x_i x_{i+2}$.

5

04A7

Let $a_1, a_2, \dots, a_n$ be positive real numbers, $n > 1$. Denote by g_n their geometric mean, and by $A_1, A_2, \dots, A_n$ the sequence of arithmetic means defined by

$$A_k = \frac{a_1 + a_2 + \cdots + a_k}{k}, \quad k = 1, 2, \dots, n.$$

Let G_n be the geometric mean of $A_1, A_2, \dots, A_n$. Prove the inequality

$$n \sqrt[n]{\frac{G_n}{A_n}} + \frac{g_n}{G_n} \leq n + 1 \text{ and establish the cases of equality.}$$

5

15A3

Let n be a fixed positive integer. Find the maximum possible value of

$$\sum_{1 \leq r < s \leq 2n} (s - r - n) x_r x_s, \text{ where } -1 \leq x_i \leq 1 \text{ for all } i = 1, \dots, 2n.$$

8

03A6

Let n be a positive integer and let $(x_1, \dots, x_n)$, $(y_1, \dots, y_n)$ be two sequences of positive real numbers. Suppose $(z_2, \dots, z_{2n})$ is a sequence of positive real numbers such that $z_{i+j}^2 \geq x_i y_j$ for all $1 \leq i, j \leq n$.

Let $M = \max\{z_2, \dots, z_{2n}\}$. Prove that

$$\left(\frac{M + z_2 + \dots + z_{2n}}{2n} \right)^2 \geq \left(\frac{x_1 + \dots + x_n}{n} \right) \left(\frac{y_1 + \dots + y_n}{n} \right).$$

Advanced Algebraic Combin

5

00A5

Let $n \geq 2$ be a positive integer and λ a positive real number. Initially there are n fleas on a horizontal line, not all at the same point. We define a move as choosing two fleas at some points A and B , with A to the left of B , and letting the flea from A jump over the flea from B to the point C so that $\frac{BC}{AB} = \lambda$. Determine all values of λ such that, for any point M on the line and for any initial position of the n fleas, there exists a sequence of moves that will take them all to the position right of M .

5

15C5

The sequence $a_1, a_2, \dots$ of integers satisfies the conditions:

$$1 \leq a_j \leq 2015 \text{ for all } j \geq 1,$$

$$k + a_k \neq \ell + a_\ell \text{ for all } 1 \leq k < \ell.$$

Prove that there exist two positive integers b and N for which

$$\left| \sum_{j=m+1}^n (a_j - b) \right| \leq 1007^2 \text{ for all integers } m \text{ and } n \text{ such that } n > m \geq N.$$

6

12A6

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function, and let f^m be f applied m times. Suppose that for every $n \in \mathbb{N}$ there exists a $k \in \mathbb{N}$ such that $f^{2k}(n) = n + k$, and let k_n be the smallest such k . Prove that the sequence $k_1, k_2, \dots$ is unbounded.

7

09A6

Suppose that $s_1, s_2, s_3, \dots$ is a strictly increasing sequence of positive integers such that the sub-sequences $s_{s_1}, s_{s_2}, s_{s_3}, \dots$ and $s_{s_1+1}, s_{s_2+1}, s_{s_3+1}, \dots$ are both arithmetic progressions. Prove that the sequence $s_1, s_2, s_3, \dots$ is itself an arithmetic progression.

7

10A7

Let $a_1, a_2, a_3, \dots$ be a sequence of positive real numbers, and s be a positive integer, such that $a_n = \max\{a_k + a_{n-k} \mid 1 \leq k \leq n-1\}$ for all $n > s$.

Prove there exist positive integers $\ell \leq s$ and N , such that

$$a_n = a_\ell + a_{n-\ell} \text{ for all } n \geq N.$$

.

Insights

Analytic Algebra

Analytic Algebra problems make use of heavy ideas from analysis. Here are a few general useful analysis idea: 1. Monotone Convergence Theorem. A strictly increasing sequence that is bounded must converge. This theorem is useful for proving many non-convergence 2. Denseness of real numbers: If $a < b$ are two real numbers, then there must exist a number between it. 3. Continuity, which enable one to proves $a \geq b$ without explicitly doing so, as long as $a \geq b' \forall b' < b$. This gives an alternate pathway to proofs with pure algebra.

Advanced Integer Functional ; Advanced Inequalities; Insane Functional

Same as their 'normal' counterparts , but demands mastery over the relevant techniques, as well as general problem solving techniques. Be prepared for the length one must go before discovering a full solution. Let's just say, it demands **GOLD** qualities.

Miraculous inequality

These miraculous inequalities is a set-up questions. Therefore in this particular case, wishful thinking is very effective. Wishful thinking describes thought process that, instead of observing and figuring out what techniques would be appropriate, one instead make wishful guesses "can I transform the inequality to this, which immediately solves the problem".

Advanced Algebraic Combin

As with the case of difficult combinatorics, all questions take on a wide range of techniques, and no two questions share the same idea. Therefore, the most important techniques are general problem solving skills.

C

Combinatorics

Introductory Problems

Description

Collection of combinatorics Problems that explore the elementary ideas around combinatorics

Main Idea

The fundamental skills of combin is to not fixate on anything but instead follow the flow of the question and attempt various skills from your toolbox. General tool box items include appropriate case splitting, induction, counting, colouring, bounding, invariant.

Ways To Improve

The most consistent way is simply do more problems, while actively try to apply problem solving strategies (available in a future document) to build them in your brain. Afterwards, you will apply the methods one by one subconsciously. To improve the ability to pick up correct ideas, think deeply why would a particular idea work or fail for a specific question.

Logical Thinking Problems

1
04C1

There are 10001 students at an university. Some students join together to form several clubs (a student may belong to different clubs). Some clubs join together to form several societies (a club may belong to different societies). There are a total of k societies. Suppose that the following conditions hold:

- i.) Each pair of students are in exactly one club.
- ii.) For each student and each society, the student is in exactly one club of the society.
- iii.) Each club has an odd number of students. In addition, a club with $2m + 1$ students (m is a positive integer) is in exactly m societies.

Find all possible values of k .

1
16C1

The leader of an IMO team chooses positive integers n and k with $n > k$, and announces them to the deputy leader and a contestant. The leader then secretly tells the deputy leader an n -digit binary string, and the deputy leader writes down all n -digit binary strings which differ from the leader's in exactly k positions. (For example, if $n = 3$ and $k = 1$, and if the leader chooses 101 , the deputy leader would write down 001,111 and 100 .) The contestant is allowed to look at the strings written by the deputy leader and guess the leader's string. What is the minimum number of guesses (in terms of n and k) needed to guarantee the correct answer?

1
00C1

A magician has one hundred cards numbered 1 to 100. He puts them into three boxes, a red one, a white one and a blue one, so that each box contains at least one card. A member of the audience draws two cards from two different boxes and announces the sum of numbers on those cards. Given this information, the magician locates the box from which no card has been drawn.

How many ways are there to put the cards in the three boxes so that the trick works?

2

07C1

Let $n > 1$ be an integer. Find all sequences $a_1, a_2, \dots, a_{n^2+n}$ satisfying the following conditions:

(a) $a_i \in \{0,1\}$ for all $1 \leq i \leq n^2 + n$;

(b) $a_{i+1} + a_{i+2} + \dots + a_{i+n} < a_{i+n+1} + a_{i+n+2} + \dots + a_{i+2n}$ for all $0 \leq i \leq n^2 - n$.

2

01C5

Find all finite sequences $(x_0, x_1, \dots, x_n)$ such that for every j , $0 \leq j \leq n$, x_j equals the number of times j appears in the sequence.

2

15C2

We say that a finite set $\mathcal{S}$ of points in the plane is balanced if, for any two different points A and B in $\mathcal{S}$, there is a point C in $\mathcal{S}$ such that $AC = BC$. We say that $\mathcal{S}$ is centre-free if for any three different points A , B and C in $\mathcal{S}$, there is no points P in $\mathcal{S}$ such that $PA = PB = PC$.

(a) Show that for all integers $n \geq 3$, there exists a balanced set consisting of n points.

(b) Determine all integers $n \geq 3$ for which there exists a balanced centre-free set consisting of n points.

2

05N2

Let $a_1, a_2, \dots$ be a sequence of integers with infinitely many positive and negative terms. Suppose that for every positive integer n the numbers $a_1, a_2, \dots, a_n$ leave n different remainders upon division by n .

Prove that every integer occurs exactly once in the sequence $a_1, a_2, \dots$.

2

05C2

Let k be a nonnegative integer. A forest consists of rooted (i. e. oriented) trees. Each vertex of the forest is either a leaf or has two successors. A vertex v is called an extended successor of a vertex u if there is a chain of vertices $u_0 = u, u_1, u_2, \dots, u_{t-1}, u_t = v$ with $t > 0$ such that the vertex u_{i+1} is a successor of the vertex u_i for every integer i with $0 \leq i \leq t-1$. A vertex is called dynastic if it has two successors and each of these successors has at least k extended successors.

Prove that if the forest has n vertices, then there are at most $\frac{n}{k+2}$ dynastic vertices.

3

06C5

An (n, k) – tournament is a contest with n players held in k rounds such that:

(i) Each player plays in each round, and every two players meet at most once.

(ii) If player A meets player B in round i , player C meets player D in round i , and player A meets player C in round j , then player B meets player D in round j .

Determine all pairs (n, k) for which there exists an (n, k) – tournament.

3

10C3

2500 chess kings have to be placed on a 100×100 chessboard so that

(i) no king can capture any other one (i.e. no two kings are placed in two squares sharing a common vertex);

(ii) each row and each column contains exactly 25 kings.

Find the number of such arrangements. (Two arrangements differing by rotation or symmetry are supposed to be different.)

4

10C4

Each of the six boxes $B_1, B_2, B_3, B_4, B_5, B_6$ initially contains one coin. The following operations are allowed

Type 1) Choose a non-empty box B_j , $1 \leq j \leq 5$, remove one coin from B_j and add two coins to B_{j+1} ;

Type 2) Choose a non-empty box B_k , $1 \leq k \leq 4$, remove one coin from B_k and swap the contents (maybe empty) of the boxes B_{k+1} and B_{k+2} .

Determine if there exists a finite sequence of operations of the allowed types, such that the five boxes B_1, B_2, B_3, B_4, B_5 become empty, while box B_6 contains exactly $2010^{2010^{2010}}$ coins.

Maximum-Possible Problems

1

03C3

Let $n \geq 5$ be an integer. Find the maximal integer k such that there exists a polygon with n vertices (non-self intersecting or overlapping, but not necessarily convex) having k internal 90° angles.

2

00C4

Let n and k be positive integers such that $\frac{1}{2}n < k \leq \frac{2}{3}n$. Find the least number m for which it is possible to place m pawns on m squares of an $n \times n$ chessboard so that no column or row contains a block of k adjacent unoccupied squares.

2

09C2

For any integer $n \geq 2$, let $N(n)$ be the maximal number of triples (a_i, b_i, c_i) , $i = 1, \dots, N(n)$, consisting of nonnegative integers a_i , b_i and c_i such that the following two conditions are satisfied:

$$a_i + b_i + c_i = n \text{ for all } i = 1, \dots, N(n),$$

If $i \neq j$ then $a_i \neq a_j$, $b_i \neq b_j$ and $c_i \neq c_j$

Determine $N(n)$ for all $n \geq 2$.

2

10C2

On some planet, there are 2^N countries ($N \geq 4$). Each country has a flag N units wide and one unit high composed of N fields of size 1×1 , each field being either yellow or blue. No two countries have the same flag. We say that a set of N flags is diverse if these flags can be arranged into an $N \times N$ square so that all N fields on its main diagonal will have the same color. Determine the smallest positive integer M such that among any M distinct flags, there exist N flags forming a diverse set.

2

11C4

Determine the greatest positive integer k that satisfies the following property: The set of positive integers can be partitioned into k subsets $A_1, A_2, \dots, A_k$ such that for all integers $n \geq 15$ and all $i \in \{1, 2, \dots, k\}$ there exist two distinct elements of A_i whose sum is n .

2

12C2

Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \dots, n\}$ such that the sums of the different pairs are different integers not exceeding n ?

3

14C3

Let $n \geq 2$ be an integer. Consider an $n \times n$ chessboard consisting of n^2 unit squares. A configuration of n rooks on this board is peaceful if every row and every column contains exactly one rook. Find the greatest positive integer k such that, for each peaceful configuration of n rooks, there is a $k \times k$ square which does not contain a rook on any of its k^2 unit squares.

3

08C1

In the plane we consider rectangles whose sides are parallel to the coordinate axes and have positive length. Such a rectangle will be called a box. Two boxes intersect if they have a common point in their interior or on their boundary. Find the largest n for which there exist n boxes $B_1, \dots, B_n$ such that B_i and B_j intersect if and only if $i \not\equiv j \pm 1 \pmod{n}$.

3

15C3

For a finite set A of positive integers, a partition of A into two disjoint nonempty subsets A_1 and A_2 is good if the least common multiple of the elements in A_1 is equal to the greatest common divisor of the elements in A_2 . Determine the minimum value of n such that there exists a set of n positive integers with exactly 2015 good partitions.

3

04C2

Let n and k be positive integers. There are given n circles in the plane. Every two of them intersect at two distinct points, and all points of intersection they determine are pairwise distinct. Each intersection point must be colored with one of n distinct colors so that each color is used at least once and exactly k distinct colors occur on each circle. Find all values of $n \geq 2$ and k for which such a coloring is possible.

3

04C4

Consider a matrix of size $n \times n$ whose entries are real numbers of absolute value not exceeding 1. The sum of all entries of the matrix is 0. Let n be an even positive integer. Determine the least number C such that every such matrix necessarily has a row or a column with the sum of its entries not exceeding C in absolute value.

4

04C6

For an $n \times n$ matrix A , let X_i be the set of entries in row i , and Y_j the set of entries in column j , $1 \leq i, j \leq n$. We say that A is golden if $X_1, \dots, X_n, Y_1, \dots, Y_n$ are distinct sets. Find the least integer n such that there exists a 2004×2004 golden matrix with entries in the set $\{1, 2, \dots, n\}$.

Pure Combinatorial Counting Problems

1 Let $n > 0$ be an integer. We are given a balance and n weights of weight $2^0, 2^1, \dots, 2^{n-1}$. We are to place each of the n weights on the balance, one after another, in such a way that the right pan is never heavier than the left pan. At each step we choose one of the weights that has not yet been placed on the balance, and place it on either the left pan or the right pan, until all of the weights have been placed.

11C1

Determine the number of ways in which this can be done.

2 Let n be a positive integer. How many permutations $(a_1, \dots, a_n)$ of the set $\{1, 2, \dots, n\}$ satisfy $k \mid 2(a_1 + \dots + a_k)$, for all $1 \leq k \leq n$?

08C2

3 Let n be a positive integer. A sequence of n positive integers (not necessarily distinct) is called full if it satisfies the following condition: for each positive integer $k \geq 2$, if the number k appears in the sequence then so does the number $k - 1$, and moreover the first occurrence of $k - 1$ comes before the last occurrence of k . For each n , how many full sequences are there?

02C3

3 Let $n \geq 3$ be a fixed integer. Each side and each diagonal of a regular n -gon is labelled with a number from the set $\{1; 2; \dots; r\}$ in a way such that the following two conditions are fulfilled:

05C4

Each number from the set $\{1; 2; \dots; r\}$ occurs at least once as a label.

In each triangle formed by three vertices of the n -gon, two of the sides are labelled with the same number, and this number is greater than the label of the third side.

(a) Find the maximal r for which such a labelling is possible.

(b) For this maximal value of r , how many such labelings are there?

3 Find all positive integers n for which the numbers in the set $S = \{1, 2, \dots, n\}$ can be colored red and blue, with the following condition being satisfied: The set $S \times S \times S$ contains exactly 2007 ordered triples (x, y, z) such that:

07C3 the numbers x, y, z are of the same color,

and the number $x + y + z$ is divisible by n .

5 Let $\alpha < \frac{3 - \sqrt{5}}{2}$ be a positive real number. Prove that there exist positive integers n and $p > \alpha \cdot 2^n$ for which one can select $2 \cdot p$ pairwise distinct subsets $S_1, \dots, S_p, T_1, \dots, T_p$ of the set $\{1, 2, \dots, n\}$ such that $S_i \cap T_j \neq \emptyset$ for all $1 \leq i, j \leq p$

07C7

5 Let p and q be relatively prime positive integers. A subset S of $\{0, 1, 2, \dots\}$ is called ideal if $0 \in S$ and for each element $n \in S$, the integers $n + p$ and $n + q$ belong to S . Determine the number of ideal subsets of $\{0, 1, 2, \dots\}$.

00C6

Insights

Logical Thinking Problems

General problems that requires accurate and useful logical deductions. In a sense, they are skinned sudoku, with highly advanced rulesets. Frequently, a certain technique from standard toolbox (case splitting, induction, counting, colouring, bounding, invariant) will give the solution. However, which exactly is not apparent from the start. Thus, general problem solving techniques will aid you to discover what lies behind the problem. General problem solving techniques include small case explorations, principle of abuse (Actively think of results that can make use of an observed property).

Maximum Possible Problem

A heavy subclass of combinatorial problems, These problems require the maximum possible element to be determined, which include the following three aspects: Guessing the answer, Provide a proof, likely a construction, that this is possible, and show that this is the maximum. The catch is flow for a proof in both directions is very different

Therefore one have to keep asking himself: Why am I convinced that this answer must be it? Could I have improved through a better construction/ counter example? For if the guessed answer is wrong, one can never prove the statement. Hence, if one feels stuck at a question , try to improve the results from both directions - both look for a better constructions and a stronger proof.

Pure Combinatorial Counting Problems

Combinatorial problems that asks for no. of ways of certain items, similar to local prelim problems , but with more complicated settings. Frequently. one can obtain the answer through small case explorations. It remains to give a correct interpretation of the formula to solve the problem,

Besides general combinatorial skills, counting techniques and formula are also necessary. Binomial coefficients and their summing formula will be useful.

Core Problems

Description

Collection of mid-tier problems that have a more-complex settings.

Main Idea

More difficult variants of what seen in Tier 1.

Ways To Improve

Readers should be familiar with a wide range of settings and their respective techniques, such as tilings and games. Afterwards, practicing seems to be the only reliable way to improve.

Proof Problem

1
07C2

A rectangle D is partitioned in several (≥ 2) rectangles with sides parallel to those of D . Given that any line parallel to one of the sides of D , and having common points with the interior of D , also has common interior points with the interior of at least one rectangle of the partition; prove that there is at least one rectangle of the partition having no common points with D 's boundary.

1
12C1

Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers x and y such that $x > y$ and x is to the left of y , and replaces the pair (x, y) by either $(y + 1, x)$ or $(x - 1, x)$. Prove that she can perform only finitely many such iterations.

1
15C1

In Lineland there are $n \geq 1$ towns, arranged along a road running from left to right. Each town has a left bulldozer (put to the left of the town and facing left) and a right bulldozer (put to the right of the town and facing right). The sizes of the $2n$ bulldozers are distinct. Every time when a left and right bulldozer confront each other, the larger bulldozer pushes the smaller one off the road. On the other hand, bulldozers are quite unprotected at their rears; so, if a bulldozer reaches the rear-end of another one, the first one pushes the second one off the road, regardless of their sizes.

Let A and B be two towns, with B to the right of A . We say that town A can sweep town B away if the right bulldozer of A can move over to B pushing off all bulldozers it meets. Similarly town B can sweep town A away if the left bulldozer of B can move over to A pushing off all bulldozers of all towns on its way.

Prove that there is exactly one town that cannot be swept away by any other one.

2
01C2

Let n be an odd integer greater than 1 and let $c_1, c_2, \dots, c_n$ be integers. For each permutation $a = (a_1, a_2, \dots, a_n)$ of $\{1, 2, \dots, n\}$, define $S(a) = \sum_{i=1}^n c_i a_i$. Prove that there exist permutations $a \neq b$ of $\{1, 2, \dots, n\}$ such that $n!$ is a divisor of $S(a) - S(b)$.

2

16C4

Find all integers n for which each cell of $n \times n$ table can be filled with one of the letters I, M and O in such a way that:

(a) in each row and each column, one third of the entries are I , one third are M and one third are O ; and

(b) In any diagonal, if the number of entries on the diagonal is a multiple of three, then one third of the entries are I , one third are M and one third are O .

2

11C2

Suppose that 1000 students are standing in a circle. Prove that there exists an integer k with $100 \leq k \leq 300$ such that in this circle there exists a contiguous group of $2k$ students, for which the first half contains the same number of girls as the second half.

3

16C3

Let n be a positive integer relatively prime to 6. We paint the vertices of a regular n -gon with three colours so that there is an odd number of vertices of each colour. Show that there exists an isosceles triangle whose three vertices are of different colours.

3

01C3

Define a k -clique to be a set of k people such that every pair of them are acquainted with each other. At a certain party, every pair of 3-cliques has at least one person in common, and there are no 5-cliques. Prove that there are two or fewer people at the party whose departure leaves no 3-clique remaining.

3

14N3

For each positive integer n , the Bank of Cape Town issues coins of denomination $\frac{1}{n}$. Given a finite collection of such coins (of not necessarily different denominations) with total value at most $99 + \frac{1}{2}$, prove that it is possible to split this collection into 100 or fewer groups, such that each group has total value at most 1

3

03C4

Given n real numbers $x_1, x_2, \dots, x_n$, and n further real numbers $y_1, y_2, \dots, y_n$. The entries a_{ij} (with $1 \leq i, j \leq n$) of an $n \times n$ matrix A are defined as follows: $a_{ij} = 1$ if $x_i + y_j \geq 0$; 0 if $x_i + y_j < 0$. Further, let B be an $n \times n$ matrix whose elements are numbers from the set $\{0, 1\}$ satisfying the following condition: The sum of all elements of each row of B equals the sum of all elements of the corresponding row of A ; the sum of all elements of each column of B equals the sum of all elements of the corresponding column of A . Show that in this case, $A = B$.

3

04C3

The following operation is allowed on a finite graph: Choose an arbitrary cycle of length 4 (if there is any), choose an arbitrary edge in that cycle, and delete it from the graph. For a fixed integer $n \geq 4$, find the least number of edges of a graph that can be obtained by repeated applications of this operation from the complete graph on n vertices (where each pair of vertices are joined by an edge).

3

13C3

A crazy physicist discovered a new kind of particle which he called an imon, after some of them mysteriously appeared in his lab. Some pairs of imons in the lab can be entangled, and each imon can participate in many entanglement relations. The physicist has found a way to perform the following two kinds of operations with these particles, one operation at a time.

(i) If some imon is entangled with an odd number of other imons in the lab, then the physicist can destroy it.

(ii) At any moment, he may double the whole family of imons in the lab by creating a copy I' of each imon I . During this procedure, the two copies I' and J' become entangled if and only if the original imons I and J are entangled, and each copy I' becomes entangled with its original imon I ; no other entanglements occur or disappear at this moment.

Prove that the physicist may apply a sequence of such operations resulting in a family of imons, no two of which are entangled.

4

05C6

In a mathematical competition, in which 6 problems were posed to the participants, every two of these problems were solved by more than $\frac{2}{5}$ of the contestants. Moreover, no contestant solved all the 6 problems. Show that there are at least 2 contestants who solved exactly 5 problems each.

5

07C5

In the Cartesian coordinate plane define the strips $S_n = \{(x, y) \mid n \leq x < n + 1\}$, $n \in \mathbb{Z}$ and color each strip black or white. Prove that any rectangle which is not a square can be placed in the plane so that its vertices have the same color.

5

00A6

For a set X of real numbers, $D(X)$ denotes the difference set $\{|x - y| \mid x, y \in X\}$. A nonempty set A of real numbers is called a B_3 -set if the conditions $a_1, a_2, a_3, a_4, a_5, a_6 \in A$ and $a_1 + a_2 + a_3 = a_4 + a_5 + a_6$ imply that the sequences (a_1, a_2, a_3) and (a_4, a_5, a_6) are identical up to a permutation. Let $A = \{a_0 = 0 < a_1 < a_2 < \dots\}$, $B = \{b_0 = 0 < b_1 < b_2 < \dots\}$ be infinite sequences of real numbers with $D(A) = D(B)$. Prove that if A is a B_3 -set, then $A = B$.

5

10C7

Let $P_1, \dots, P_s$ be arithmetic progressions of integers, the following conditions being satisfied:

- (i) each integer belongs to at least one of them;
- (ii) each progression contains a number which does not belong to other progressions.

Denote by n the least common multiple of steps of these progressions; let $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ be its prime factorization.

Prove that $s \geq 1 + \sum_{i=1}^k \alpha_i(p_i - 1)$.

Tiling Problems

A staircase-brick with 3 steps of width 2 is made of 12 unit cubes. Determine all integers n for which it is possible to build a cube of side n using such bricks.

1

00C2

2

02C2

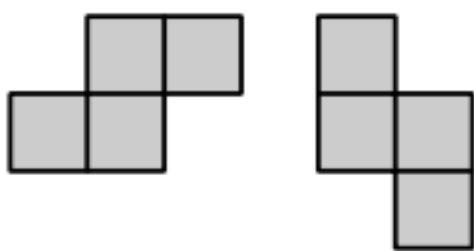
For $A = B$ an odd positive integer, the unit squares of an $A = B$ chessboard are coloured alternately black and white, with the four corners coloured black. A tromino is an L -shape formed by three connected unit squares. For which values of L is it possible to cover all the black squares with non-overlapping trominos? When it is possible, what is the minimum number of trominos needed?

3

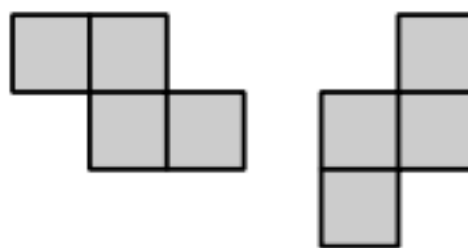
14C4

Construct a tetromino by attaching two 2×1 dominoes along their longer sides such that the midpoint of the longer side of one domino is a corner of the other domino. This construction yields two kinds of tetrominoes with opposite orientations. Let us call them S - and Z -tetrominoes, respectively.

Assume that a lattice polygon P can be tiled with S -tetrominoes. Prove that no matter how we tile P using only S - and Z -tetrominoes, we always use an even number of Z -tetrominoes.



S-tetrominoes



Z-tetrominoes

5

06C6

A holey triangle is an upward equilateral triangle of side length n with n upward unit triangular holes cut out. A diamond is a $60^\circ - 120^\circ$ unit rhombus.

Prove that a holey triangle T can be tiled with diamonds if and only if the following condition holds: Every upward equilateral triangle of side length k in T contains at most k holes, for $1 \leq k \leq n$.

Combinatorial Geom

1

16C5

Let $n \geq 3$ be a positive integer. Find the maximum number of diagonals in a regular n -gon one can select, so that any two of them do not intersect in the interior or they are perpendicular to each other.

1

00C3

Let $n \geq 4$ be a fixed positive integer. Given a set $S = \{P_1, P_2, \dots, P_n\}$ of n points in the plane such that no three are collinear and no four concyclic, let a_t , $1 \leq t \leq n$, be the number of circles $P_i P_j P_k$ that contain P_t in their interior, and let $m(S) = a_1 + a_2 + \dots + a_n$. Prove that there exists a positive integer $f(n)$, depending only on n , such that the points of S are the vertices of a convex polygon if and only if $m(S) = f(n)$.

2

03C2

Let $D_1, D_2, \dots, D_n$ be closed discs in the plane. (A closed disc is the region limited by a circle, taken jointly with this circle.) Suppose that every point in the plane is contained in at most 2003 discs D_i . Prove that there exists a disc D_k which intersects at most $7 \cdot 2003 - 1 = 14020$ other discs D_i .

2

13C2

A configuration of 4027 points in the plane is called Colombian if it consists of 2013 red points and 2014 blue points, and no three of the points of the configuration are collinear. By drawing some lines, the plane is divided into several regions. An arrangement of lines is good for a Colombian configuration if the following two conditions are satisfied:

- i) No line passes through any point of the configuration.
- ii) No region contains points of both colors.

Find the least value of k such that for any Colombian configuration of 4027 points, there is a good arrangement of k lines.

2

Let P be a regular 2006-gon. A diagonal is called good if its endpoints divide the boundary of P into two parts, each composed of an odd number of sides of P . The sides of P are also called good.

06C2 Suppose P has been dissected into triangles by 2003 diagonals, no two of which have a common point in the interior of P . Find the maximum number of isosceles triangles having two good sides that could appear in such a configuration

4

00G7 Ten gangsters are standing on a flat surface, and the distances between them are all distinct. At twelve o'clock, when the church bells start chiming, each of them fatally shoots the one among the other nine gangsters who is the nearest. At least how many gangsters will be killed?

4

16C7 There are $n \geq 2$ line segments in the plane such that every two segments cross and no three segments meet at a point. Geoff has to choose an endpoint of each segment and place a frog on it facing the other endpoint. Then he will clap his hands $n - 1$ times. Every time he claps, each frog will immediately jump forward to the next intersection point on its segment. Frogs never change the direction of their jumps. Geoff wishes to place the frogs in such a way that no two of them will ever occupy the same intersection point at the same time.

(a) Prove that Geoff can always fulfill his wish if n is odd.

(b) Prove that Geoff can never fulfill his wish if n is even.

5

11C3 Let $\mathcal{S}$ be a finite set of at least two points in the plane. Assume that no three points of $\mathcal{S}$ are collinear. A windmill is a process that starts with a line ℓ going through a single point $P \in \mathcal{S}$. The line rotates clockwise about the pivot P until the first time that the line meets some other point belonging to $\mathcal{S}$. This point, Q , takes over as the new pivot, and the line now rotates clockwise about Q , until it next meets a point of $\mathcal{S}$. This process continues indefinitely.

Show that we can choose a point P in $\mathcal{S}$ and a line ℓ going through P such that the resulting windmill uses each point of $\mathcal{S}$ as a pivot infinitely many times.

5

17Q3
[C5]

A hunter and an invisible rabbit play a game in the Euclidean plane. The rabbit's starting point, A_0 , and the hunter's starting point, B_0 are the same. After $n - 1$ rounds of the game, the rabbit is at point A_{n-1} and the hunter is at point B_{n-1} . In the n^{th} round of the game, three things occur in order:

(1) The rabbit moves invisibly to a point A_n such that the distance between A_{n-1} and A_n is exactly 1.

(2) A tracking device reports a point P_n to the hunter. The only guarantee provided by the tracking device to the hunter is that the distance between P_n and A_n is at most 1.

(3) The hunter moves visibly to a point B_n such that the distance between B_{n-1} and B_n is exactly B_n .

Is it always possible, no matter how the rabbit moves, and no matter what points are reported by the tracking device, for the hunter to choose her moves so that after 10^9 rounds, she can ensure that the distance between her and the rabbit is at most 100?

Games

We have $n \geq 2$ lamps $L_1, \dots, L_n$ in a row, each of them being either on or off. Every second we simultaneously modify the state of each lamp as follows: if the lamp L_i and its neighbours (only one neighbour for $i = 1$ or $i = n$, two neighbours for other i) are in the same state, then L_i is switched off; – otherwise, L_i is switched on.

Initially all the lamps are off except the leftmost one which is on.

(a) Prove that there are infinitely many integers n for which all the lamps will eventually be off.

(b) Prove that there are infinitely many integers n for which the lamps will never be all off.

1

06C1

2

09C1

Consider 2009 cards, each having one gold side and one black side, lying in parallel on a long table. Initially all cards show their gold sides. Two players, standing by the same long side of the table, play a game with alternating moves. Each move consists of choosing a block of 50 consecutive cards, the leftmost of which is showing gold, and turning them all over, so those which showed gold now show black and vice versa. The last player who can make a legal move wins.

(a) Does the game necessarily end?

(b) Does there exist a winning strategy for the starting player?

2

04C5

A and B play a game, given an integer N , A writes down 1 first, then every player sees the last number written and if it is n then in his turn he writes $n + 1$ or $2n$, but his number cannot be bigger than N . The player who writes N wins. For which values of N does B win?

3

05C5

There are n markers, each with one side white and the other side black. In the beginning, these n markers are aligned in a row so that their white sides are all up. In each step, if possible, we choose a marker whose white side is up (but not one of the outermost markers), remove it, and reverse the closest marker to the left of it and also reverse the closest marker to the right of it. Prove that, by a finite sequence of such steps, one can achieve a state with only two markers remaining if and only if $n - 1$ is not divisible by 3.

3

13N5

Fix an integer $k \geq 2$. Two players, called Ana and Banana, play the following game of numbers. Initially, some integer $n \geq k$ gets written on the blackboard. Then they take moves in turn, with Ana beginning. A player making a move erases the number m just written on the blackboard and replaces it by some number m' with $k \leq m' < m$ that is coprime to m . The first player who cannot move anymore loses.

An integer $n \geq k$ is called good if Banana has a winning strategy when the initial number is n , and bad otherwise.

Consider two integers $n, n' \geq k$ with the property that each prime number $p \leq k$ divides n if and only if it divides n' . Prove that either both n and n' are good or both are bad.

4

02C4

Let T be the set of ordered triples (x, y, z) , where x, y, z are integers with $0 \leq x, y, z \leq 9$. Players A and B play the following guessing game. Player A chooses a triple (x, y, z) in T , and Player B has to discover A 's triple in as few moves as possible. A move consists of the following: B gives A a triple (a, b, c) in T , and A replies by giving B the number $|x + y - a - b| + |y + z - b - c| + |z + x - c - a|$. Find the minimum number of moves that A needs to be sure of determining A 's triple.

4

12C4

Players A and B play a game with $N \geq 2012$ coins and 2012 boxes arranged around a circle. Initially A distributes the coins among the boxes so that there is at least 1 coin in each box. Then the two of them make moves in the order $B, A, B, A, \dots$ by the following rules:

On every move of his B passes 1 coin from every box to an adjacent box.

On every move of hers A chooses several coins that were not involved in B 's previous move and are in different boxes. She passes every chosen coin to an adjacent box.

Player A 's goal is to ensure at least 1 coin in each box after every move of hers, regardless of how B plays and how many moves are made. Find the least N that enables her to succeed.

4

13C8

Players A and B play a "paintful" game on the real line. Player A has a pot of paint with infinite black ink. A quantity p of this ink suffices to blacken a (closed) real interval of length p . In every round, player A picks some positive integer m and provides $1/2^m$ units of ink from the pot. Player B then picks an integer k and blackens the interval from $k/2^m$ to $(k+1)/2^m$ (some parts of this interval may have been blackened before). The goal of player A is to reach a situation where at least 4 ink from the pot is used and the interval $[0,1]$ is not completely blackened.

Decide whether there exists a strategy for player A to win in a finite number of moves.

4

14C8

A card deck consists of 1024 cards. On each card, a set of distinct decimal digits is written in such a way that no two of these sets coincide (thus, one of the cards is empty). Two players alternately take cards from the deck, one card per turn. After the deck is empty, each player checks if he can throw out one of his cards so that each of the ten digits occurs on an even number of his remaining cards. If one player can do this but the other one cannot, the one who can is the winner; otherwise a draw is declared.

Determine all possible first moves of the first player after which he has a winning strategy.

4

15C4

Let n be a positive integer. Two players A and B play a game in which they take turns choosing positive integers $k \leq n$. The rules of the game are:

(i) A player cannot choose a number that has been chosen by either player on any previous turn.

(ii) A player cannot choose a number consecutive to any of those the player has already chosen on any previous turn.

(iii) The game is a draw if all numbers have been chosen; otherwise the player who cannot choose a number anymore loses the game.

The player A takes the first turn. Determine the outcome of the game, assuming that both players use optimal strategies.

4

07C4

Let $A_0 = (a_1, \dots, a_n)$ be a finite sequence of real numbers. For each $k \geq 0$, from the sequence $A_k = (x_1, \dots, x_n)$ we construct a new sequence A_{k+1} in the following way.

We choose a partition $\{1, \dots, n\} = I \cup J$, where I and J are two disjoint sets, such that the expression

$$\left| \sum_{i \in I} x_i - \sum_{j \in J} x_j \right|$$

attains the smallest value. (We allow I or J to be empty; in this case the corresponding sum is 0.) If there are several such partitions, one is chosen arbitrarily.

We set $A_{k+1} = (y_1, \dots, y_n)$ where $y_i = x_i + 1$ if $i \in I$, and $y_i = x_i - 1$ if $i \in J$.

Prove that for some k , the sequence A_k contains an element x such that $|x| \geq \frac{n}{2}$

5

09C5

Five identical empty buckets of 2-liter capacity stand at the vertices of a regular pentagon. Cinderella and her wicked Stepmother go through a sequence of rounds: At the beginning of every round, the Stepmother takes one liter of water from the nearby river and distributes it arbitrarily over the five buckets. Then Cinderella chooses a pair of neighbouring buckets, empties them into the river and puts them back. Then the next round begins. The Stepmother's goal's is to make one of these buckets overflow. Cinderella's goal is to prevent this. Can the wicked Stepmother enforce a bucket overflow?

Constant Sensitive Problems

Let A be a 101-element subset of the set $S = \{1, 2, \dots, 1000000\}$. Prove that there exist numbers $t_1, t_2, \dots, t_{100}$ in S such that the sets $A_j = \{x + t_j \mid x \in A\}$, $j = 1, 2, \dots, 100$ are pairwise disjoint.

1

03C1

2 Twenty-one girls and twenty-one boys took part in a mathematical competition. It turned out that each contestant solved at most six problems, and for each pair of a girl and a boy, there was at least one problem that was solved by both the girl and the boy. Show that there is a problem that was solved by at least three girls and at least three boys.

01C8

2 In a concert, 20 singers will perform. For each singer, there is a (possibly empty) set of other singers such that he wishes to perform later than all the singers from that set. Can it happen that there are exactly 2010 orders of the singers such that all their wishes are satisfied?

10C1

4 Let n be a positive integer, and let A be a subset of $\{1, \dots, n\}$. An A -partition of n into k parts is a representation of n as a sum $n = a_1 + \dots + a_k$, where the parts $a_1, \dots, a_k$ belong to A and are not necessarily distinct. The number of different parts in such a partition is the number of (distinct) elements in the set $\{a_1, a_2, \dots, a_k\}$.

13C4

We say that an A -partition of n into k parts is optimal if there is no A -partition of n into r parts with $r < k$. Prove that any optimal A -partition of n contains at most $\sqrt[3]{6n}$ different parts.

6 Is it possible to find 100 positive integers not exceeding 25,000, such that all pairwise sums of them are different?

01N6

Insights

Proof Problem

Problems that demand a proof of a simple result. The dynamics are different from logical thinking problem in that one should actively look for auxiliary results and intermediate steps. General proof techniques applies: Attempt to construct a configuration that can fail the statement as much as possible, and think about what limits: Why would it fail?

Tiling problems

A rather predictable subset of problems. The major technique used is colouring and invariants. As seen from 90% of problems of these type, one try to colour the board in a certain way, then derive useful information through local features (frequently, how each tile acts on the colours) and global features (frequently, how much tiles are present).

Combinatorial Geom

Combinatorial problems that set on the euclidean plane. These problems frequently concern distances of points, intersections of objects, orientations of lines.

Indeed, there are no general theorems or techniques on these classes of problems. The two fringe techniques are given below: (1) Existence of convex hulls: a rigorous definition of “out-most boundary” for a set of points. (2) “Sweeping”: that one can scan all points one by one by taking a point and a ray while spinning it.

Games

Problems of games are generally have more cases and conditions, as there are many moves that a player can play. However, the settings are generally simpler.

The most essential concept is the simple one:

If A could win, he must be able to counter every single move that B had played.

If it is A to move and A wish to win, it suffices to consider one move by A such that he wins afterwards.

For problems that do not state the wining player, remember to avoid fixating on a certain answer.

Complex Problems

Description

Collection of mid-tier problems with complicated and less common settings

Main Idea

More difficult and complex variants of what seen in Tier 1.

Ways To Improve

Readers should be familiar with a wide range of settings and their respective techniques, such as tilings and games. Afterwards, practicing seems to be the only reliable way to improve.

Complex Deduction Problems

3

15N4

Suppose that $a_0, a_1, \dots$ and $b_0, b_1, \dots$ are two sequences of positive integers such that $a_0, b_0 \geq 2$ and $a_{n+1} = \gcd(a_n, b_n) + 1$, $b_{n+1} = \text{lcm}(a_n, b_n) - 1$. Show that the sequence a_n is eventually periodic; in other words, there exist integers $N \geq 0$ and $t > 0$ such that $a_{n+t} = a_n$ for all $n \geq N$.

4

01C7

A pile of n pebbles is placed in a vertical column. This configuration is modified according to the following rules. A pebble can be moved if it is at the top of a column which contains at least two more pebbles than the column immediately to its right. (If there are no pebbles to the right, think of this as a column with 0 pebbles.) At each stage, choose a pebble from among those that can be moved (if there are any) and place it at the top of the column to its right. If no pebbles can be moved, the configuration is called a final configuration. For each n , show that, no matter what choices are made at each stage, the final configuration obtained is unique. Describe that configuration in terms of n .

5

02C5

Let $r \geq 2$ be a fixed positive integer, and let F be an infinite family of sets, each of size r , no two of which are disjoint. Prove that there exists a set of size $r - 1$ that meets each set in F .

5

09C6

On a 999×999 board a limp rook can move in the following way: From any square it can move to any of its adjacent squares, i.e. a square having a common side with it, and every move must be a turn, i.e. the directions of any two consecutive moves must be perpendicular. A non-intersecting route of the limp rook consists of a sequence of pairwise different squares that the limp rook can visit in that order by an admissible sequence of moves. Such a non-intersecting route is called cyclic, if the limp rook can, after reaching the last square of the route, move directly to the first square of the route and start over. How many squares does the longest possible cyclic, non-intersecting route of a limp rook visit?

5

12C5

The columns and the row of a $3n \times 3n$ square board are numbered $1, 2, \dots, 3n$. Every square (x, y) with $1 \leq x, y \leq 3n$ is colored asparagus, byzantium or citrine according as the modulo 3 remainder of $x + y$ is 0, 1 or 2 respectively.

One token colored asparagus, byzantium or citrine is placed on each square, so that there are $3n^2$ tokens of each color.

Suppose that one can permute the tokens so that each token is moved to a distance of at most d from its original position, each asparagus token replaces a byzantium token, each byzantium token replaces a citrine token, and each citrine token replaces an asparagus token. Prove that it is possible to permute the tokens so that each token is moved to a distance of at most $d + 2$ from its original position, and each square contains a token with the same color as the square.

5

16C6

There are $n \geq 3$ islands in a city. Initially, the ferry company offers some routes between some pairs of islands so that it is impossible to divide the islands into two groups such that no two islands in different groups are connected by a ferry route.

After each year, the ferry company will close a ferry route between some two islands X and Y . At the same time, in order to maintain its service, the company will open new routes according to the following rule: for any island which is connected to a ferry route to exactly one of X and Y , a new route between this island and the other of X and Y is added.

Suppose at any moment, if we partition all islands into two nonempty groups in any way, then it is known that the ferry company will close a certain route connecting two islands from the two groups after some years. Prove that after some years there will be an island which is connected to all other islands by ferry routes.

We are given an infinite deck of cards, each with a real number on it. For every real number x , there is exactly one card in the deck that has x written on it. Now two players draw disjoint sets A and B of 100 cards each from this deck. We would like to define a rule that declares one of them a winner. This rule should satisfy the following conditions:

The winner only depends on the relative order of the 200 cards: if the cards are laid down in increasing order face down and we are told which card belongs to which player, but not what numbers are written on them, we can still decide the winner.

If we write the elements of both sets in increasing order as $A = \{a_1, a_2, \dots, a_{100}\}$ and $B = \{b_1, b_2, \dots, b_{100}\}$, and $a_i > b_i$ for all i , then A beats B .

If three players draw three disjoint sets A, B, C from the deck, A beats B and B beats C then A also beats C .

How many ways are there to define such a rule? Here, we consider two rules as different if there exist two sets A and B such that A beats B according to one rule, but B beats A according to the other.

Combinatorial Inequalities

1 Let $A = (a_1, a_2, \dots, a_{2001})$ be a sequence of positive integers. Let m be the number of 3-element subsequences (a_i, a_j, a_k) with $1 \leq i < j < k \leq 2001$, such that $a_j = a_i + 1$ and $a_k = a_j + 1$. Considering all such sequences A , find the greatest value of m .

01C1

2 In some country several pairs of cities are connected by direct two-way flights. It is possible to go from any city to any other by a sequence of flights. The distance between two cities is defined to be the least possible numbers of flights required to go from one of them to the other. It is known that for any city there are at most 100 cities at distance exactly three from it. Prove that there is no city such that more than 2550 other cities have distance exactly four from it.

13C6

2

14C2

We have 2^m sheets of paper, with the number 1 written on each of them. We perform the following operation. In every step we choose two distinct sheets; if the numbers on the two sheets are a and b , then we erase these numbers and write the number $a + b$ on both sheets. Prove that after $m2^{m-1}$ steps, the sum of the numbers on all the sheets is at least 4^m .

3

01C6

For a positive integer n define a sequence of zeros and ones to be balanced if it contains n zeros and n ones. Two balanced sequences a and b are neighbors if you can move one of the $2n$ symbols of a to another position to form b . For instance, when $n = 4$, the balanced sequences 01101001 and 00110101 are neighbors because the third (or fourth) zero in the first sequence can be moved to the first or second position to form the second sequence. Prove that there is a set S of at most $\frac{1}{n+1} \binom{2n}{n}$ balanced sequences such that every balanced sequence is equal to or is a neighbor of at least one sequence in S .

3

12C3

In a 999×999 square table some cells are white and the remaining ones are red. Let T be the number of triples (C_1, C_2, C_3) of cells, the first two in the same row and the last two in the same column, with C_1, C_3 white and C_2 red. Find the maximum value T can attain.

5

04C8

For a finite graph G , let $f(G)$ be the number of triangles and $g(G)$ the number of tetrahedra formed by edges of G . Find the least constant c such that $g(G)^3 \leq c \cdot f(G)^4$ for every graph G .

5

08C5

Let $S = \{x_1, x_2, \dots, x_{k+l}\}$ be a $(k+l)$ -element set of real numbers contained in the interval $[0,1]$; k and l are positive integers. A k -element subset $A \subset S$ is called nice if

$$\left| \frac{1}{k} \sum_{x_i \in A} x_i - \frac{1}{l} \sum_{x_j \in S \setminus A} x_j \right| \leq \frac{k+l}{2kl}$$

Prove that the number of nice subsets is at least $\frac{2}{k+l} \binom{k+l}{k}$.

5 $n \geq 4$ players participated in a tennis tournament. Any two players have played exactly one game, and there was no tie game. We call a company of four players *bad* if one player was defeated by the other three players, and each of these three players won a game and lost another game among themselves. Suppose that there is no bad company in this tournament. Let w_i and l_i be respectively the number of wins and losses of the i -th player. Prove that $\sum_{i=1}^n (w_i - l_i)^3 \geq 0$.

10C5

6 Among a group of 120 people, some pairs are friends. A weak quartet is a set of four people containing exactly one pair of friends. What is the maximum possible number of weak quartets ?

02C7

Geometrical Counting Problems

1 Let n points be given inside a rectangle R such that no two of them lie on a line parallel to one of the sides of R . The rectangle R is to be dissected into smaller rectangles with sides parallel to the sides of R in such a way that none of these rectangles contains any of the given points in its interior. Prove that we have to dissect R into at least $n + 1$ smaller rectangles.

14C1

3 Given is a convex polygon P with n vertices. Triangle whose vertices lie on vertices of P is called good if all its sides are of unit length. Prove that there are at most $\frac{2n}{3}$ good triangles.

07C8

3 Regard a plane with a Cartesian coordinate system; for each point with integer coordinates, draw a circular disk centered at this point and having the radius $\frac{1}{1000}$.

03C5

- Prove the existence of an equilateral triangle whose vertices lie in the interior of different disks;
- Show that every equilateral triangle whose vertices lie in the interior of different disks has a sidelength > 96 .

3

06C3

Let M be a finite point set in the plane and no three points are collinear. A subset A of M will be called round if its elements is the set of vertices of a convex A -gon $V(A)$. For each round subset let $r(A)$ be the number of points from M which are exterior from the convex A -gon $V(A)$. Subsets with 0,1 and 2 elements are always round, its corresponding polygons are the empty set, a point or a segment, respectively (for which all other points that are not vertices of the polygon are exterior). For each round subset A of M construct the polynomial

$$P_A(x) = x^{|A|}(1 - x)^{r(A)}.$$

Show that the sum of polynomials for all round subsets is exactly the polynomial $P(x) = 1$.

4

09C4

For an integer $m \geq 1$, we consider partitions of a $2^m \times 2^m$ chessboard into rectangles consisting of cells of chessboard, in which each of the 2^m cells along one diagonal forms a separate rectangle of side length 1. Determine the smallest possible sum of rectangle perimeters in such a partition.

4

00C5

In the plane we have n rectangles with parallel sides. The sides of distinct rectangles lie on distinct lines. The boundaries of the rectangles cut the plane into connected regions. A region is nice if it has at least one of the vertices of the n rectangles on the boundary. Prove that the sum of the numbers of the vertices of all nice regions is less than $40n$. (There can be nonconvex regions as well as regions with more than one boundary curve.)

4

08G5

Let k and n be integers with $0 \leq k \leq n - 2$. Consider a set L of n lines in the plane such that no two of them are parallel and no three have a common point. Denote by I the set of intersections of lines in L . Let O be a point in the plane not lying on any line of L . A point $X \in I$ is colored red if the open line segment OX intersects at most k lines in L . Prove that I contains at least $\frac{1}{2}(k+1)(k+2)$ red points.

5

05C8

Suppose we have an n -gon. Some $n - 3$ diagonals are coloured black and some other $n - 3$ diagonals are coloured red (a side is not a diagonal), so that no two diagonals of the same colour can intersect strictly inside the polygon, although they can share a vertex. Find the maximum number of intersection points between diagonals coloured differently strictly inside the polygon, in terms of n .

6

06C7

Consider a convex polyhedron without parallel edges and without an edge parallel to any face other than the two faces adjacent to it. Call a pair of points of the polyhedron antipodal if there exist two parallel planes passing through these points and such that the polyhedron is contained between these planes. Let A be the number of antipodal pairs of vertices, and let B be the number of antipodal pairs of midpoints of edges. Determine the difference $A - B$ in terms of the numbers of vertices, edges, and faces.

Combinatorial Counting-Identity Problems

1

02C1

Let n be a positive integer. Each point (x, y) in the plane, where x and y are non-negative integers with $x + y < n$, is coloured red or blue, subject to the following condition: if a point (x, y) is red, then so are all points (x', y') with $x' \leq x$ and $y' \leq y$. Let A be the number of ways to choose n blue points with distinct x -coordinates, and let B be the number of ways to choose n blue points with distinct y -coordinates. Prove that $A = B$.

3

03C6

Let $f(k)$ be the number of all non-negative integers n satisfying the following conditions:

The integer n has exactly k digits in the decimal representation (where the first digit is not necessarily non-zero!), i. e. we have $0 \leq n < 10^k$.

These k digits of n can be permuted in such a way that the resulting number is divisible by 11.

Show that for any positive integer number m , we have $f(2m) = 10f(2m - 1)$.

3

05C3

Consider an $m \times n$ rectangular board consisting of mn unit squares. Two of its unit squares are called adjacent if they have a common edge, and a path is a sequence of unit squares in which any two consecutive squares are adjacent. Two paths are called non-intersecting if they don't share any common squares. Each unit square of the rectangular board can be colored black or white. We speak of a coloring of the board if all its mn unit squares are colored.

Let N be the number of colorings of the board such that there exists at least one black path from the left edge of the board to its right edge. Let M be the number of colorings of the board for which there exist at least two non-intersecting black paths from the left edge of the board to its right edge.

Prove that $N^2 \geq M \cdot 2^{mn}$.

4

08C4

Let n and k be positive integers with $k \geq n$ and $k - n$ an even number. Let $2n$ lamps labelled $1, 2, \dots, 2n$ be given, each of which can be either on or off. Initially all the lamps are off. We consider sequences of steps: at each step one of the lamps is switched (from on to off or from off to on).

Let N be the number of such sequences consisting of k steps and resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off.

Let M be number of such sequences consisting of k steps, resulting in the state where lamps 1 through n are all on, and lamps $n + 1$ through $2n$ are all off, but where none of the lamps $n + 1$ through $2n$ is ever switched on.

Determine $\frac{N}{M}$.

“Always-Exist” Problem

2

A house has an even number of lamps distributed among its rooms in such a way that there are at least three lamps in every room. Each lamp shares a switch with exactly one other lamp, not necessarily from the same room.

05C1

Each change in the switch shared by two lamps changes their states simultaneously. Prove that for every initial state of the lamps there exists a sequence of changes in some of the switches at the end of which each room contains lamps which are on as well as lamps which are off.

3

A set of three nonnegative integers $\{x, y, z\}$ with $x < y < z$ is called historic if $\{z - y, y - x\} = \{1776, 2001\}$. Show that the set of all nonnegative integers can be written as the union of pairwise disjoint historic sets.

01C4

4

A cake has the form of an $n \times n$ square composed of n^2 unit squares. Strawberries lie on some of the unit squares so that each row or column contains exactly one strawberry; call this arrangement $\mathcal{A}$.

06C4

Let B be another such arrangement. Suppose that every grid rectangle with one vertex at the top left corner of the cake contains no fewer strawberries of arrangement B than of arrangement $\mathcal{A}$. Prove that arrangement B can be obtained from $\mathcal{A}$ by performing a number of switches, defined as follows:

A switch consists in selecting a grid rectangle with only two strawberries, situated at its top right corner and bottom left corner, and moving these two strawberries to the other two corners of that rectangle.

4

17Q5
[C4]

An integer $N \geq 2$ is given. A collection of $N(N + 1)$ soccer players, no two of whom are of the same height, stand in a row. Sir Alex wants to remove $N(N - 1)$ players from this row leaving a new row of $2N$ players in which the following N conditions hold:

(1) no one stands between the two tallest players,

(2) no one stands between the third and fourth tallest players,

$\vdots$

(N) no one stands between the two shortest players.

Show that this is always possible.

4

02C6

Let n be an even positive integer. Show that there is a permutation $(x_1, x_2, \dots, x_n)$ of $(1, 2, \dots, n)$ such that for every $i \in \{1, 2, \dots, n\}$, the number x_{i+1} is one of the numbers $2x_i$, $2x_i - 1$, $2x_i - n$, $2x_i - n - 1$. Hereby, we use the cyclic subscript convention, so that x_{n+1} means x_1 .

5

11C6

Let n be a positive integer, and let $W = \dots x_{-1}x_0x_1x_2\dots$ be an infinite periodic word, consisting of just letters a and/or b . Suppose that the minimal period N of W is greater than 2^n .

A finite nonempty word U is said to appear in W if there exist indices $k \leq \ell$ such that $U = x_kx_{k+1}\dots x_\ell$. A finite word U is called ubiquitous if the four words Ua , Ub , aU , and bU all appear in W . Prove that there are at least n ubiquitous finite nonempty words.

Insights

Complex Deduction Problems

Similar to simple logic deduction problems. However, these may make use of special ideas and theorems, and may include multiple steps

Combinatorial Inequalities

These problem ask for a some counting that is very symmetric, such that it is algebraic in nature. Combinatorial ideas are useful, but also include some ideas of inequality from algebra. Some elementary inequalities such as AM-GM will be useful. The main goal should be simplifying and reducing the problem statement combinatorially, to a point where simple algebra and inequalities skills will finish.

Geometrical Counting Problems

Counting problem with a geometric settings. General techniques apply.

Combinatorial Counting-Identity Problems

These problem asks for proof of an relationship of counts. With this specific settings, in general it is not feasible to determine the exact no. of configurations. Instead, one should aim for directions to link two sides of the equations together. Specifically, through a mapping, bijective or otherwise, or construct objects that resemble two sides of the equations.

“Always-Exist” Problem

You are required to prove the existence of something, over all possible initial configurations. In general there are two ways to show existence. 1) By explicit construction. A direct construction is not possible to cover all variations of the initial conditions. Construction here means algorithm, some delicate ways to obtain a construction in every single circumstances. 2) By contradiction. To show something would fail if there is absolutely no such configurations.

Grand Finale

A collection of the hardest problems in combinatorics. To solve any one of them , one must use everything they have. Contestants must experiment with every single concepts and techniques.

A heavy element of combinatorial problems is explorations. Therefore, the problems are not arranged by what ideas are used. Instead, it is up to the readers to find out what lies behind each of them.

Equip yourself with **POST-GOLD** strength before proceeding. They are NOT to be tempered with. Nothing can be learnt from them unless you are of this level.

Way To Improve

If you are not ready for these problems, training on mid-tier (4/5) problems will be a good idea. Even if you can solve a few and are trying to reduce solving time, solving mid-tier problems is still a better idea. Focusing thoughts and actively try to reduce time that you are not thinking. You will steadily pick up speed.

5

11C5

Let m be a positive integer, and consider an $m \times m$ checkerboard consisting of unit squares. At the centre of some of these unit squares there is an ant. At time 0, each ant starts moving with speed 1 parallel to some edge of the checkerboard. When two ants moving in the opposite directions meet, they both turn 90° clockwise and continue moving with speed 1. When more than 2 ants meet, or when two ants moving in perpendicular directions meet, the ants continue moving in the same direction as before they met. When an ant reaches one of the edges of the checkerboard, it falls off and will not re-appear.

Considering all possible starting positions, determine the latest possible moment at which the last ant falls off the checkerboard, or prove that such a moment does not necessarily exist.

6

13C7

Let $n \geq 3$ be an integer, and consider a circle with $n + 1$ equally spaced points marked on it. Consider all labelings of these points with the numbers $0, 1, \dots, n$ such that each label is used exactly once; two such labelings are considered to be the same if one can be obtained from the other by a rotation of the circle. A labelling is called beautiful if, for any four labels $a < b < c < d$ with $a + d = b + c$, the chord joining the points labelled a and d does not intersect the chord joining the points labelled b and c .

Let M be the number of beautiful labelings, and let N be the number of ordered pairs (x, y) of positive integers such that $x + y \leq n$ and $\gcd(x, y) = 1$. Prove that $M = N + 1$.

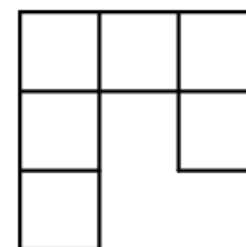
6

04C7

Define a "hook" to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotations and reflections to this figure.

Determine all $m \times n$ rectangles that can be covered without gaps and without overlaps with hooks such that

- the rectangle is covered without gaps and without overlaps
- no part of a hook covers area outside the rectangle.



6

14C5

A set of lines in the plane is in general position if no two are parallel and no three pass through the same point. A set of lines in general position cuts the plane into regions, some of which have finite area; we call these its finite regions. Prove that for all sufficiently large n , in any set of n lines in general position it is possible to colour at least $\sqrt{n}$ lines blue in such a way that none of its finite regions has a completely blue boundary.

6

09C8

For any integer $n \geq 2$, we compute the integer $h(n)$ by applying the following procedure to its decimal representation. Let r be the rightmost digit of n .

If $r = 0$, then the decimal representation of $h(n)$ results from the decimal representation of n by removing this rightmost digit 0.

If $1 \leq r \leq 9$ we split the decimal representation of n into a maximal right part R that solely consists of digits not less than r and into a left part L that either is empty or ends with a digit strictly smaller than r . Then the decimal representation of $h(n)$ consists of the decimal representation of L , followed by two copies of the decimal representation of $R - 1$. For instance, for the number 17,151,345,543, we will have $L = 17,151$, $R = 345,543$ and $h(n) = 17,151,345,542,345,542$.

Prove that, starting with an arbitrary integer $n \geq 2$, iterated application of h produces the integer 1 after finitely many steps.

6

06N6

Let $a > b > 1$ be relatively prime positive integers. Define the weight of an integer c , denoted by $w(c)$ to be the minimal possible value of $|x| + |y|$ taken over all pairs of integers x and y such that $ax + by = c$. An integer c is called a local champion if $w(c) \geq w(c \pm a)$ and $w(c) \geq w(c \pm b)$.

Find all local champions and determine their number.

6

13C5

Let r be a positive integer, and let $a_0, a_1, \dots$ be an infinite sequence of real numbers. Assume that for all nonnegative integers m and s there exists a positive integer $n \in [m+1, m+r]$ such that

$$a_m + a_{m+1} + \dots + a_{m+s} = a_n + a_{n+1} + \dots + a_{n+s}$$

Prove that the sequence is periodic, i.e. there exists some $p \geq 1$ such that $a_{n+p} = a_n$ for all $n \geq 0$.

6

10C6

Given a positive integer k and other two integers $b > w > 1$. There are two strings of pearls, a string of b black pearls and a string of w white pearls. The length of a string is the number of pearls on it. One cuts these strings in some steps by the following rules. In each step:

The strings are ordered by their lengths in a non-increasing order. If there are some strings of equal lengths, then the white ones precede the black ones. Then k first ones (if they consist of more than one pearl) are chosen; if there are less than k strings longer than 1, then one chooses all of them.

Next, one cuts each chosen string into two parts differing in length by at most one. (For instance, if there are strings of 5,4,4,2 black pearls, strings of 8,4,3 white pearls and $k = 4$, then the strings of 8 white, 5 black, 4 white and 4 black pearls are cut into the parts (4,4), (3,2), (2,2) and (2,2) respectively.) The process stops immediately after the step when a first isolated white pearl appears.

Prove that at this stage, there will still exist a string of at least two black pearls.

7

12C7

There are given 2^{500} points on a circle labeled $1, 2, \dots, 2^{500}$ in some order. Prove that one can choose 100 pairwise disjoint chords joining some of these points so that the 100 sums of the pairs of numbers at the endpoints of the chosen chords are equal.

7

The liar's guessing game is a game played between two players A and B . The rules of the game depend on two positive integers k and n which are known to both players.

12C6

At the start of the game A chooses integers x and N with $1 \leq x \leq N$. Player A keeps x secret, and truthfully tells N to player B . Player B now tries to obtain information about x by asking player A questions as follows: each question consists of B specifying an arbitrary set S of positive integers (possibly one specified in some previous question), and asking A whether x belongs to S . Player B may ask as many questions as he wishes. After each question, player A must immediately answer it with yes or no, but is allowed to lie as many times as she wants; the only restriction is that, among any $k + 1$ consecutive answers, at least one answer must be truthful.

After B has asked as many questions as he wants, he must specify a set X of at most n positive integers. If x belongs to X , then B wins; otherwise, he loses. Prove that:

(a) If $n \geq 2^k$, then B can guarantee a win.

(b) For all sufficiently large k , there exists an integer $n \geq (1.99)^k$ such that B cannot guarantee a win.

7

Suppose that $a_1, a_2, \dots, a_n$ are integers such that $n \mid a_1 + a_2 + \dots + a_n$.

05C7

Prove that there exist two permutations $(b_1, b_2, \dots, b_n)$ and $(c_1, c_2, \dots, c_n)$ of $(1, 2, \dots, n)$ such that for each integer i with $1 \leq i \leq n$, we have

$$n \mid a_i - b_i - c_i$$

7

Find all positive integers n for which there exist non-negative integers $a_1, a_2, \dots, a_n$ such that

12N7

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \dots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \dots + \frac{n}{3^{a_n}} = 1.$$

7 A triangulation of a convex polygon Π is a partitioning of Π into triangles by diagonals having no common points other than the vertices of the polygon. We say that a triangulation is a Thaiangulation if all triangles in it have the same area.

15G8

Prove that any two different Thaiangulations of a convex polygon Π differ by exactly two triangles. (In other words, prove that it is possible to replace one pair of triangles in the first Thaiangulation with a different pair of triangles so as to obtain the second Thaiangulation.)

7 Let S be a nonempty set of positive integers. We say that a positive integer n is clean if it has a unique representation as a sum of an odd number of distinct elements from S . Prove that there exist infinitely many positive integers that are not clean.

15C6

7 Let n be a positive integer. Determine the smallest positive integer k with the following property: it is possible to mark k cells on a $2n \times 2n$ board so that there exists a unique partition of the board into 1×2 and 2×1 dominoes, none of which contain two marked cells.

16C8

7 Assign to each side b of a convex polygon P the maximum area of a triangle that has P as a side and is contained in P . Show that the sum of the areas assigned to the sides of P is at least twice the area of P

06G10

7 For $n \geq 2$, let $S_1, S_2, \dots, S_{2^n}$ be 2^n subsets of $A = \{1, 2, 3, \dots, 2^{n+1}\}$ that satisfy the following property: There do not exist indices a and b with $a < b$ and elements $x, y, z \in A$ with $x < y < z$ and $y, z \in S_a$, and $x, z \in S_b$. Prove that at least one of the sets $S_1, S_2, \dots, S_{2^n}$ contains no more than $4n$ elements.

08C6

7 Let M be a set of $n \geq 4$ points in the plane, no three of which are collinear. Initially these points are connected with n segments so that each point in M is the endpoint of exactly two segments. Then, at each step, one may choose two segments AB and CD sharing a common interior point and replace them by the segments AC and BD if none of them is present at this moment. Prove that it is impossible to perform $n^3/4$ or more such moves.

14C7

8

09C7

Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n - 1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen in such a way that the grasshopper never lands on any point in M .

8

11C7

On a square table of 2011 by 2011 cells we place a finite number of napkins that each cover a square of 52 by 52 cells. In each cell we write the number of napkins covering it, and we record the maximal number k of cells that all contain the same nonzero number. Considering all possible napkin configurations, what is the largest value of k ?

8

15C7

In a company of people some pairs are enemies. A group of people is called unsociable if the number of members in the group is odd and at least 3, and it is possible to arrange all its members around a round table so that every two neighbors are enemies. Given that there are at most 2015 unsociable groups, prove that it is possible to partition the company into 11 parts so that no two enemies are in the same part.

Insight

Always try to keep in mind of all problem solving skills you have learnt up to this point and actively apply them. Look for special statements, think about if standard techniques are useful.

We mention two techniques that are only useful for contestants at this level. Incorrect uses will be very harmful.

After you found a result, think about in essence what you did: Why did these steps lead to something useful, and try to generalize it fully. This could lead to a stronger results.

Rejecting ideas. If an idea do not seem to work, look for justification. Think about a counter-example with which a certain direction can never handle. Think about if a certain idea fails to account for something or make use of conditions, Think about what this method fails to control. Afterwards, you can throw the entire idea out of the window and more likely to find one that works.

G

Geometry

Introductory Problems

Description

The easiest set of problems in geometry.

Main Idea

For simple geometry, the most important and useful skill is simple angle chasing, then, once in a while one may use parallelism, appropriate length chasing, constructing parallelograms, isosceles trapezium, radical axis/center and power.

Frequently the problems will be over within 2 or 3 steps.

How To Improve

For problems on this level, the following techniques will be useful.

1. Draw a good diagram. You can observe collinear/similarity/parallel more easily.
2. Wishful Thinking. Wish about what if something is true, for example a concyclic condition, then you can solve. The chance for wishes to come true are not low.
3. Backward deduction: Treat the problem statement as a result, since it is true. Then try to derive extra information about existing points
4. Reduction: Look for alternate conditions about required statements that can be intuitively easier. For correct intuition, this requires a better training.

Symmetric Construct

1
08G1

Let H be the orthocenter of an acute-angled triangle ABC . The circle Γ_A centered at the midpoint of BC and passing through H intersects the sideline BC at points A_1 and A_2 . Similarly, define the points B_1, B_2, C_1 and C_2 .

Prove that the six points A_1, A_2, B_1, B_2, C_1 and C_2 are concyclic.

1
01G1

Let A_1 be the center of the square inscribed in acute triangle ABC with two vertices of the square on side BC . Thus one of the two remaining vertices of the square is on side AB and the other is on AC . Points B_1, C_1 are defined in a similar way for inscribed squares with two vertices on sides AC and AB , respectively. Prove that lines AA_1, BB_1, CC_1 are concurrent.

1
05G2

Six points are chosen on the sides of an equilateral triangle ABC : A_1, A_2 on BC , B_1, B_2 on CA and C_1, C_2 on AB , such that they are the vertices of a convex hexagon $A_1A_2B_1B_2C_1C_2$ with equal side lengths.

Prove that the lines A_1B_2, B_1C_2 and C_1A_2 are concurrent.

Triangle Centers

Let ABC be triangle with incenter I . A point P in the interior of the triangle satisfies $\angle PBA + \angle PCA = \angle PBC + \angle PCB$. Show that $AP \geq AI$, and that equality holds if and only if $P = I$.

1
06G1

The circle S has centre O , and BC is a diameter of S . Let A be a point of S such that $\angle AOB < 120^\circ$. Let D be the midpoint of the arc AB which does not contain C . The line through O parallel to DA meets the line AC at I . The perpendicular bisector of OA meets S at E and at F . Prove that I is the incentre of the triangle CEF .

1
02G3

Let $\triangle ABC$ be an acute-angled triangle with $AB \neq AC$. Let H be the orthocenter of triangle ABC , and let M be the midpoint of the side BC . Let D be a point on the side AB and E a point on the side AC such that $AE = AD$ and the points D, H, E are on the same line. Prove that the line HM is perpendicular to the common chord of the circumscribed circles of triangle $\triangle ABC$ and triangle $\triangle ADE$.

2

05G5

Special Angle Problems

2

09G1

Let ABC be a triangle with $AB = AC$. The angle bisectors of $\angle CAB$ and $\angle ABC$ meet the sides BC and CA at D and E , respectively. Let K be the incentre of triangle ADC . Suppose that $\angle BEK = 45^\circ$. Find all possible values of $\angle CAB$.

2

04G3

Let O be the circumcenter of an acute-angled triangle ABC with $\angle B < \angle C$. The line AO meets the side BC at D . The circumcenters of the triangles ABD and ACD are E and F , respectively. Extend the sides BA and CA beyond A , and choose on the respective extensions points G and H such that $AG = AC$ and $AH = AB$. Prove that the quadrilateral $EFGH$ is a rectangle if and only if $\angle ACB - \angle ABC = 60^\circ$.

Core Problems

Let B be a point on a circle S_1 , and let A be a point distinct from B on the tangent at B to S_1 . Let C be a point not on S_1 such that the line segment AC meets S_1 at two distinct points. Let S_2 be the circle touching AC at C and touching S_1 at a point D on the opposite side of AC from B . Prove that the circumcentre of triangle BCD lies on the circumcircle of triangle ABC .

1

02G1

Let ABC be an acute-angled triangle with $AB \neq AC$. The circle with diameter BC intersects the sides AB and AC at M and N respectively. Denote by O the midpoint of the side BC . The bisectors of the angles $\angle BAC$ and $\angle MON$ intersect at R . Prove that the circumcircles of the triangles BMR and CNR have a common point lying on the side BC .

1

04G1

1 Let ABC be an acute triangle with D, E, F the feet of the altitudes lying on BC, CA, AB respectively. One of the intersection points of the line EF and the circumcircle is P . The lines BP and DF meet at point Q . Prove that $AP = AQ$.

10G1

1 Let $ABCD$ be a cyclic quadrilateral. Let P, Q, R be the feet of the perpendiculars from D to the lines BC, CA, AB , respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with AC .

03G1

1 Given a triangle ABC satisfying $AC + BC = 3 \cdot AB$. The incircle of triangle ABC has center I and touches the sides BC and CA at the points D and E , respectively. Let K and L be the reflections of the points D and E with respect to I . Prove that the points A, B, K, L lie on one circle.

05G1

1 In triangle ABC the bisector of angle BCA intersects the circumcircle again at R , the perpendicular bisector of BC at P , and the perpendicular bisector of AC at Q . The midpoint of BC is K and the midpoint of AC is L . Prove that the triangles RPK and RQL have the same area.

07G1

1 Given triangle ABC the point J is the centre of the excircle opposite the vertex A . This excircle is tangent to the side BC at M , and to the lines AB and AC at K and L , respectively. The lines LM and BJ meet at F , and the lines KM and CJ meet at G . Let S be the point of intersection of the lines AF and BC , and let T be the point of intersection of the lines AG and BC . Prove that M is the midpoint of ST .

12G1

1 Let P and Q be on segment BC of an acute triangle ABC such that $\angle PAB = \angle BCA$ and $\angle CAQ = \angle ABC$. Let M and N be the points on AP and AQ , respectively, such that P is the midpoint of AM and Q is the midpoint of AN . Prove that the intersection of BM and CN is on the circumcircle of triangle ABC .

14G1

1 Let ABC be an acute triangle with orthocenter H . Let G be the point such that the quadrilateral $ABGH$ is a parallelogram. Let I be the point on the line GH such that AC bisects HI . Suppose that the line AC intersects the circumcircle of the triangle GCI at C and J . Prove that $IJ = AH$.

15G1

1 Triangle ABC has circumcircle Ω and circumcenter O . A circle Γ with center A intersects the segment BC at points D and E , such that B, D, E , and C are all different and lie on line BC in this order. Let F and G be the points of intersection of Γ and Ω , such that A, F, B, C , and G lie on Ω in this order. Let K be the second point of intersection of the circumcircle of triangle BDF and the segment AB . Let L be the second point of intersection of the circumcircle of triangle CGE and the segment CA .

15G2

Suppose that the lines FK and GL are different and intersect at the point X . Prove that X lies on the line AO .

2 Two circles G_1 and G_2 intersect at two points M and N . Let AB be the line tangent to these circles at A and B , respectively, so that M lies closer to AB than N . Let CD be the line parallel to AB and passing through the point M , with C on G_1 and D on G_2 . Lines AC and BD meet at E ; lines AN and CD meet at P ; lines BN and CD meet at Q . Show that $EP = EQ$.

00G2

2 Let ABC be an acute triangle with orthocenter H , and let W be a point on the side BC , lying strictly between B and C . The points M and N are the feet of the altitudes from B and C , respectively. Denote by ω_1 is the circumcircle of BWN , and let X be the point on ω_1 such that WX is a diameter of ω_1 . Analogously, denote by ω_2 the circumcircle of triangle CWM , and let Y be the point such that WY is a diameter of ω_2 . Prove that X, Y and H are collinear.

13G1

2 Denote by M midpoint of side BC in an isosceles triangle $\triangle ABC$ with $AC = AB$. Take a point X on a smaller arc $\widehat{MA}$ of circumcircle of triangle $\triangle ABM$. Denote by T point inside of angle BMA such that $\angle TMX = 90$ and $TX = BX$.

07G2

Prove that $\angle MTB - \angle CTM$ does not depend on choice of X .

2

05G3

Let $ABCD$ be a parallelogram. A variable line g through the vertex A intersects the rays BC and DC at the points X and Y , respectively. Let K and L be the A -excenters of the triangles ABX and ADY . Show that the angle $\angle KCL$ is independent of the line g .

2

06G2

Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD , respectively, so that $AK/KB = DL/LC$. Suppose that there are points P and Q on the line segment KL satisfying $\angle APB = \angle BCD$ and $\angle CQD = \angle ABC$. Prove that the points P , Q , B and C are concyclic.

2

06G3

Let $ABCDE$ be a convex pentagon such that

$\angle BAC = \angle CAD = \angle DAE$ and $\angle ABC = \angle ACD = \angle ADE$. The diagonals BD and CE meet at P . Prove that the line AP bisects the side CD .

2

13G2

Let ω be the circumcircle of a triangle ABC . Denote by M and N the midpoints of the sides AB and AC , respectively, and denote by T the midpoint of the arc BC of ω not containing A . The circumcircles of the triangles AMT and ANT intersect the perpendicular bisectors of AC and AB at points X and Y , respectively; assume that X and Y lie inside the triangle ABC . The lines MN and XY intersect at K . Prove that $KA = KT$.

2

09G2

Let ABC be a triangle with circumcentre O . The points P and Q are interior points of the sides CA and AB respectively. Let K , L and M be the midpoints of the segments BP , CQ and PQ , respectively, and let Γ be the circle passing through K , L and M . Suppose that the line PQ is tangent to the circle Γ . Prove that $OP = OQ$.

2

10G2

Let P be a point interior to triangle ABC (with $CA \neq CB$). The lines AP , BP and CP meet again its circumcircle Γ at K , L , respectively M . The tangent line at C to Γ meets the line AB at S . Show that from $SC = SP$ follows $MK = ML$.

2

16G1

Triangle BCF has a right angle at B . Let A be the point on line CF such that $FA = FB$ and F lies between A and C . Point D is chosen so that $DA = DC$ and AC is the bisector of $\angle DAB$. Point E is chosen so that $EA = ED$ and AD is the bisector of $\angle EAC$. Let M be the midpoint of CF . Let X be the point such that $AMXE$ is a parallelogram. Prove that BD, FX and ME are concurrent.

217Q4
[G2]

Let R and S be different points on a circle Ω such that RS is not a diameter. Let ℓ be the tangent line to Ω at R . Point T is such that S is the midpoint of the line segment RT . Point J is chosen on the shorter arc RS of Ω so that the circumcircle Γ of triangle JST intersects ℓ at two distinct points. Let A be the common point of Γ and ℓ that is closer to R . Line AJ meets Ω again at K . Prove that the line KT is tangent to Γ .

2

09G3

Let ABC be a triangle. The incircle of ABC touches the sides AB and AC at the points Z and Y , respectively. Let G be the point where the lines BY and CZ meet, and let R and S be points such that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelogram.

Prove that $GR = GS$.

2

08G4

In an acute triangle ABC segments BE and CF are altitudes. Two circles passing through the points A and F and tangent to the line BC at the points P and Q so that B lies between C and Q . Prove that lines PE and QF intersect on the circumcircle of triangle AEF .

2

14G3

Let Ω and O be the circumcircle and the circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM . The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q , respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that $BR \parallel AC$.

2

12G3

In an acute triangle ABC the points D, E and F are the feet of the altitudes through A, B and C respectively. The incenters of the triangles AEF and BDF are I_1 and I_2 respectively; the circumcenters of the triangles ACI_1 and BCI_2 are O_1 and O_2 respectively. Prove that I_1I_2 and O_1O_2 are parallel.

2

08G2

Given trapezoid $ABCD$ with parallel sides AB and CD , assume that there exist points E on line BC outside segment BC , and F inside segment AD such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF , and by J the point of intersection of AB and EF . Let K be the midpoint of segment EF , assume it does not lie on line AB . Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ .

3

04G5

Let $A_1A_2A_3\dots A_n$ be a regular n -gon. Let B_1 and B_n be the midpoints of its sides A_1A_2 and $A_{n-1}A_n$. Also, for every $i \in \{2, 3, 4, \dots, n-1\}$. Let S be the point of intersection of the lines A_1A_{i+1} and A_nA_i , and let B_i be the point of intersection of the angle bisector of the angle $\angle A_iSA_{i+1}$ with the segment A_iA_{i+1} .

Prove that $\sum_{i=1}^{n-1} \angle A_1B_iA_n = 180^\circ$.

3

03G3

Let ABC be a triangle, and P a point in the interior of this triangle. Let D, E, F be the feet of the perpendiculars from the point P to the lines BC, CA, AB , respectively. Assume that

$$AP^2 + PD^2 = BP^2 + PE^2 = CP^2 + PF^2.$$

Furthermore, let I_a, I_b, I_c be the excenters of triangle ABC . Show that the point P is the circumcenter of triangle $I_aI_bI_c$.

3

04G4

In a convex quadrilateral $ABCD$, the diagonal BD bisects neither the angle ABC nor the angle CDA . The point P lies inside $ABCD$ and satisfies $\angle PBC = \angle DBA$ and $\angle PDC = \angle BDA$. Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$.

Insights

Symmetric Construct

Problem statements may seem symmetric and nice. There are almost no ways to deal with the entire all at once. Instead, one should break down the statement smaller statements that concern 2 of the directions and attempt to prove so, then combine the symmetric results to give one statement.

Triangle Centers

For these very well-known points, being aware of their properties is very important. Understanding interactions between mid-point of arcs and incenter, circumcircle and orthocenter is a must for , in fact, all problems. At this level, incenter and orthocenter will be sufficient for our purposes.

Special Angle Problem

Special angles that are not 90 degrees. Unfortunately, there are no consistent way to make use of them in pure geometry. Therefore, do not be afraid to use trigonometric functions and length to get what you want.

High-level geometry Problem

Description

The medium difficulty problems in geometry.

Main Idea

As one progresses through the higher tiers, more advanced techniques, facts and concepts are necessary. False position, Homothety, Radical axis, simson's line, anti-parallel, isogonal, parallelism, pascal and much more. If one spend some time to learn projective geometry and inversion, they can be of assistance a lot of the time, although elementary geometry skills are all still necessary.

Ways To Improve

More training, more training, unfortunately due to the mechanical nature of geometry, this is the only way.

Learn supplement methods: Complex and coordinate, and length chasing. It is generally a poor attitude to depend yourself on these techniques, yet if you are one step away from done, and within the useful scope of these techniques, having these techniques as an emergency solution is reasonable.

Invariant Points

1

00G1

In the plane we are given two circles intersecting at X and Y . Prove that there exist four points with the following property:

(P) For every circle touching the two given circles at A and B , and meeting the line XY at C and D , each of the lines AC , AD , BC , BD passes through one of these points.

2

00G3

Let O be the circumcenter and H the orthocenter of an acute triangle ABC . Show that there exist points D , E , and F on sides BC , CA , and AB respectively such that $OD + DH = OE + EH = OF + FH$ and the lines AD , BE , and CF are concurrent.

2

02G4

Circles S_1 and S_2 intersect at points P and Q . Distinct points A_1 and B_1 (not at P or Q) are selected on S_1 . The lines A_1P and B_1P meet S_2 again at A_2 and B_2 respectively, and the lines A_1B_1 and A_2B_2 meet at C . Prove that, as A_1 and B_1 vary, the circumcentres of triangles A_1A_2C all lie on one fixed circle.

2

03G2

Given three fixed pairwise distinct points A , B , C lying on one straight line in this order. Let G be a circle passing through A and C whose center does not lie on the line AC . The tangents to G at A and C intersect each other at a point P . The segment PB meets the circle G at Q .

Show that the point of intersection of the angle bisector of the angle AQC with the line AC does not depend on the choice of the circle G .

3

05G4

Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and BC not parallel with DA . Let two variable points E and F lie on the sides BC and DA , respectively and satisfy $BE = DF$. The lines AC and BD meet at P , the lines BD and EF meet at Q , the lines EF and AC meet at R .

Prove that the circumcircles of the triangles PQR , as E and F vary, have a common point other than P .

Consider a fixed circle Γ with three fixed points A, B , and C on it. Also, let us fix a real number $\lambda \in (0,1)$. For a variable point $P \notin \{A, B, C\}$ on Γ , let M be the point on the segment CP such that $CM = \lambda \cdot CP$. Let Q be the second point of intersection of the circumcircles of the triangles AMP and BMC . Prove that as P varies, the point Q lies on a fixed circle.

3

14G4

For a given triangle ABC , let X be a variable point on the line BC such that C lies between B and X and the incircles of the triangles ABX and ACX intersect at two distinct points P and Q . Prove that the line PQ passes through a point independent of X .

4

04G7

Simple but deadly

2

12G2

Let $ABCD$ be a cyclic quadrilateral whose diagonals AC and BD meet at E . The extensions of the sides AD and BC beyond A and B meet at F . Let G be the point such that $ECGD$ is a parallelogram, and let H be the image of E under reflection in AD . Prove that D, H, F, G are concyclic.

3

16G4

Let ABC be a triangle with $AB = AC \neq BC$ and let I be its incentre. The line BI meets AC at D , and the line through D perpendicular to AC meets AI at E . Prove that the reflection of I in AC lies on the circumcircle of triangle BDE .

3

07G3

The diagonals of a trapezoid $ABCD$ intersect at point P . Point Q lies between the parallel lines BC and AD such that $\angle A Q D = \angle C Q B$, and line CD separates points P and Q . Prove that $\angle B Q P = \angle D A Q$.

4

12G4

Let ABC be a triangle with $AB \neq AC$ and circumcenter O . The bisector of $\angle BAC$ intersects BC at D . Let E be the reflection of D with respect to the midpoint of BC . The lines through D and E perpendicular to BC intersect the lines AO and AD at X and Y respectively. Prove that the quadrilateral $BXC Y$ is cyclic.

4

Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F . The midpoints of AB and CD are G and H , respectively. Show that EF is tangent at E to the circle through the points E , G and H .

Projective Nature

3

Let Γ be a circle and let d be a line such that Γ and d have no common points. Further, let AB be a diameter of the circle Γ ; assume that this diameter AB is perpendicular to the line d , and the point B is nearer to the line d than the point A . Let C be an arbitrary point on the circle Γ , different from the points A and B . Let D be the point of intersection of the lines AC and d . One of the two tangents from the point D to the circle Γ touches this circle Γ at a point E ; hereby, we assume that the points B and E lie in the same halfplane with respect to the line AC . Denote by F the point of intersection of the lines BE and d . Let the line AF intersect the circle Γ at a point G , different from A .

Prove that the reflection of the point G in the line AB lies on the line CF .

Length Evaluation

Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let O_1 and r_1 be the circumcentre and the circumradius of the triangle $A_2A_3A_4$. Define O_2, O_3, O_4 and r_2, r_3, r_4 in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

2

11G2

Let $\Gamma_1, \Gamma_2, \Gamma_3, \Gamma_4$ be distinct circles such that Γ_1, Γ_3 are externally tangent at P , and Γ_2, Γ_4 are externally tangent at the same point P . Suppose that Γ_1 and Γ_2 ; Γ_2 and Γ_3 ; Γ_3 and Γ_4 ; Γ_4 and Γ_1 meet at A, B, C, D , respectively, and that all these points are different from P . Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

2

03G4

Let ABC be an acute triangle. Let DAC , EAB , and FBC be isosceles triangles exterior to ABC , with $DA = DC$, $EA = EB$, and $FB = FC$, such that

$$\angle ADC = 2\angle BAC, \quad \angle BEA = 2\angle ABC, \quad \angle CFB = 2\angle ACB.$$

Let D' be the intersection of lines DB and EF , let E' be the intersection of EC and DF , and let F' be the intersection of FA and DE . Find, with proof, the value of the sum

$$\frac{DB}{DD'} + \frac{EC}{EE'} + \frac{FA}{FF'}.$$

Simple Geometric Inequality

1
01G2

Consider an acute-angled triangle ABC . Let P be the foot of the altitude of triangle ABC issuing from the vertex A , and let O be the circumcenter of triangle ABC . Assume that $\angle C \geq \angle B + 30^\circ$. Prove that $\angle A + \angle COP < 90^\circ$.

1
14G2

Let ABC be a triangle. The points K , L , and M lie on the segments BC , CA , and AB , respectively, such that the lines AK , BL , and CM intersect in a common point. Prove that it is possible to choose two of the triangles ALM , BMK , and CKL whose inradii sum up to at least the inradius of the triangle ABC .

2
02G2

Let ABC be a triangle for which there exists an interior point F such that $\angle AFB = \angle BFC = \angle CFA$. Let the lines BF and CF meet the sides AC and AB at D and E respectively. Prove that $AB + AC \geq 4DE$.

2
11G1

Let ABC be an acute triangle. Let ω be a circle whose centre L lies on the side BC . Suppose that ω is tangent to AB at B' and AC at C' . Suppose also that the circumcentre O of triangle ABC lies on the shorter arc $B'C'$ of ω . Prove that the circumcircle of ABC and ω meet at two points.

Let ABC be a triangle with centroid G . Determine, with proof, the position of the point P in the plane of ABC such that $AP \cdot AG + BP \cdot BG + CP \cdot CG$ is a minimum, and express this minimum value in terms of the side lengths of ABC .

2

01G3

Core Problems

3

08G3

Let $ABCD$ be a convex quadrilateral and let P and Q be points in $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point E on the line segment PQ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic.

3

06G4

A point D is chosen on the side AC of a triangle ABC with $\angle C < \angle A < 90^\circ$ in such a way that $BD = BA$. The incircle of ABC is tangent to AB and AC at points K and L , respectively. Let J be the incenter of triangle BCD . Prove that the line KL intersects the line segment AJ at its midpoint.

4

04G8

Given a cyclic quadrilateral $ABCD$, let M be the midpoint of the side CD , and let N be a point on the circumcircle of triangle ABM . Assume that the point N is different from the point M and satisfies $\frac{AN}{BN} = \frac{AM}{BM}$. Prove that the points E , F , N are collinear, where $E = AC \cap BD$ and $F = BC \cap DA$.

4

10G5

Let $ABCDE$ be a convex pentagon such that $BC \parallel AE$, $AB = BC + AE$, and $\angle ABC = \angle CDE$. Let M be the midpoint of CE , and let O be the circumcenter of triangle BCD . Given that $\angle DMO = 90^\circ$, prove that $2\angle BDA = \angle CDE$.

4

13G4

Let ABC be a triangle with $\angle B > \angle C$. Let P and Q be two different points on line AC such that $\angle PBA = \angle QBA = \angle ACB$ and A is located between P and C . Suppose that there exists an interior point D of segment BQ for which $PD = PB$. Let the ray AD intersect the circle ABC at $R \neq A$. Prove that $QB = QR$.

4

06G5

In triangle ABC , let J be the center of the excircle tangent to side BC at A_1 and to the extensions of the sides AC and AB at B_1 and C_1 respectively. Suppose that the lines A_1B_1 and AB are perpendicular and intersect at D . Let E be the foot of the perpendicular from C_1 to line DJ . Determine the angles $\angle BEA_1$ and $\angle AEB_1$.

4

00G6

Let $ABCD$ be a convex quadrilateral. The perpendicular bisectors of its sides AB and CD meet at Y . Denote by X a point inside the quadrilateral $ABCD$ such that $\angle ADX = \angle BCX < 90^\circ$ and $\angle DAX = \angle CBX < 90^\circ$. Show that $\angle AYB = 2 \cdot \angle ADX$.

4

10G4

Given a triangle ABC , with I as its incenter and Γ as its circumcircle, AI intersects Γ again at D . Let E be a point on the arc BDC , and F a point on the segment BC , such that $\angle BAF = \angle CAE < \frac{1}{2}\angle BAC$. If G is the midpoint of IF , prove that the meeting point of the lines EI and DG lies on Γ .

4

11G3

Let $ABCD$ be a convex quadrilateral whose sides AD and BC are not parallel. Suppose that the circles with diameters AB and CD meet at points E and F inside the quadrilateral. Let ω_E be the circle through the feet of the perpendiculars from E to the lines AB , BC and CD . Let ω_F be the circle through the feet of the perpendiculars from F to the lines CD , DA and AB . Prove that the midpoint of the segment EF lies on the line through the two intersections of ω_E and ω_F .

4

07G4

Consider five points A , B , C , D and E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let ℓ be a line passing through A . Suppose that ℓ intersects the interior of the segment DC at F and intersects line BC at G . Suppose also that $EF = EG = EC$. Prove that ℓ is the bisector of angle DAB .

4

12G5

Let ABC be a triangle with $\angle BCA = 90^\circ$, and let D be the foot of the altitude from C . Let X be a point in the interior of the segment CD . Let K be the point on the segment AX such that $BK = BC$. Similarly, let L be the point on the segment BX such that $AL = AC$. Let M be the point of intersection of AL and BK . Show that $MK = ML$.

4

02G8

Let two circles S_1 and S_2 meet at the points A and B . A line through A meets S_1 again at C and S_2 again at D . Let M, N, K be three points on the line segments CD, BC, BD respectively, with MN parallel to BD and MK parallel to BC . Let E and F be points on those arcs BC of S_1 and BD of S_2 respectively that do not contain A . Given that EN is perpendicular to BC and FK is perpendicular to BD prove that $\angle EMF = 90^\circ$.

4

11G4

Let ABC be an acute triangle with circumcircle Ω . Let B_0 be the midpoint of AC and let C_0 be the midpoint of AB . Let D be the foot of the altitude from A and let G be the centroid of the triangle ABC . Let ω be a circle through B_0 and C_0 that is tangent to the circle Ω at a point $X \neq A$. Prove that the points D, G and X are collinear.

4

11G5

Let ABC be a triangle with incentre I and circumcircle ω . Let D and E be the second intersection points of ω with AI and BI , respectively. The chord DE meets AC at a point F , and BC at a point G . Let P be the intersection point of the line through F parallel to AD and the line through G parallel to BE . Suppose that the tangents to ω at A and B meet at a point K . Prove that the three lines AE, BD and KP are either parallel or concurrent.

4

07G5

Let ABC be a fixed triangle, and let A_1, B_1, C_1 be the midpoints of sides BC, CA, AB , respectively. Let P be a variable point on the circumcircle. Let lines PA_1, PB_1, PC_1 meet the circumcircle again at A', B', C' , respectively. Assume that the points A, B, C, A', B', C' are distinct, and lines AA', BB', CC' form a triangle. Prove that the area of this triangle does not depend on P .

4

02G7

The incircle Ω of the acute-angled triangle ABC is tangent to its side BC at a point K . Let AD be an altitude of triangle ABC , and let M be the midpoint of the segment AD . If N is the common point of the circle Ω and the line KM (distinct from K), then prove that the incircle Ω and the circumcircle of triangle BCN are tangent to each other at the point N .

4

15G4

Let ABC be an acute triangle and let M be the midpoint of AC . A circle ω passing through B and M meets the sides AB and BC at points P and Q respectively. Let T be the point such that $BPTQ$ is a parallelogram. Suppose that T lies on the circumcircle of ABC . Determine all possible values of $\frac{BT}{BM}$.

4

05G6

Let ABC be a triangle, and M the midpoint of its side BC . Let γ be the incircle of triangle ABC . The median AM of triangle ABC intersects the incircle γ at two points K and L . Let the lines passing through K and L , parallel to BC , intersect the incircle γ again in two points X and Y . Let the lines AX and AY intersect BC again at the points P and Q . Prove that $BP = CQ$.

5

03G5

Let ABC be an isosceles triangle with $AC = BC$, whose incentre is I . Let P be a point on the circumcircle of the triangle AIB lying inside the triangle ABC . The lines through P parallel to CA and CB meet AB at D and E , respectively. The line through P parallel to AB meets CA and CB at F and G , respectively. Prove that the lines DF and EG intersect on the circumcircle of the triangle ABC .

5

08G6

There is given a convex quadrilateral $ABCD$. Prove that there exists a point P inside the quadrilateral such that

$$\angle PAB + \angle PDC = \angle PBC + \angle PAD = \angle PCD + \angle PBA = \angle PDA + \angle PCB = 90^\circ$$

if and only if the diagonals AC and BD are perpendicular.

5

11G6

Let ABC be a triangle with $AB = AC$ and let D be the midpoint of AC . The angle bisector of $\angle BAC$ intersects the circle through D, B and C at the point E inside the triangle ABC . The line BD intersects the circle through A, E and B in two points B and F . The lines AF and BE meet at a point I , and the lines CI and BD meet at a point K . Show that I is the incentre of triangle KAB .

Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω . 15G3

Geometrical Combin

2

16G3

Let $B = (-1,0)$ and $C = (1,0)$ be fixed points on the coordinate plane. A nonempty, bounded subset S of the plane is said to be nice if

(i) there is a point T in S such that for every point Q in S , the segment TQ lies entirely in S ; and

(ii) for any triangle $P_1P_2P_3$, there exists a unique point A in S and a permutation σ of the indices $\{1,2,3\}$ for which triangles ABC and $P_{\sigma(1)}P_{\sigma(2)}P_{\sigma(3)}$ are similar.

Prove that there exist two distinct nice subsets S and S' of the set $\{(x,y) : x \geq 0, y \geq 0\}$ such that if $A \in S$ and $A' \in S'$ are the unique choices of points in (ii), then the product $BA \cdot BA'$ is a constant independent of the triangle $P_1P_2P_3$.

3

10G3

Let $A_1A_2\dots A_n$ be a convex polygon. Point P inside this polygon is chosen so that its projections $P_1, \dots, P_n$ onto lines $A_1A_2, \dots, A_nA_1$ respectively lie on the sides of the polygon. Prove that for arbitrary points $X_1, \dots, X_n$ on sides $A_1A_2, \dots, A_nA_1$ respectively,

$$\max \left\{ \frac{X_1X_2}{P_1P_2}, \dots, \frac{X_nX_1}{P_nP_1} \right\} \geq 1.$$

4

01G4

Let M be a point in the interior of triangle ABC . Let A' lie on BC with MA' perpendicular to BC . Define B' on CA and C' on AB similarly. Define

$$p(M) = \frac{MA' \cdot MB' \cdot MC'}{MA \cdot MB \cdot MC}.$$

Determine, with proof, the location of M such that $p(M)$ is maximal. Let $\mu(ABC)$ denote this maximum value. For which triangles ABC is the value of $\mu(ABC)$ maximal?

4

02G5

For any set S of five points in the plane, no three of which are collinear, let $M(S)$ and $m(S)$ denote the greatest and smallest areas, respectively, of triangles determined by three points from S . What is the minimum possible value of $M(S)/m(S)$?

4

02G6

Let $n \geq 3$ be a positive integer. Let $C_1, C_2, C_3, \dots, C_n$ be unit circles in the plane, with centres $O_1, O_2, O_3, \dots, O_n$ respectively. If no line meets more than two of the circles, prove that
$$\sum_{1 \leq i < j \leq n} \frac{1}{O_i O_j} \leq \frac{(n-1)\pi}{4}.$$

5

04G6

Let P be a convex polygon. Prove that there exists a convex hexagon that is contained in P and whose area is at least $\frac{3}{4}$ of the area of the polygon P .

5

09G5

Let P be a polygon that is convex and symmetric to some point O . Prove that for some parallelogram R satisfying $P \subset R$ we have
$$\frac{|R|}{|P|} \leq \sqrt{2}$$

where $|R|$ and $|P|$ denote the area of the sets R and P , respectively.

6

03G6

Each pair of opposite sides of a convex hexagon has the following property: the distance between their midpoints is equal to $\frac{\sqrt{3}}{2}$ times the sum of their lengths. Prove that all the angles of the hexagon are equal.

Insights

Invariant Points

These problems require the proof of existence of a certain point, without asking what it is. However, in most cases it is impossible to prove so without giving an explicit definition for this point. Therefore, the first step to these problems is to identify the points, then follow up with standard techniques.

Simple but deadly

Problems having so few points. Simple? Not at all.

The necessary information and properties can't possibly be contained within the points themselves. Therefore it is pointless to focus only on the existing points, say , look for concyclic points within the given points. Instead, one must actively construct new points creatively to make use of the conditions.

Projective Nature

This problem sets in projective geometry. Using simple geometry is fine, but do miss the point of the problem. I hereby encourage the readers to learn projective geometry.

Length Evaluation

Tedious expressions, which leads to a different dynamic. Instead of looking for certain results, one should try to derive as much information about the required points as possible, and think at each point of time , whether the derived information determined the quantity good as to finish the problem.

Simple Geometric Inequality

Proceed similar to normal geometry problems, except the finishing touch will be inequality. Triangle inequality, Ptolmey, or algebraic classifications (i.e. length chasing) are all essential.

Core Problems

Apply the following general strategies: Creatively Construct points. At this level, not all information are contained by the points that already exist. Counter-intuitively, extra points will derive additional information about existing problems, and will can be removed form the field once the information is gained.

Focus on results you are trying to prove, and draw a new diagram removing all likely unnecessary points and lines.

Draw a VERY GOOD diagram. To quickly confirm any properties, particularly about constructed points, an accurate diagram is very essential.

Geometrical Combin

Sorry, nothing to offer....

Highly-Advanced Geometry Problems

Description

The very high difficulty problems in geometry.

Main Idea

There are no further technique requirements from tier 2. However, problems on this level are generally very long and unintuitive. Most problems simply start by constructing almost random points. Previous techniques such as backward reasonings are sometimes ineffective, for there are 3 or 4 hidden connections. In general, there are many more ways to solve an advanced problems of this scale. Therefore, it comes down to using whatever you are most familiar with, and have expertise over your own toolbox.

Ways To Improve

Improve your tool box. In particularly , projective geometry and inversion are almost must have at this level. They provide useful insight and lemmas, along with their own useful deductive power.

Learn more facts, “Definitions and facts on Plane geometry” is a good start. Do not ignore difficult theorems, such as mixtilinear circles and isogonal lines.

Advanced Geometric Inequalities

3
13G3

In a triangle ABC , let D and E be the feet of the angle bisectors of angles A and B , respectively. A rhombus is inscribed into the quadrilateral $AEDB$ (all vertices of the rhombus lie on different sides of $AEDB$). Let φ be the non-obtuse angle of the rhombus. Prove that $\varphi \leq \max\{\angle BAC, \angle ABC\}$

6
07G6

Determine the smallest positive real number k with the following property. Let $ABCD$ be a convex quadrilateral, and let points A_1 , B_1 , C_1 , and D_1 lie on sides AB , BC , CD , and DA , respectively. Consider the areas of triangles AA_1D_1 , BB_1A_1 , CC_1B_1 and DD_1C_1 ; let S be the sum of the two smallest ones, and let S_1 be the area of quadrilateral $A_1B_1C_1D_1$. Then we always have $kS_1 \geq S$.

6
01G7

Let O be an interior point of acute triangle ABC . Let A_1 lie on BC with OA_1 perpendicular to BC . Define B_1 on CA and C_1 on AB similarly. Prove that O is the circumcenter of ABC if and only if the perimeter of $A_1B_1C_1$ is not less than any one of the perimeters of AB_1C_1 , BC_1A_1 , and CA_1B_1 .

6
07G7

Given an acute triangle ABC with $\angle B > \angle C$. Point I is the incenter, and R the circumradius. Point D is the foot of the altitude from vertex A . Point K lies on line AD such that $AK = 2R$, and D separates A and K . Lines DI and KI meet sides AC and BC at E, F respectively. Let $IE = IF$.

Prove that $\angle B \leq 3\angle C$.

6
10G6

The vertices X, Y, Z of an equilateral triangle XYZ lie respectively on the sides BC, CA, AB of an acute-angled triangle ABC . Prove that the incenter of triangle ABC lies inside triangle XYZ .

6

05G7

In an acute triangle ABC , let D, E, F be the feet of the perpendiculars from the points A, B, C to the lines BC, CA, AB , respectively, and let P, Q, R be the feet of the perpendiculars from the points A, B, C to the lines EF, FD, DE , respectively.

Prove that $p(ABC)p(PQR) \geq (p(DEF))^2$, where $p(T)$ denotes the perimeter of triangle T .

6

06G8

Let $ABCD$ be a convex quadrilateral. A circle passing through the points A and D and a circle passing through the points B and C are externally tangent at a point P inside the quadrilateral. Suppose that

$\angle PAB + \angle PDC \leq 90^\circ$ and $\angle PBA + \angle PCD \leq 90^\circ$. Prove that $AB + CD \geq BC + AD$.

7

16G8

Let A_1, B_1 and C_1 be points on sides BC, CA and AB of an acute triangle ABC respectively, such that AA_1, BB_1 and CC_1 are the internal angle bisectors of triangle ABC . Let I be the incentre of triangle ABC , and H be the orthocentre of triangle $A_1B_1C_1$. Show that $AH + BH + CH \geq AI + BI + CI$.

Incircle and Tangents

3

06G6

Circles w_1 and w_2 with centres O_1 and O_2 are externally tangent at point D and internally tangent to a circle w at points E and F respectively. Line t is the common tangent of w_1 and w_2 at D . Let AB be the diameter of w perpendicular to t , so that A, E, O_1 are on the same side of t . Prove that lines AO_1, BO_2, EF and t are concurrent.

4

08G7

Let $ABCD$ be a convex quadrilateral with $BA \neq BC$. Denote the incircles of triangles ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle ω tangent to ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents to ω_1 and ω_2 intersect on ω .

6

12G7

Let $ABCD$ be a convex quadrilateral with non-parallel sides BC and AD . Assume that there is a point E on the side BC such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point F on the side AD such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if AB is parallel to CD .

6

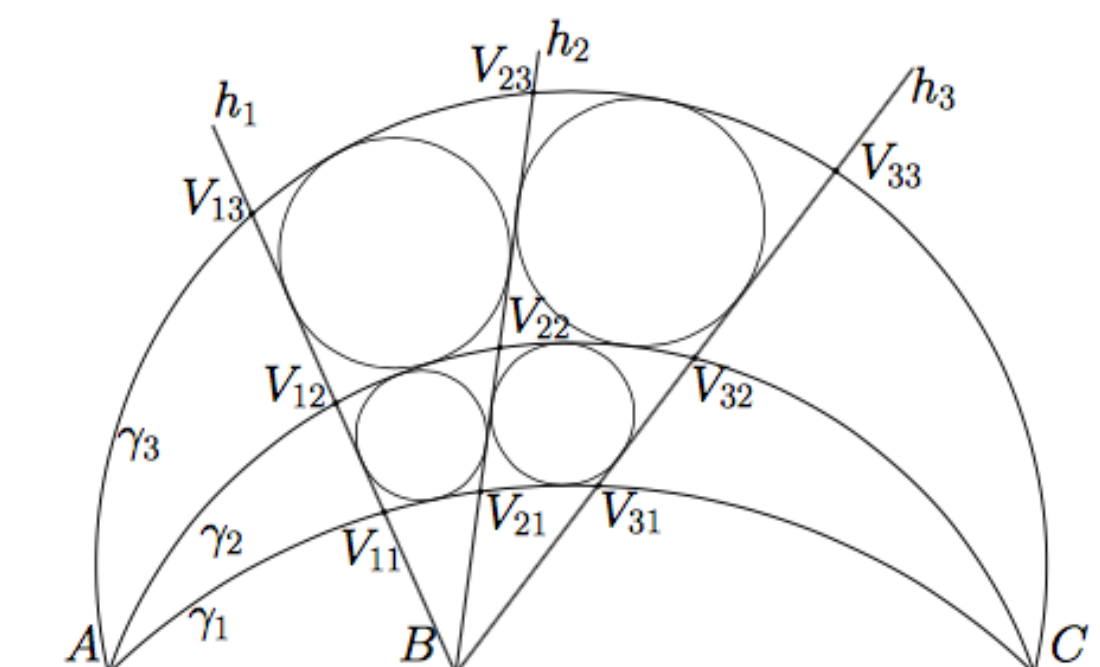
07G8

Point P lies on side AB of a convex quadrilateral $ABCD$. Let ω be the incircle of triangle CPD , and let I be its incenter. Suppose that ω is tangent to the incircles of triangles APD and BPC at points K and L , respectively. Let lines AC and BD meet at E , and let lines AK and BL meet at F . Prove that points E , I , and F are collinear.

7

10G7

Three circular arcs γ_1, γ_2 , and γ_3 connect the points A and C . These arcs lie in the same half-plane defined by line AC in such a way that arc γ_2 lies between the arcs γ_1 and γ_3 . Point B lies on the segment AC . Let h_1, h_2 , and h_3 be three rays starting at B , lying in the same half-plane, h_2 being between h_1 and h_3 . For $i, j = 1, 2, 3$, denote by V_{ij} the point of intersection of h_i and γ_j (see the Figure below). Denote by $\widehat{V_{ij}V_{kj}} \widehat{V_{kl}V_{il}}$ the curved quadrilateral, whose sides are the segments $V_{ij}V_{il}$, $V_{kj}V_{kl}$ and arcs $V_{ij}V_{kj}$ and $V_{il}V_{kl}$. We say that this quadrilateral is *circumscribed* if there exists a circle touching these two segments and two arcs. Prove that if the curved quadrilaterals $\widehat{V_{11}V_{21}V_{22}V_{12}}, \widehat{V_{12}V_{22}V_{23}V_{13}}, \widehat{V_{21}V_{31}V_{32}V_{22}}$ are circumscribed, then the curved quadrilateral $\widehat{V_{22}V_{32}V_{33}V_{23}}$ is circumscribed, too.



Let $ABCD$ be a convex quadrilateral, and let P , Q , R , and S be points on the sides AB , BC , CD , and DA , respectively. Let the line segment PR and QS meet at O . Suppose that each of the quadrilaterals $APOS$, $BQOP$, $CROQ$, and $DSOR$ has an incircle. Prove that the lines AC , PQ , and RS are either concurrent or parallel to each other.

Advanced Symmetric Constructions

4

00G8

Let AH_1 , BH_2 , CH_3 be the altitudes of an acute angled triangle ABC . Its incircle touches the sides BC , AC and AB at T_1 , T_2 and T_3 respectively. Consider the symmetric images of the lines H_1H_2 , H_2H_3 and H_3H_1 with respect to the lines T_1T_2 , T_2T_3 and T_3T_1 . Prove that these images form a triangle whose vertices lie on the incircle of ABC .

4

01G6

Let ABC be a triangle and P an exterior point in the plane of the triangle. Suppose the lines AP , BP , CP meet the sides BC , CA , AB (or extensions thereof) in D , E , F , respectively. Suppose further that the areas of triangles PBD , PCE , PAF are all equal. Prove that each of these areas is equal to the area of triangle ABC itself.

5

06G9

Points A_1 , B_1 , C_1 are chosen on the sides BC , CA , AB of a triangle ABC respectively. The circumcircles of triangles AB_1C_1 , BC_1A_1 , CA_1B_1 intersect the circumcircle of triangle ABC again at points A_2 , B_2 , C_2 respectively ($A_2 \neq A$, $B_2 \neq B$, $C_2 \neq C$). Points A_3 , B_3 , C_3 are symmetric to A_1 , B_1 , C_1 with respect to the midpoints of the sides BC , CA , AB respectively. Prove that the triangles $A_2B_2C_2$ and $A_3B_3C_3$ are similar.

5

06G7

In a triangle ABC , let M_a , M_b , M_c be the midpoints of the sides BC , CA , AB , respectively, and T_a , T_b , T_c be the midpoints of the arcs BC , CA , AB of the circumcircle of ABC , not containing the vertices A , B , C , respectively. For $i \in \{a, b, c\}$, let w_i be the circle with M_iT_i as diameter. Let p_i be the common external tangent to the circles w_j and w_k (for all $\{i, j, k\} = \{a, b, c\}$) such that w_i lies on the opposite side of p_i than w_j and w_k do.

Prove that the lines p_a , p_b , p_c form a triangle similar to ABC and find the ratio of similitude

516G7
(a)

Let I be the incentre of a non-equilateral triangle ABC , I_A be the A -excentre, I'_A be the reflection of I_A in BC , and l_A be the reflection of line AI'_A in AI . Define points I_B , I'_B and line l_B analogously. Let P be the intersection point of l_A and l_B . Prove that P lies on line OI where O is the circumcentre of triangle ABC .

6

13G6

Let the excircle of triangle ABC opposite the vertex A be tangent to the side BC at the point A_1 . Define the points B_1 on CA and C_1 on AB analogously, using the excircles opposite B and C , respectively. Suppose that the circumcentre of triangle $A_1B_1C_1$ lies on the circumcircle of triangle ABC . Prove that triangle ABC is right-angled.

7

11G8

Let ABC be an acute triangle with circumcircle Γ . Let ℓ be a tangent line to Γ , and let ℓ_a, ℓ_b and ℓ_c be the lines obtained by reflecting ℓ in the lines BC , CA and AB , respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b and ℓ_c is tangent to the circle Γ .

7

12G8

Let ABC be a triangle with circumcircle ω and ℓ a line without common points with ω . Denote by P the foot of the perpendicular from the center of ω to ℓ . The side-lines BC, CA, AB intersect ℓ at the points X, Y, Z different from P . Prove that the circumcircles of the triangles AXP , BYP and CZP have a common point different from P or are mutually tangent at P .

General Problems

5

16G5

Let D be the foot of perpendicular from A to the Euler line (the line passing through the circumcentre and the orthocentre) of an acute scalene triangle ABC . A circle ω with centre S passes through A and D , and it intersects sides AB and AC at X and Y respectively. Let P be the foot of altitude from A to BC , and let M be the midpoint of BC . Prove that the circumcentre of triangle $XS Y$ is equidistant from P and M .

5

00G5

The tangents at B and A to the circumcircle of an acute angled triangle ABC meet the tangent at C at T and U respectively. AT meets BC at P , and Q is the midpoint of AP ; BU meets CA at R , and S is the midpoint of BR . Prove that $\angle ABQ = \angle BAS$. Determine, in terms of ratios of side lengths, the triangles for which this angle is a maximum.

5

09G8

Let $ABCD$ be a circumscribed quadrilateral. Let g be a line through A which meets the segment BC in M and the line CD in N . Denote by I_1 , I_2 and I_3 the incenters of $\triangle ABM$, $\triangle MNC$ and $\triangle NDA$, respectively. Prove that the orthocenter of $\triangle I_1I_2I_3$ lies on g .

5

13G5

Let $ABCDEF$ be a convex hexagon with $AB = DE$, $BC = EF$, $CD = FA$, and $\angle A - \angle D = \angle C - \angle F = \angle E - \angle B$. Prove that the diagonals AD , BE , and CF are concurrent.

5

11G7

Let $ABCDEF$ be a convex hexagon all of whose sides are tangent to a circle ω with centre O . Suppose that the circumcircle of triangle ACE is concentric with ω . Let J be the foot of the perpendicular from B to CD . Suppose that the perpendicular from B to DF intersects the line EO at a point K . Let L be the foot of the perpendicular from K to DE . Prove that $DJ = DL$.

6

16G6

Let $ABCD$ be a convex quadrilateral with $\angle ABC = \angle ADC < 90^\circ$. The internal angle bisectors of $\angle ABC$ and $\angle ADC$ meet AC at E and F respectively, and meet each other at point P . Let M be the midpoint of AC and let ω be the circumcircle of triangle BPD . Segments BM and DM intersect ω again at X and Y respectively. Denote by Q the intersection point of lines XE and YF . Prove that $PQ \perp AC$.

6

15G6

Let ABC be an acute triangle with $AB > AC$. Let Γ be its circumcircle, H its orthocenter, and F the foot of the altitude from A . Let M be the midpoint of BC . Let Q be the point on Γ such that $\angle HQA = 90^\circ$ and let K be the point on Γ such that $\angle HKQ = 90^\circ$. Assume that the points A , B , C , K and Q are all different and lie on Γ in this order. Prove that the circumcircles of triangles KQH and FKM are tangent to each other

6

15G5

Let ABC be a triangle with $CA \neq CB$. Let D , F , and G be the midpoints of the sides AB , AC , and BC respectively. A circle Γ passing through C and tangent to AB at D meets the segments AF and BG at H and I , respectively. The points H' and I' are symmetric to H and I about F and G , respectively. The line $H'I'$ meets CD and FG at Q and M , respectively. The line CM meets Γ again at P . Prove that $CQ = QP$.

6

14G5

Convex quadrilateral $ABCD$ has $\angle ABC = \angle CDA = 90^\circ$. Point H is the foot of the perpendicular from A to BD . Points S and T lie on sides AB and AD , respectively, such that H lies inside triangle SCT and $\angle CHS - \angle CSB = 90^\circ$, $\angle THC - \angle DTC = 90^\circ$. Prove that line BD is tangent to the circumcircle of triangle TSH .

6

14G6

Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB , respectively, and let M be the midpoint of EF . Let the perpendicular bisector of EF intersect the line BC at K , and let the perpendicular bisector of MK intersect the lines AC and AB at S and T , respectively. We call the pair (E, F) interesting, if the quadrilateral $KSAT$ is cyclic.

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that

$$\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}$$

7

09G6

Let the sides AD and BC of the quadrilateral $ABCD$ (such that AB is not parallel to CD) intersect at point P . Points O_1 and O_2 are circumcenters and points H_1 and H_2 are orthocenters of triangles ABP and CDP , respectively. Denote the midpoints of segments O_1H_1 and O_2H_2 by E_1 and E_2 , respectively. Prove that the perpendicular from E_1 on CD , the perpendicular from E_2 on AB and the line H_1H_2 are concurrent.

7

Let ABC be a triangle with circumcircle Γ and incenter I and let M be the midpoint of $\overline{BC}$. The points D, E, F are selected on sides $\overline{BC}, \overline{CA}, \overline{AB}$ such that $\overline{ID} \perp \overline{BC}$, $\overline{IE} \perp \overline{AC}$, and $\overline{IF} \perp \overline{AB}$. Suppose that the circumcircle of $\triangle AEF$ intersects Γ at a point X other than A . Prove that lines XD and AM meet on Γ .

7

14G7

Let ABC be a triangle with circumcircle Ω and incentre I . Let the line passing through I and perpendicular to CI intersect the segment BC and the arc BC (not containing A) of Ω at points U and V , respectively. Let the line passing through U and parallel to AI intersect AV at X , and let the line passing through V and parallel to AI intersect AB at Y . Let W and Z be the midpoints of AX and BC , respectively. Prove that if the points I, X , and Y are collinear, then the points I, W , and Z are also collinear.

Insights

Advanced Geometric Inequalities

Sorry, nothing to offer....

Incircle and Tangents

A subclass of problems that very much feels like projective geometry. In fact, typical uses of lengths (besides a few lemmas) are almost useless. The center concept is homothety: Understanding what homothety sends each point will derive some collinearity. Monge's theorem is essential: Pairwise homothetic centers of three circles are collinear.

Advanced Symmetric Constructions

As with the 'normal' ones. Even though the statements are symmetric, one must still attempt to break the symmetry but focus on 1 or 2 sides of three on a time.

N

Number Theory

Introductory Problems

Description

Illustrate the most elementary ideas of number theory.

Main Idea

Number Theory is heavily knowledge dependent. Awareness of all necessary theorems, tricks, concepts and ideas are essential to solving all NT problems. Conversely, Number theory problems are comparatively less complex to other aspect given these knowledge as one only need to consider a few ideas. Once in a while, combinatorial ideas will be useful.

Ways To Improve

It is essential to understand all relevant ideas. Pay attention to trainings, find Elementary number theory books, and online notes to get an idea. At the very least, one must be comfortable with divisibility , mod, primes, factorisation.

Sequences

1
06N2

For $x \in (0,1)$ let $y \in (0,1)$ be the number whose n -th digit after the decimal point is the 2^n -th digit after the decimal point of x . Show that if x is rational then so is y .

1
15N1

Determine all positive integers M such that the sequence $a_0, a_1, a_2, \dots$ defined by $a_0 = M + \frac{1}{2}$ and $a_{k+1} = a_k \lfloor a_k \rfloor$ for $k = 0, 1, 2, \dots$ contains at least one integer term.

2
17Q1
[N1]

For each integer $a_0 > 1$, define the sequence $a_0, a_1, a_2, \dots$ for $n \geq 0$ as

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

Determine all values of a_0 such that there exists a number A such that $a_n = A$ for infinitely many values of n .

2
05N1

Determine all positive integers relatively prime to all the terms of the infinite sequence $a_n = 2^n + 3^n + 6^n - 1$, $n \geq 1$.

2
09N1

Let n be a positive integer and let $a_1, a_2, a_3, \dots, a_k$ ($k \geq 2$) be distinct integers in the set $1, 2, \dots, n$ such that n divides $a_i(a_{i+1} - 1)$ for $i = 1, 2, \dots, k-1$. Prove that n does not divide $a_k(a_1 - 1)$.

2
03N1

Let m be a fixed integer greater than 1. The sequence $x_0, x_1, x_2, \dots$ is defined as follows:

$$x_i = 2^i \text{ if } 0 \leq i \leq m-1 \text{ and } x_i = \sum_{j=1}^m x_{i-j}, \text{ if } i \geq m.$$

Find the greatest k for which the sequence contains k consecutive terms divisible by m .

Let $a_0, a_1, a_2, \dots$ be a sequence of positive integers such that the greatest common divisor of any two consecutive terms is greater than the preceding term; in symbols, $\gcd(a_i, a_{i+1}) > a_{i-1}$. Prove that $a_n \geq 2^n$ for all $n \geq 0$.

3

08N3

Typical Diophantine and divisibility equations

1

00N5

Prove that there exist infinitely many positive integers n such that $p = nr$, where p and r are respectively the semi-perimeter and the in-radius of a triangle with integer side lengths.

1

06N1

Determine all pairs (x, y) of integers such that $1 + 2^x + 2^{2x+1} = y^2$.

1

07N1

Find all pairs of natural numbers (a, b) such that $7^a - 3^b$ divides $a^4 + b^2$.

2

10N2

Find all pairs (m, n) of nonnegative integers for which $m^2 + 2 \cdot 3^n = m(2^{n+1} - 1)$.

2

16N4

Let n, m, k and l be positive integers with $n \neq 1$ such that $n^k + mn^l + 1$ divides $n^{k+l} - 1$. Prove that

$m = 1$ and $l = 2k$; or

$l \mid k$ and $m = \frac{n^{k-l} - 1}{n^l - 1}$.

2

12N2

Find all triples (x, y, z) of positive integers such that $x \leq y \leq z$ and

$x^3(y^3 + z^3) = 2012(xyz + 2)$.

2

14N2

Determine all pairs (x, y) of positive integers such that

$$\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1.$$

2

02N2

Let $n \geq 2$ be a positive integer, with divisors $1 = d_1 < d_2 < \dots < d_k = n$. Prove that $d_1d_2 + d_2d_3 + \dots + d_{k-1}d_k$ is always less than n^2 , and determine when it is a divisor of n^2 .

3

08N1

Let n be a positive integer and let p be a prime number. Prove that if a, b, c are integers (not necessarily positive) satisfying the equations $a^n + pb = b^n + pc = c^n + pa$ then $a = b = c$.

3

12N4(b)

Determine if $(m^2 + n)(n^2 + m) = 2(m - n)^3$ has a solution over the positive integers.

3

15N5

Find all positive integers (a, b, c) such that

$ab - c, \quad bc - a, \quad ca - b$ are all powers of 2.

4

01N2

Consider the system $x + y = z + u, \quad 2xy = zu$. Find the greatest value of the real constant m such that $m \leq x/y$ for any positive integer solution (x, y, z, u) of the system, with $x \geq y$.

4

02N4

Is there a positive integer m such that the equation

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{abc} = \frac{m}{a + b + c}$$

has infinitely many solutions in positive integers a, b, c ?

4

03N3

Determine all pairs of positive integers (a, b) such that $\frac{a^2}{2ab^2 - b^3 + 1}$ is a positive integer.

5

09N4

Find all positive integers n such that there exists a sequence of positive integers $a_1, a_2, \dots, a_n$ satisfying: $a_{k+1} = \frac{a_k^2 + 1}{a_{k-1} + 1} - 1$ for every k with $2 \leq k \leq n - 1$.

Unique Problems

1

00N1

Determine all positive integers $n \geq 2$ that satisfy the following condition: for all a and b relatively prime to n we have $a \equiv b \pmod{n}$ if and only if $ab \equiv 1 \pmod{n}$.

1

02N1

What is the smallest positive integer t such that there exist integers $x_1, x_2, \dots, x_t$ with $x_1^3 + x_2^3 + \dots + x_t^3 = 2002^{2002}$?

2

16C2

Find all positive integers n for which all positive divisors of n can be put into the cells of a rectangular table such that each cell contains a distinct divisor, the sums of all rows are equal; and the sums of all columns are equal.

2

01N1

Prove that there is no positive integer n such that, for $k = 1, 2, \dots, 9$, the leftmost digit (in decimal notation) of $(n + k)!$ equals k .

2

09N2

A positive integer N is called balanced, if $N = 1$ or if N can be written as a product of an even number of not necessarily distinct primes. Given positive integers a and b , consider the polynomial P defined by $P(x) = (x + a)(x + b)$. Prove that if $P(n)$ is balanced for all positive integers n , then $a = b$.

2

05N3

Let a, b, c, d, e, f be positive integers and let $S = a + b + c + d + e + f$. Suppose that the number S divides $abc + def$ and $ab + bc + ca - de - ef - df$. Prove that S is composite.

3

00N3

Does there exist a positive integer n such that n has exactly 2000 prime divisors and n divides $2^n + 1$?

3

04N5

We call a positive integer alternating if every two consecutive digits in its decimal representation are of different parity.

Find all positive integers n such that n has a multiple which is alternating.

3

Find the least positive integer n for which there exists a set $\{s_1, s_2, \dots, s_n\}$ consisting of n distinct positive integers such that

$$\left(1 - \frac{1}{s_1}\right) \left(1 - \frac{1}{s_2}\right) \cdots \left(1 - \frac{1}{s_n}\right) = \frac{51}{2010}.$$

3

13N2

Prove that for any positive integers k and n , there exist positive integers

$$m_1, \dots, m_k \text{ such that } 1 + \frac{2^k - 1}{n} = \left(1 + \frac{1}{m_1}\right) \cdots \left(1 + \frac{1}{m_k}\right).$$

Insights

Sequences

Sequences is a fairly common settings for number theory problems. There is nothing special about sequences however, except the deterministic nature: Same input leads to same result. It is up to readers to understand how this concept could apply to sequence questions.

Typical Diophantine and divisibility equations

These problems are the “Pure” number theory questions. General techniques include considering the equation under some mod. if there is no solution, then an appropriate modding could be enough to contradict. If there are solutions however, then mod itself can’t reject all incorrect solutions, since the actual solutions is equal to some other solutions under mod. Mod can still be used to gain more information on the terms/ variables.

The second technique to follow up on the above is bounding. Any divisibility equation naturally gives a bounding: $a|b, b \neq 0, \implies |a| \leq |b|$, used alongside with obvious bounds, such as exponential < Polynomial failing for slightly large values. These techniques combined is generally enough to give a full solution.

Unique Problems

These problems may require the use of techniques that is only ever useful once. No two problems share the same idea. The only way to prepare for these problems are (1) Improve general problem solving skills (2) Learn some extra topics in Number Theory.

Core Problems

Description

Collection of classic number theory problems.

Main Idea

Same carries from elementary problems, Number Theory is heavily knowledge dependent. Awareness of all necessary theorems, tricks, concepts and ideas are essential to solving all NT problems. In these problems, contestants have to make use of more advanced ideas, such as order, groups structure mod prime, lifting the exponent.

Ways To Improve

The same carries over. Pay attention to trainings, find Elementary number theory books, and online notes to get an idea. At this level, order, groups, LTE are all regarded as fundamentals.

Pure Evaluation NT

2

04N6

Given an integer $n > 1$, denote by P_n the product of all positive integers x less than n and such that n divides $x^2 - 1$. For each $n > 1$, find the remainder of P_n on division by n .

4

07N4

For every integer $k \geq 2$, prove that 2^{3k} divides the number

$$\binom{2^{k+1}}{2^k} - \binom{2^k}{2^{k-1}} \text{ but } 2^{3k+1} \text{ does not.}$$

4

08N4

Let n be a positive integer. Show that the numbers

$$\binom{2^n - 1}{0}, \binom{2^n - 1}{1}, \binom{2^n - 1}{2}, \dots, \binom{2^n - 1}{2^{n-1} - 1}$$

are congruent modulo 2^n to $1, 3, 5, \dots, 2^n - 1$ in some order

NT-Functional

Let $\mathbb{Z}_{>0}$ be the set of positive integers. Find all functions $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ such that

$$m^2 + f(n) \mid mf(m) + n$$

for all positive integers m and n .

1

13N1

Call admissible a set A of integers that has the following property:

If $x, y \in A$ (possibly $x = y$) then $x^2 + kxy + y^2 \in A$ for every integer k .

Determine all pairs m, n of nonzero integers such that the only admissible set containing both m and n is the set of all integers.

1

12N1

Find all functions $f : \mathbb{N}^* \rightarrow \mathbb{N}^*$ satisfying

$$((f(m))^2 + f(n)) \mid (m^2 + n)^2$$

for any two positive integers m and n .

3

04N3

4

07N5

Find all surjective functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m + n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p .

4

09N3

Let f be a non-constant function from the set of positive integers into the set of positive integers, such that $a - b$ divides $f(a) - f(b)$ for all distinct positive integers a, b . Prove that there exist infinitely many primes p such that p divides $f(c)$ for some positive integer c .

4

10N5

Find all functions $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $(g(m) + n)(g(n) + m)$ is a perfect square for all $m, n \in \mathbb{N}$.

4

11N3

Let $n \geq 1$ be an odd integer. Determine all functions f from the set of integers to itself, such that for all integers x and y the difference $f(x) - f(y)$ divides $x^n - y^n$.

4

16N6

Denote by $\mathbb{N}$ the set of all positive integers. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for all positive integers m and n , the integer $f(m) + f(n) - mn$ is nonzero and divides $mf(m) + nf(n)$.

5

11N5

Let f be a function from the set of integers to the set of positive integers. Suppose that, for any two integers m and n , the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.

5

12N5

For a nonnegative integer n define $\text{rad}(n) = 1$ if $n = 0$ or $n = 1$, and $\text{rad}(n) = p_1 p_2 \cdots p_k$ where $p_1 < p_2 < \cdots < p_k$ are all prime factors of n . Find all polynomials $f(x)$ with nonnegative integer coefficients such that $\text{rad}(f(n))$ divides $\text{rad}(f(n^{\text{rad}(n)}))$ for every nonnegative integer n .

For every $n \in \mathbb{N}$ let $d(n)$ denote the number of (positive) divisors of n . Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ with the following properties:

$$d(f(x)) = x \text{ for all } x \in \mathbb{N}.$$

$$f(xy) \text{ divides } (x-1)y^{xy-1}f(x) \text{ for all } x, y \in \mathbb{N}.$$

5

08N5

High power diophantine and divisibility

2

00N4

Find all triplets of positive integers (a, m, n) such that $a^m + 1 \mid (a+1)^n$.

2

05N6

Let a, b be positive integers such that $b^n + n$ is a multiple of $a^n + n$ for all positive integers n . Prove that $a = b$.

3

06N5

Find all integer solutions of the equation $\frac{x^7 - 1}{x - 1} = y^5 - 1$.

3

06N7

For all positive integers n , show that there exists a positive integer m such that n divides $2^m + m$.

5

12N6

Let x and y be positive integers. If $x^{2^n} - 1$ is divisible by $2^n y + 1$ for every positive integer n , prove that $x = 1$.

5

14N5

Find all triples (p, x, y) consisting of a prime number p and two positive integers x and y such that $x^{p-1} + y$ and $x + y^{p-1}$ are both powers of p .

Existence

Let $\tau(n)$ denote the number of positive divisors of the positive integer n . Prove that there exist infinitely many positive integers a such that the equation $\tau(an) = n$ does not have a positive integer solution n .

2
04N1

A positive integer N is called balanced, if $N = 1$ or if N can be written as a product of an even number of not necessarily distinct primes. Given positive integers a and b , consider the polynomial P defined by $P(x) = (x + a)(x + b)$. Prove that there exist distinct positive integers a and b such that all the number $P(1), P(2), \dots, P(50)$ are balanced.

2
09N2

We define a sequence $(a_1, a_2, a_3, \dots)$ by

$$a_n = \frac{1}{n} \left(\left\lfloor \frac{n}{1} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor + \dots + \left\lfloor \frac{n}{n} \right\rfloor \right), \text{ where } \lfloor x \rfloor \text{ denotes the integer part of } x.$$

2
06N3

a) Prove that $a_{n+1} > a_n$ infinitely often.

b) Prove that $a_{n+1} < a_n$ infinitely often.

3

13N3

Prove that there exist infinitely many positive integers n such that the largest prime divisor of $n^4 + n^2 + 1$ is equal to the largest prime divisor of $(n + 1)^4 + (n + 1)^2 + 1$.

3

12N4

(a)

An integer a is called friendly if the equation $(m^2 + n)(n^2 + m) = a(m - n)^3$ has a solution over the positive integers. Prove that there are at least 500 friendly integers in the set $\{1, 2, \dots, 2012\}$.

3

01N4

Let $p \geq 5$ be a prime number. Prove that there exists an integer a with $1 \leq a \leq p - 2$ such that neither $a^{p-1} - 1$ nor $(a + 1)^{p-1} - 1$ is divisible by p^2 .

Let $n > 1$ be a given integer. Prove that infinitely many terms of the sequence $(a_k)_{k \geq 1}$, defined by $a_k = \left\lfloor \frac{n^k}{k} \right\rfloor$, are odd. (For a real number x , $\lfloor x \rfloor$ denotes the largest integer not exceeding x .)

3

14N4

Let m and n be positive integers such that $m > n$. Define $x_k = \frac{m+k}{n+k}$ for $k = 1, 2, \dots, n+1$. Prove that if all the numbers $x_1, x_2, \dots, x_{n+1}$ are integers, then $x_1 x_2 \dots x_{n+1} - 1$ is divisible by an odd prime.

3

15N3

Prove that there are infinitely many positive integers n such that $n^2 + 1$ has a prime divisor greater than $2n + \sqrt{2n}$.

4

08N6

Special Functions

1

For any integer $d > 0$, let $f(d)$ be the smallest possible integer that has exactly d positive divisors (so for example we have $f(1) = 1, f(5) = 16$, and $f(6) = 12$). Prove that for every integer $k \geq 0$ the number $f(2^k)$ divides $f(2^{k+1})$.

11N1

2

Let a and b be positive integers such that $a! + b!$ divides $a!b!$. Prove that $3a \geq 2b + 2$.

15N2

2

For a positive integer n , let $d(n)$ be the number of all positive divisors of n . Find all positive integers n such that $d(n)^3 = 4n$.

00N2

2

04N2

The function f from the set $\mathbb{N}$ of positive integers into itself is defined by the equality $f(n) = \sum_{k=1}^n \gcd(k, n)$, $n \in \mathbb{N}$.

- a) Prove that $f(mn) = f(m)f(n)$ for every two relatively prime $m, n \in \mathbb{N}$.
- b) Prove that for each $a \in \mathbb{N}$ the equation $f(x) = ax$ has a solution.
- c) Find all $a \in \mathbb{N}$ such that the equation $f(x) = ax$ has a unique solution.

2

16N2

Let $\tau(n)$ be the number of positive divisors of n . Let $\tau_1(n)$ be the number of positive divisors of n which have remainders 1 when divided by 3. Find all positive integral values of the fraction $\frac{\tau(10n)}{\tau_1(10n)}$.

3

03N2

Each positive integer a is subjected to the following procedure, yielding the number $d = d(a)$:

- (a) The last digit of a is moved to the first position. The resulting number is called b .
- (b) The number b is squared. The resulting number is called c .
- (c) The first digit of c is moved to the last position. The resulting number is called d .

(All numbers are considered in the decimal system.) For instance, $a = 2003$ gives $b = 3200$, $c = 10240000$ and $d = 02400001 = 2400001 = d(2003)$.

Find all integers a such that $d(a) = a^2$.

3

11N4

For each positive integer k , let $t(k)$ be the largest odd divisor of k . Determine all positive integers a for which there exists a positive integer n , such that all the differences

$t(n+a) - t(n); t(n+a+1) - t(n+1), \dots, t(n+2a-1) - t(n+a-1)$ are divisible by 4.

3

05N4

Find all positive integers n such that there exists a unique integer a such that $0 \leq a < n!$ with the following property:

$$n! \mid a^n + 1$$

4

05N5

Denote by $d(n)$ the number of divisors of the positive integer n . A positive integer n is called highly divisible if $d(n) > d(m)$ for all positive integers $m < n$.

Two highly divisible integers m and n with $m < n$ are called consecutive if there exists no highly divisible integer s satisfying $m < s < n$.

(a) Show that there are only finitely many pairs of consecutive highly divisible integers of the form (a, b) with $a \mid b$.

(b) Show that for every prime number p there exist infinitely many positive highly divisible integers r such that pr is also highly divisible.

Insights

Pure Evaluation Number Theory

The problem requires some expressions to be evaluated. It may be a surprise that there may not necessarily exist a nice and simple transformation. Therefore, the quality for these problems is to be patient, and apply standard ideas from the book to proceed.

NT-Functional

Functionals that have divisibility in condition. They share the same general ideas with functional equations in algebra: appropriate substitution of values. However, dynamics are quite different. In general, $a|b$ cannot provide some very useful properties about a nor b .

Applying the principle of “Power Tap”, we should look for the most powerful way to make use of a divisibility condition. With a slight investigation, it is the most powerful in the following case: If b is a product of a few items, then having a being prime is the most powerful, as one of the items must be divisible by p . Same conclusion is not possible with composites. Therefore, in general it is useful to consider primes first in these functionals. If all prime values are confirmed, then we have infinitely known values. It is not difficult to confirm all non-prime values from here.

High power diophantine and divisibility

These diophantine makes use of high power. Therefore a proper knowledge of them is necessary. The most important idea in this is Lifting the Exponent, a frequently seen lemma that describes the behaviour of $a^n - b^n \pmod{p^n}$ ($n > 1$). The lemma gives an instinctive image of the structure in brain, which in general aid the solving of these questions.

Existence

The problems ask for existence of something. In general, there are two methods to show this: 1) By explicit construction. If one can give a formula or value such that the statement always hold, then existence is clearly proven. 2) By contradiction. To show something would fail if there is absolutely no such objects. To do this, try to “Power Tap”, think of a condition that makes use of non-existence everywhere completely.

Special Functions

These problems ask for properties of some special functions. There are not much known properties about these functions, except by their definition. Therefore, the most nature idea is follow the definition and attempt to apply general problem solving techniques on the way.

Unconventional Number Theory

Description

Collection of number theory problems that have less seen ideas.

Main Idea

Same advice: Awareness of all necessary theorems, tricks, concepts and ideas are essential to solving all NT problems.

Ways To Improve

After making sure you have the knowledge requirements, doing more problems is a consistent way to improve.

Advanced Sequences

3

01N3

Let $a_1 = 11^{11}$, $a_2 = 12^{12}$, $a_3 = 13^{13}$, and $a_n = |a_{n-1} - a_{n-2}| + |a_{n-2} - a_{n-3}|$, $n \geq 4$. Determine $a_{14^{14}}$.

4

04N4

Let k be a fixed integer greater than 1, and let $m = 4k^2 - 5$. Show that there exist positive integers a and b such that the sequence (x_n) defined by $x_0 = a$, $x_1 = b$, $x_{n+2} = x_{n+1} + x_n$ for $n = 0, 1, 2, \dots$, has all of its terms relatively prime to m .

4

14N7

Let $c \geq 1$ be an integer. Define a sequence of positive integers by $a_1 = c$ and $a_{n+1} = a_n^3 - 4c \cdot a_n^2 + 5c^2 \cdot a_n + c$ for all $n \geq 1$. Prove that for each integer $n \geq 2$ there exists a prime number p dividing a_n but none of the numbers $a_1, \dots, a_{n-1}$.

5

10N6

The rows and columns of a $2^n \times 2^n$ table are numbered from 0 to $2^n - 1$. The cells of the table have been coloured with the following property being satisfied: for each $0 \leq i, j \leq 2^n - 1$, the j -th cell in the i -th row and the $(i + j)$ -th cell in the j -th row have the same colour. (The indices of the cells in a row are considered modulo 2^n .) Prove that the maximal possible number of colours is 2^n .

6

09N6

Let k be a positive integer. Show that if there exists a sequence $a_0, a_1, \dots$ of integers satisfying the condition $a_n = \frac{a_{n-1} + n^k}{n}$ for all $n \geq 1$, then $k - 2$ is divisible by 3.

Sufficiently large and Infinity

Consider a polynomial $P(x) = \prod_{j=1}^9 (x + d_j)$, where $d_1, d_2, \dots, d_9$ are nine distinct integers. Prove that there exists an integer N , such that for all integers $x \geq N$ the number $P(x)$ is divisible by a prime number greater than 20.

2

11N2

Let $b, n > 1$ be integers.

Suppose that for each $k > 1$ there exists an integer a_k such that $b - a_k^n$ is divisible by k . Prove that $b = A^n$ for some integer A .

2

07N2

Determine whether there exists an infinite sequence of nonzero digits $a_1, a_2, a_3, \dots$ and a positive integer N such that for every integer $k > N$, the number $\overline{a_k a_{k-1} \dots a_1}$ is a perfect square.

4

13N4

Let b be an integer greater than 5. For each positive integer n , consider the number $x_n = \underbrace{11 \dots 1}_{n-1} \underbrace{22 \dots 2}_n 5$, written in base b .

5

o3N4

Prove that the following condition holds if and only if $b = 10$: there exists a positive integer M such that for any integer n greater than M , the number x_n is a perfect square.

Integer Polynomial

1

For any positive integer k , denote the sum of digits of k in its decimal representation by $S(k)$. Find all polynomials $P(x)$ with integer coefficients such that for any positive integer $n \geq 2016$, the integer $P(n)$ is positive and

16N1 $S(P(n)) = P(S(n))$.

4

09N5

Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function T from the set of integers into the set of integers such that the number of integers x with $T^n(x) = x$ is equal to $P(n)$ for every $n \geq 1$, where T^n denotes the n -fold application of T .

5

06N4

Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients and let k be a positive integer. Consider the polynomial $Q(x) = P(P(\dots P(P(x)) \dots))$, where P occurs k times. Prove that there are at most n integers t such that $Q(t) = t$.

Combin-NT

2

14N1

Let $n \geq 2$ be an integer, and let A_n be the set $A_n = \{2^n - 2^k \mid k \in \mathbb{Z}, 0 \leq k < n\}$. Determine the largest positive integer that cannot be written as the sum of one or more (not necessarily distinct) elements of A_n .

3

07N3

Let X be a set of 10,000 integers, none of them is divisible by 47. Prove that there exists a 2007-element subset Y of X such that $a - b + c - d + e$ is not divisible by 47 for any $a, b, c, d, e \in Y$.

3

16N3

A set of positive integers is called fragrant if it contains at least two elements and each of its elements has a prime factor in common with at least one of the other elements. Let $P(n) = n^2 + n + 1$. What is the least possible positive integer value of b such that there exists a non-negative integer a for which the set $\{P(a+1), P(a+2), \dots, P(a+b)\}$ is fragrant?

5

02N5

Let $m, n \geq 2$ be positive integers, and let $a_1, a_2, \dots, a_n$ be integers, none of which is a multiple of m^{n-1} . Show that there exist integers $e_1, e_2, \dots, e_n$, not all zero, with $|e_i| < m$ for all i , such that $e_1 a_1 + e_2 a_2 + \dots + e_n a_n$ is a multiple of m^n .

5

03N8

Let p be a prime number and let A be a set of positive integers that satisfies the following conditions: (1) the set of prime divisors of the elements in A consists of $p - 1$ elements; (2) for any nonempty subset of A , the product of its elements is not a perfect p -th power. What is the largest possible number of elements in A ?

5

07N7

For a prime p and a given integer n let $\nu_p(n)$ denote the exponent of p in the prime factorisation of $n!$. Given $d \in \mathbb{N}$ and $\{p_1, p_2, \dots, p_k\}$ a set of k primes, show that there are infinitely many positive integers n such that $d \mid \nu_{p_i}(n)$ for all $1 \leq i \leq k$.

Show that the set of positive integers that cannot be represented as a sum of distinct perfect squares is finite.

6

00N6

Advanced unique problems

3

02N3

Let $p_1, p_2, \dots, p_n$ be distinct primes greater than 3. Show that $2^{p_1 p_2 \cdots p_n} + 1$ has at least 4^n divisors.

4

10N3

Find the smallest number n such that there exist polynomials $f_1, f_2, \dots, f_n$ with rational coefficients satisfying $x^2 + 7 = f_1(x)^2 + f_2(x)^2 + \dots + f_n(x)^2$.

4

10N4

Let a, b be integers, and let $P(x) = ax^3 + bx$. For any positive integer n we say that the pair (a, b) is n -good if $n \mid P(m) - P(k)$ implies $n \mid m - k$ for all integers m, k . We say that (a, b) is *very good* if (a, b) is n -good for infinitely many positive integers n .

(a) Find a pair (a, b) which is 51-good, but not very good.

(b) Show that all 2010-good pairs are very good.

4

16N5

Let a be a positive integer which is not a perfect square, and consider the equation $k = \frac{x^2 - a}{x^2 - y^2}$. Let A be the set of positive integers k for which the equation admits a solution with $x > \sqrt{a}$, and let B be the set of positive integers for which the equation admits a solution with $0 \leq x < \sqrt{a}$. Show that $A = B$.

5

01N5

Let $a > b > c > d$ be positive integers and suppose that $ac + bd = (b + d + a - c)(b + d - a + c)$. Prove that $ab + cd$ is not prime.

Insights

Advanced Sequences

Always be prepared for new and fresh ideas, although simple ideas for sequences mentioned in Introductory problems is always helpful.

Sufficiently large and Infinity

The main idea is to realise infinity is a large number, indeed! One can make very large jumps, consider very large numbers, consider an infinite collection etc.

Pigeon-hole is a way to make use of infinity: If infinitely many items take value from a finite set, then a certain value is taken infinitely many times.

Integer Polynomial

Integer polynomial is a rare topic, since there is one single theorem that is very powerful, too powerful in fact. If $P(x)$ is a polynomial with integer coefficients, then $a - b \mid P(a) - P(b)$. This is the most powerful and useful property. Powerful in the sense that the converse is also true, with the assumption that the function is bounded by polynomial-growth. Therefore, this simple statement defines inter-conversion of a global feature (Integer coefficients) to local features (Above statement), giving amazing perspectives to work on.

Combin-NT

Combine with NT settings. Keep an eye on opportunities to use combinatorial techniques.

Advanced unique problems

They are too unique and pairwise distinct for me to offer anything.

Advanced Number Theory

Description

Highly advanced number theory questions.

Main Idea

Adding onto Tier1/2 Number theory problems, they require a creative , somewhat wishful plan. Come out with a plan and see if it is feasible. If not, reject and try again. Indeed, some statements are open - impossible to reject or proof in contest conditions. It is an essential ability to realise some of the plans is not quite right, and do not consider them.

Insights

I have no insight to provide for this collection of problems, for they are advanced versions of previous problems, and would have given away the main idea had I point out something useful.

Advanced NT functional

6
13N6

Determine all functions $f : \mathbb{Q} \rightarrow \mathbb{Z}$ satisfying

$$f\left(\frac{f(x) + a}{b}\right) = f\left(\frac{x + a}{b}\right)$$

for all $x \in \mathbb{Q}$, $a \in \mathbb{Z}$, and $b \in \mathbb{Z}_{>0}$. (Here, $\mathbb{Z}_{>0}$ denotes the set of positive integers.)

6
15N6

Let $\mathbb{Z}_{>0}$ denote the set of positive integers. Consider a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$. For any $m, n \in \mathbb{Z}_{>0}$ we write $f^n(m) = \underbrace{f(f(\dots f(m)\dots))}_n$. Suppose that f has the following

two properties:

(i) if $m, n \in \mathbb{Z}_{>0}$, then $\frac{f^n(m) - m}{n} \in \mathbb{Z}_{>0}$;

(ii) The set $\mathbb{Z}_{>0} \setminus \{f(n) \mid n \in \mathbb{Z}_{>0}\}$ is finite.

Prove that the sequence $f(1) - 1, f(2) - 2, f(3) - 3, \dots$ is periodic.

6
15N7

Let $\mathbb{Z}_{>0}$ denote the set of positive integers. For any positive integer k , a function $f : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0}$ is called k -good if $\gcd(f(m) + n, f(n) + m) \leq k$ for all $m \neq n$. Find all k such that there exists a k -good function.

7
15N8

For every positive integer n with prime factorization $n = \prod_{i=1}^k p_i^{\alpha_i}$, define

$\mathfrak{U}(n) = \sum_{i: p_i > 10^{100}} \alpha_i$. That is, $\mathfrak{U}(n)$ is the number of prime factors of n greater than

10^{100} , counted with multiplicity.

Find all strictly increasing functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$\mathfrak{U}(f(a) - f(b)) \leq \mathfrak{U}(a - b)$ for all integers a and b with $a > b$.

Advanced Evaluation Problems

7

03N7

The sequence $a_0, a_1, a_2, \dots$ is defined as follows: $a_0 = 2$, $a_{k+1} = 2a_k^2 - 1$ for $k \geq 0$. Prove that if an odd prime p divides a_n , then 2^{n+3} divides $p^2 - 1$.

7

11N7

Let p be an odd prime number. For every integer a , define the number $S_a = \sum_{j=1}^{p-1} \frac{a^j}{j}$. Let $m, n \in \mathbb{Z}$, such that $S_3 + S_4 - 3S_2 = \frac{m}{n}$. Prove that p divides m .

7

14N6

Let $a_1 < a_2 < \dots < a_n$ be pairwise coprime positive integers with a_1 being prime and $a_1 \geq n + 2$. On the segment $I = [0, a_1 a_2 \dots a_n]$ of the real line, mark all integers that are divisible by at least one of the numbers $a_1, \dots, a_n$. These points split I into a number of smaller segments. Prove that the sum of the squares of the lengths of these segments is divisible by a_1 .

Number Theory and geometry

3

08C3

In the coordinate plane consider the set S of all points with integer coordinates. For a positive integer k , two distinct points $A, B \in S$ will be called k -friends if there is a point $C \in S$ such that the area of the triangle ABC is equal to k . A set $T \subset S$ will be called k -clique if every two points in T are k -friends. Find the least positive integer k for which there exists a k -clique with more than 200 elements.

7

04N7

Let p be an odd prime and n a positive integer. In the coordinate plane, eight distinct points with integer coordinates lie on a circle with diameter of length p^n . Prove that there exists a triangle with vertices at three of the given points such that the squares of its side lengths are integers divisible by p^{n+1} .

7

16N7

Let $P = A_1 A_2 \dots A_k$ be a convex polygon in the plane. The vertices $A_1, A_2, \dots, A_k$ have integral coordinates and lie on a circle. Let S be the area of P . An odd positive integer n is given such that the squares of the side lengths of P are integers divisible by n . Prove that $2S$ is an integer divisible by n .

Mythical unique problems

6

07N6

Let k be a positive integer. Prove that the number $(4 \cdot k^2 - 1)^2$ has a positive divisor of the form $8kn - 1$ if and only if k is even.

6

03N6

Let p be a prime number. Prove that there exists a prime number q such that for every integer n , the number $n^p - p$ is not divisible by q .

6

11N6

Let $P(x)$ and $Q(x)$ be two polynomials with integer coefficients, such that no non-constant polynomial with rational coefficients divides both $P(x)$ and $Q(x)$. Suppose that for every positive integer n the integers $P(n)$ and $Q(n)$ are positive, and $2^{Q(n)} - 1$ divides $3^{P(n)} - 1$. Prove that $Q(x)$ is a constant polynomial.

7

17Q6
[N7]

An ordered pair (x, y) of integers is a primitive point if the greatest common divisor of x and y is 1. Given a finite set S of primitive points, prove that there exist a positive integer n and integers $a_0, a_1, \dots, a_n$ such that, for each (x, y) in S , we have:

$$a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \dots + a_{n-1}xy^{n-1} + a_ny^n = 1.$$

8

11N8

Let $k \in \mathbb{Z}^+$ and set $n = 2^k + 1$. Prove that n is a prime number if and only if the following holds: there is a permutation $a_1, \dots, a_{n-1}$ of the numbers $1, 2, \dots, n-1$ and a sequence of integers $g_1, \dots, g_{n-1}$, such that n divides $g_i^{a_i} - a_{i+1}$ for every $i \in \{1, 2, \dots, n-1\}$, where we set $a_n = a_1$.

8

13N7

Let ν be an irrational positive number, and let m be a positive integer. A pair (a, b) of positive integers is called good if

$a \lceil b\nu \rceil - b \lfloor a\nu \rfloor = m$. A good pair (a, b) is called excellent if neither of the pairs $(a - b, b)$ and $(a, b - a)$ is good.

Prove that the number of excellent pairs is equal to the sum of the positive divisors of m .

8

12N8

Prove that for every prime $p > 100$ and every integer r , there exist two integers a and b such that p divides $a^2 + b^5 - r$.

Uninspiring Problems

Less elegant problems that it likely not involved in a contest. But then again, one can never be sure. Indeed, 17Q6 represented a tedious Algebra

These problems test mostly the calculation fortitudes, that one can catch onto a certain direction and finish with pure calculation

In general, one should still attempt these problems, but is of no concern if they cannot be solved.

Violent Problems

3

16A5

Consider fractions $\frac{a}{b}$ where a and b are positive integers.

(a) Prove that for every positive integer n , there exists such a fraction $\frac{a}{b}$ such that $\sqrt{n} \leq \frac{a}{b} \leq \sqrt{n+1}$ and $b \leq \sqrt{n+1}$.

(b) Show that there are infinitely many positive integers n such that no such fraction $\frac{a}{b}$ satisfies $\sqrt{n} \leq \frac{a}{b} \leq \sqrt{n+1}$ and $b \leq \sqrt{n}$.

4

02A5

Let n be a positive integer that is not a perfect cube. Define real numbers a, b, c by $a = \sqrt[3]{n}, b = \frac{1}{a - [a]}, c = \frac{1}{b - [b]}$ where $[x]$ denotes the integer part of x .

Prove that there are infinitely many such integers n with the property that there exist integers r, s, t , not all zero, such that $ra + sb + tc = 0$.

4

03N5

An integer n is said to be good if $|n|$ is not the square of an integer.

Determine all integers m with the following property: m can be represented, in infinitely many ways, as a sum of three distinct good integers whose product is the square of an odd integer.

Tedious Problems

Let $A_1A_2\dots A_n$ be a convex polygon, $n \geq 4$. Prove that $A_1A_2\dots A_n$ is cyclic if and only if to each vertex A_j one can assign a pair (b_j, c_j) of real numbers, $j = 1, 2, \dots, n$, so that $A_iA_j = b_jc_i - b_ic_j$ for all i, j with $1 \leq i < j \leq n$.

2

00G4

The sequence $c_0, c_1, \dots, c_n, \dots$ is defined by $c_0 = 1, c_1 = 0$, and $c_{n+2} = c_{n+1} + c_n$ for $n \geq 0$. Consider the set S of ordered pairs (x, y) for which there is a finite set J of positive integers such that $x = \sum_{j \in J} c_j, y = \sum_{j \in J} c_{j-1}$. Prove that there exist real numbers α, β, m, M with the following property: An ordered pair of nonnegative integers (x, y) satisfies the inequality $m < \alpha x + \beta y < M$ if and only if $(x, y) \in S$.

5

06A3

We say that a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is a metapolynomial if, for some positive integers m and n , it can be represented in the form

$$f(x_1, \dots, x_k) = \max_{i=1, \dots, m} \min_{j=1, \dots, n} P_{i,j}(x_1, \dots, x_k),$$

where $P_{i,j}$ are multivariate polynomials. Prove that the product of two metapolynomials is also a metapolynomial.

Unfair-Inequalities

7

12A7

5

07A6

Let $a_1, a_2, \dots, a_{100}$ be nonnegative real numbers such that

$$a_1^2 + a_2^2 + \dots + a_{100}^2 = 1. \text{ Prove that } a_1^2 \cdot a_2 + a_2^2 \cdot a_3 + \dots + a_{100}^2 \cdot a_1 < \frac{12}{25}.$$

7

16A8

Find the largest real constant a such that for all $n \geq 1$ and for all real numbers $x_0, x_1, \dots, x_n$ satisfying $0 = x_0 < x_1 < x_2 < \dots < x_n$ we have

$$\frac{1}{x_1 - x_0} + \frac{1}{x_2 - x_1} + \dots + \frac{1}{x_n - x_{n-1}} \geq a \left(\frac{2}{x_1} + \frac{3}{x_2} + \dots + \frac{n+1}{x_n} \right)$$

Wild Polynomial Problems

For a polynomial P of degree 2000 with distinct real coefficients let $M(P)$ be the set of all polynomials that can be produced from P by permutation of its coefficients. A polynomial P will be called n -independent if $P(n) = 0$ and we can get from any $Q \in M(P)$ a polynomial Q_1 such that $Q_1(n) = 0$ by interchanging at most one pair of coefficients of Q . Find all integers n for which n -independent polynomials exist.

8

00A7

Let n be a fixed integer with $n \geq 2$. We say that two polynomials P and Q with real coefficients are block-similar if for each $i \in \{1, 2, \dots, n\}$ the sequences

$$P(2015i), P(2015i - 1), \dots, P(2015i - 2014) \text{ and } Q(2015i), Q(2015i - 1), \dots, Q(2015i - 2014)$$

are permutations of each other.

(a) Prove that there exist distinct block-similar polynomials of degree $n + 1$.

(b) Prove that there do not exist distinct block-similar polynomials of degree n

8

15A6

Find all polynomials $P(x)$ of odd degree d and with integer coefficients satisfying the following property: for each positive integer n , there exists n positive integers $x_1, x_2, \dots, x_n$ such that $\frac{1}{2} < \frac{P(x_i)}{P(x_j)} < 2$ and $\frac{P(x_i)}{P(x_j)}$ is the d -th power of a rational number for every pair of indices i and j with $1 \leq i, j \leq n$.

8
16N8

Wild-NT-Existence Problem

6

05N7

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, where $a_0, \dots, a_n$ are integers, $a_n > 0$, $n \geq 2$. Prove that there exists a positive integer m such that $P(m!)$ is a composite number.

7

09N7

Let a and b be distinct integers greater than 1. Prove that there exists a positive integer n such that $(a^n - 1)(b^n - 1)$ is not a perfect square.

8

14N8

For every real number x , let $||x||$ denote the distance between x and the nearest integer.

Prove that for every pair (a, b) of positive integers there exist an odd prime p

and a positive integer k satisfying $\left| \left| \frac{a}{p^k} \right| \right| + \left| \left| \frac{b}{p^k} \right| \right| + \left| \left| \frac{a+b}{p^k} \right| \right| = 1$.

Computational Geometry

Let ABC be a triangle with $\angle BAC = 60^\circ$. Let AP bisect $\angle BAC$ and let BQ bisect $\angle ABC$, with P on BC and Q on AC . If $AB + BP = AQ + QB$, what are the angles of the triangle?

1
01G8

Let I' be the image of inversion of incentre I of triangle ABC with respect to circumcircle of ABC . Let one of the tangents from I' to the incircle of triangle ABC meet the circumcircle at points X and Y . Show that $\angle XIY = 120^\circ$.

2
16G7
(b)

3

09G7

Let ABC be a triangle with incenter I and let X , Y and Z be the incenters of the triangles BIC , CIA and AIB , respectively. Let the triangle XYZ be equilateral. Prove that ABC is equilateral too.

4

03G7

Let ABC be a triangle with semiperimeter s and inradius r . The semicircles with diameters BC , CA , AB are drawn on the outside of the triangle ABC . The circle tangent to all of these three semicircles has radius t . Prove that

$$\frac{s}{2} < t \leq \frac{s}{2} + \left(1 - \frac{\sqrt{3}}{2}\right)r.$$

Insights

Violent Problems

Problems that include strange statement and the most natural solution would be of considerable violence.

Tedious Problems

Problems that have a very clear flow and direction. One only need to schedule a plan and finish the computational requirements.

Unfair Inequalities

Inequalities so **non**-intuitive that no one can realistically figure them out. Failing to solve does not indicate poor abilities, and solving means understanding of certain facts only.

Wild Polynomial Problems

There are too many wild variables that are out of control. The main strategy should be trying to catch onto something that is useful for various polynomial properties.

Wild-NT-Existence Problem

These problems require proving a seemingly obvious statement, except a lot of variables are wild and out of control. There is never an intuitive solution and one could only hope that trying random things could do the trick.

Computational Geometry

Geometry problems that resemble nothing in geometry.

Mastery

The absolute hardest problems that is almost not practically solvable in contest conditions for humans.

Do not attempt until you identified yourself as Perfect-Score-Capability. Allow up to 2 hours for each question.

Legends have that those who can finish are born to be [perfect scorers](#).

I must emphasis once again that mathematical abilities is just most of the equations. At such high performance, contest skill, physical strength preparations, and luck are all extremely important.

8

07Q6

Let n be a positive integer. Consider

$S = \{(x, y, z) \mid x, y, z \in \{0, 1, \dots, n\}, x + y + z > 0\}$ as a set of $(n + 1)^3 - 1$ points in the three-dimensional space. Determine the smallest possible number of planes, the union of which contains S but does not include $(0, 0, 0)$

8

12G6

Let ABC be a triangle with circumcenter O and incenter I . The points D, E and F on the sides BC, CA and AB respectively are such that $BD + BF = CA$ and $CD + CE = AB$. The circumcircles of the triangles BFD and CDE intersect at $P \neq D$. Prove that $OP = OI$.

8

07C6

In a mathematical competition some competitors are friends. Friendship is always mutual. Call a group of competitors a clique if each two of them are friends. (In particular, any group of fewer than two competitors is a clique.) The number of members of a clique is called its size.

Given that, in this competition, the largest size of a clique is even, prove that the competitors can be arranged into two rooms such that the largest size of a clique contained in one room is the same as the largest size of a clique contained in the other room.

8

14C9

There are n circles drawn on a piece of paper in such a way that any two circles intersect in two points, and no three circles pass through the same point. Turbo the snail slides along the circles in the following fashion. Initially he moves on one of the circles in clockwise direction. Turbo always keeps sliding along the current circle until he reaches an intersection with another circle. Then he continues his journey on this new circle and also changes the direction of moving, i.e. from clockwise to anti-clockwise or vice versa.

Suppose that Turbo's path entirely covers all circles. Prove that n must be odd.

Terminology

$\mathbb{Z}$ Set of Integers

$\mathbb{Q}$ Set of Rationals

$\mathbb{R}$ Set of Real Numbers

$\mathbb{R}^+$ Set of Positive Real numbers

(x, y) An ordered pair, or the open interval between x and y

The booklet contains all recent shortlist problems, organized for self-practices.

- Difficulty rating for each problem
- Problems grouped by general settings
- Short insight for each group of problem

An additional aim for compiling the booklet is to serves as a fine reference in preparation for training materials by trainers, hopefully enhancing training quality in the process.

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